Institut Polytechnique de Paris

Master's Degree in Interaction, Graphics and Design

Social Robot versus Computer Interface for Stress Regulation: A Pilot Study of Guided Breathing with Physiological Biofeedback

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Master's Thesis

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Abstract

This pilot study compared a zoomorphic social robot, Tyto, with a computer interface during a guided breathing exercise with biofeedback at six breaths per minute, to explore whether the robot helps regulate stress better. Nineteen healthy adults aged between 21 and 62 took part and 15 were analysed (7 in the computer condition and 8 in the robot condition). All of them went through a baseline phase, a stress-induction phase with a Stroop task and a relaxation with biofeedback phase. Stress was measured using a visual analogue scale (VAS), heart rate (HR) and EEG. Usability, workload, attitudes toward robots and perception of the robot were assessed with questionnaires and an interview.

No significant differences between conditions were found in the VAS, HR or EEG measures. Across the phases, HR, relative alpha and the beta/alpha ratio changed in the direction expected for stress during the Stroop task, and relative alpha and the beta/alpha ratio moved towards recovery during relaxation. As the study had no control phase with no guided breathing, this recovery cannot be attributed to the breathing exercise alone. Usability was high in both conditions (mean SUS grade A) and the robot was perceived positively (likeability of 4.6 out of 5). Heart rate variability could not be analysed because of the limitations of the cardiac sensor, and the small sample limits the conclusions. However, relative alpha and the beta/alpha ratio seem promising measures to follow stress during these tasks, and some trends were observed that could be worth checking in a larger study.

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1 Introduction

This project was carried out within the internship framework of the Master in Interaction, Graphics and Design of the Institut Polytechnique de Paris (IP Paris), at the Escola Politècnica Superior d'Enginyeria de Manresa (EPSEM) of the Universitat Politècnica de Catalunya (UPC), within the Smart Sustainable Resources research centre (SSR-UPC).

The motivation behind this internship stems from an interest in the field of Human-Robot Interaction (HRI), particularly in social robotics and how humans interact with this type of technology, as well as its potential to improve people's quality of life. There is also a personal interest in the application of social robots in healthcare, and more specifically in mental health. Building on this interest, and after discussing potential research directions with professors at EPSEM, it was decided to investigate the effectiveness of relaxation exercises delivered through a computer interface compared to a social robot. Since prior research already exists in this area, the study was extended by recording electroencephalography (EEG) and heart-rate-related signals to assess participants' psychological stress levels during the experiment. This took advantage of the expertise of Professor Linus Manubens Gil, who works in computational neuroscience and has experience analysing brain signals. These recordings complemented the self-reported (perceived) stress levels with a more objective, physiological measure. The project was supervised by Marta Isabel Tarrés Puertas. She has previous research experience in social robotics and could support the HRI part of the project.

Psychological stress (shortened to "stress" below) is an adaptive physiological and psychological response that can become problematic when sustained over time. Stress is a common experience in contemporary contexts, where cognitive, occupational and digital demands are high. Human-Computer Interaction (HCI) research has explored how the design of interactive systems can influence users' emotional and physiological states, as well as their ability to regulate stress through digital interventions such as guided breathing or biofeedback systems. However, many of these solutions rely on purely digital interfaces, which may limit emotional engagement and the sense of social presence during the interaction. In recent years, social robotics has become established as an emerging field within HRI, with growing applications in health and emotional well-being, as social robots can interact with humans through multimodal channels that increase the sense of social presence and engagement during interaction. This suggests that the embodied presence of an interactive agent may affect stress regulation differently than a computer interface does, a hypothesis this project sets out to explore. Stress can be measured through its effects on the body. The autonomic nervous system regulates the heart, and under stress the heart rate (HR), the number of heartbeats per minute, generally increases (Giannakakis et al., 2022). The variation in time between successive heartbeats, known as heart rate variability (HRV), generally decreases as stress increases (Kennedy & Parker, 2019). Stress is also reflected in brain activity, which can be recorded with EEG using electrodes placed on the scalp. In this study, a chest strap recorded the cardiac signal and a headset recorded the EEG. HRV was the planned cardiac measure, but it could not be analysed because of a limitation of the sensor, so mean HR was used instead (see Section 7.4). Physiological monitoring adds another dimension to this comparison. Klęczek et al. (2024) found that, while questionnaire scores alone showed no significant difference between a robot-guided and a laptop-guided breathing exercise, physiological signals revealed response patterns that self-report missed. This finding motivated the combined use of cardiac and EEG signals alongside subjective measures in the present study. However, much of the existing evidence on social robots in health and well-being contexts concentrates on specific populations, particularly children and older adults, leaving open how these approaches perform among the broader adult population targeted by this study. A detailed review of this literature is presented in Chapter 2, Related Work.

Initially, the internship was proposed as a broader experimental study. It would compare humanoid social robots (such as TIAGo or Qui-Bot) against conventional digital solutions (apps or guided videos) for the regulation of chronic stress. The project plan included biometric data collection, psychometric

assessment, and collaboration with local healthcare partners (Fundació Althaia and Fundació AMPANS) to recruit participants. As the project progressed, the scope was narrowed down to a pilot proof-of-concept study. Its aim was to explore the feasibility of a physiological biofeedback system based on interaction with a social robot, compared to a conventional computer interface. The pilot study focuses on healthy adults, to obtain an initial understanding of how different interactive technologies affect stress regulation and user experience, rather than on clinical populations. It was also originally intended to use Qui-Bot, a social robot previously developed at EPSEM and already used in research. Due to access constraints, Qui-Bot could not be used, and a new robot, named Tyto, was designed and built from scratch for this experiment. This evolution of the project scope, and the reasons behind it, is discussed further in Chapter 7, Discussion.

This report does not cover clinical populations or long-term (chronic) stress interventions, which were part of the project's initial scope but fell outside the timeframe and resources available during the internship. These aspects are revisited as future work in Chapter 9. Similarly, the in-depth engineering design and validation of Tyto, while briefly described in Chapter 3 for context, falls outside the primary scope of this report. The only evaluation of the robot itself is a short analysis of how participants perceived it, based on the Godspeed questionnaire. In this project, Tyto is used as a tool to study stress regulation through HRI.

The rest of the report is organised as follows. Chapter 2 reviews the related work. Chapter 3 describes the design and implementation of the breathing exercise, the web application and the robot Tyto. Chapters 4 and 5 present the methodology and the data analysis, and Chapter 6 reports the results. Chapter 7 discusses the results in relation to the research questions and hypotheses, together with the changes, limitations and possible improvements. Finally, Chapters 8 and 9 present the conclusions and the future work.

1.1 Objectives and Research Question

The main objective of this study is to compare the effectiveness of two interaction modalities, one based on a social robot and one based on a conventional computer interface, in the physiological regulation of stress. Specifically, the study aims to obtain an initial understanding of which of the two conditions is more effective at reducing the physiological stress response and helping participants relax during guided breathing exercises.

In this report, the computer condition refers to a conventional, non-embodied interface displayed on a laptop screen. Tyto also uses screens (eyes and belly), so what differs between the two conditions is the physical embodiment and social presence of the interface, not the presence of a screen.

The secondary objectives are to: (i) analyse preliminary differences in stress-related physiological variables between the robot and computer conditions; (ii) compare participants' subjective perception of stress with objective measures obtained from physiological signals; and (iii) explore the relationship and possible correlation between physiological indicators and perceived-stress scores.

Based on these objectives, this study is guided by the following main research question:

RQ: Does interaction with a social robot lead to greater physiological and subjective stress regulation than interaction with a conventional computer interface, during guided biofeedback-based breathing exercises?

This main question can be broken down into the following sub-questions:

- **RQ1:** Are there differences in stress-related physiological indicators, such as heart rate variability (HRV) and EEG-derived measures, between the robot condition and the computer condition?
- **RQ2:** How consistent are participants' subjective stress perception and the objective physiological measures obtained during each condition?

- **RQ3:** Is there a measurable correlation between physiological indicators and perceived-stress scores, and does this relationship differ between conditions?
- **RQ4:** Does interacting with a socially present physical agent, such as a social robot, lead to a greater sense of emotional support and engagement compared to a computer interface?

1.2 Hypotheses

Building on the research questions presented above (RQ1 to RQ4), the following null and alternative hypotheses were formulated to guide the statistical analysis of the study.

RQ1 and the hypotheses H0, H2 and H4 were formulated in terms of HRV, which was the planned cardiac measure. HRV could not be analysed because of a limitation of the heart rate sensor, and mean HR was used instead (see Section 7.4). How each research question and hypothesis was answered is summarised in Section 7.3.

H0 (Null Hypothesis): There will be no significant differences between the robot condition and the computer condition in any of the stress measures, either physiological (HRV) or subjective (perceived stress).

H1: Participants interacting with the social robot will show a greater reduction in perceived stress compared to the computer condition.

H2: The robot condition will show higher HRV values, indicating better physiological stress regulation, compared to the computer condition.

H3: A partial dissociation will occur between subjective and physiological measures, such that the reduction in stress reported by the participants will be proportionally greater than the improvement observed in physiological measures.

H4: There will be a significant correlation between HRV values and perceived-stress scores, indicating consistency between objective and subjective measures.

H5: The robot condition will generate a greater sense of emotional support and engagement during the exercise.

2 Related Work

2.1 Social Robotics for Stress Regulation

Social robots have been increasingly explored across health and well-being contexts. Their embodied presence and interactivity offer a more engaging alternative to conventional digital tools, such as laptops, tablets or smartphone apps. They have been used, for example, to support children during medical care and to keep older adults company at home.

Nichol et al. (2024) conducted an umbrella review of 35 systematic reviews on socially assistive robots across the lifespan. Most of the available evidence still concerns older adults, particularly those with dementia. Only a few reviews addressed children or broader adult populations. Pooled meta-analyses found no significant effects on depression, anxiety, or quality of life. However, the same review found significant improvements in social interaction, mood, positive affect, loneliness, and stress. The authors concluded that socially assistive robots show promise for non-psychiatric outcomes such as stress, but not for psychiatric outcomes such as depression.

A related PRISMA systematic review focused specifically on psychological interventions. Duradoni et al. (2021) reviewed 39 studies across five psychological domains. Robots targeting anxiety, most commonly animal-like robots such as PARO or the Haptic Creature, produced mixed but generally positive results. These included significant reductions in state anxiety and physiological arousal in both children and young adults. Here, arousal refers to how activated the body and mind are, from calm to excited. It is usually described together with valence, which refers to how pleasant or unpleasant an emotional state is. Together, these reviews suggest that stress is one of the more consistently supported outcomes in the literature. Even so, the populations studied remain concentrated in older adults and children, rather than in healthy adults such as those targeted by the present study.

Within this context, social robots have also been explored for stress and anxiety management, understood here as techniques that help a person reduce tension and calm down, such as breathing exercises, relaxation or mindfulness. So far, most robots have delivered a broad variety of these techniques, rather than breathing exercises alone. Ab Aziz et al. (2017) developed CAKNA, a tabletop social robot that delivers relaxation techniques. These included deep breathing, body scan, mindfulness meditation, and guided imagery. In a pilot study with 24 young adults with anxiety traits (mean age 21.5), those who interacted with the robot showed a significant reduction in state anxiety after the intervention. A comparison group using an equivalent PC-based version of the same therapies did not.

Pande et al. (2025) took a more adaptive approach. A NAO robot monitored participants' facial expressions and physiological signals during a relaxation task and a stress-inducing task. Upon detecting stress, it verbally recommended remedies such as breathing exercises, stretches, or music. Most participants found these recommendations helpful, though they also suggested the interaction could be extended with more varied activities. Across these studies, breathing exercises appear only as one technique among several. This leaves open how a robot performs specifically as a guide for breathing regulation, which is the focus of the studies discussed next.

A number of studies have developed robots specifically designed to guide breathing exercises, most relying on a tactile cue rather than a purely visual or auditory one. Sefidgar et al. (2016) developed the Haptic Creature, a furred robot with a simulated moving rib cage, and found that stroking it while it "breathed" reduced participants' anxiety, respiration rate, and heart rate. Sabinson et al. (2024) found that a soft, inflating robotic surface produced a significantly greater reduction in self-reported stress than a static control. Saood et al. (2025) fitted a humanoid robot with a soft haptic chest interface simulating heartbeat and breathing, and compared it with and without the interface after a stress-inducing task. Both conditions produced similar reductions in anxiety, but participants with the haptic

interface described the robot as more natural and supportive. Thomas et al. (2025) presented Neffy, whose torso rises and falls to simulate breathing, and reported a significant reduction in state anxiety after a single session. Neffy evolved from Ommie (Matheus et al., 2025), a robot that expands and contracts against the user's hands and deliberately omits facial features and natural language. Its usability study also found a significant reduction in state anxiety. Across these studies, the recurring design choice is a breathing-like movement or a touch-based cue. However, this reflects a common design trend rather than a demonstrated necessity. Saood et al. (2025) found no significant difference in anxiety reduction between their haptic and non-haptic conditions, suggesting that touch may not be essential. This leaves open whether similar benefits can be achieved through other means, such as the visual and sound feedback used by Tyto in the present study.

Only a few works have directly compared a robot-guided intervention against a non-embodied alternative. Klęczek et al. (2024) compared a Pepper robot to a laptop in guiding university students through a deep breathing exercise, in a between-subjects design closely resembling the present one. Besides a subjective stress questionnaire, they recorded galvanic skin response (GSR) as a measure of arousal, and EEG-derived frontal alpha asymmetry (FAA) as a measure of valence. This is methodologically close to the present study's combined use of cardiac and EEG measures. Taken individually, none of the three measures differed significantly between the robot and laptop groups. However, when arousal and valence were considered jointly, the groups showed distinguishable emotional response patterns. The laptop group tended to show low arousal and negative valence, consistent with boredom or disengagement. The robot group split into two patterns: low arousal with positive valence, consistent with calmness, and high arousal with positive valence, consistent with excitement, an effect the authors attributed in part to the novelty of the robot.

The CAKNA study described above also compared a robot with a PC-based condition. In contrast to Klęczek et al. (2024), it found a significant anxiety reduction only for the robot. Taken together, these studies give a mixed and still inconclusive picture. It remains unclear whether robot embodiment offers a measurable advantage over non-embodied alternatives when relying on self-reports alone. At the same time, physiological signals may reveal differences in the underlying emotional response that self-reports fail to capture. This motivates both the robot versus computer comparison and the multi-signal approach of the present study.

2.2 Biosignals for Stress Assessment

Biosignals are electrical or physiological signals recorded from the body. Two of them are widely used to assess stress. The first comes from the heart. Heart rate (HR) is the number of heartbeats per minute, and heart rate variability (HRV) is the variation in time between successive beats. Both can be obtained from an electrocardiogram or from a wearable sensor on the chest or wrist. The second comes from the brain. Electroencephalography (EEG) records the electrical activity of the cortex through electrodes placed on the scalp, and its signal is commonly divided into frequency bands such as alpha and beta.

HR is the simplest cardiac indicator of stress. It increases when the body is under demand, because stress activates the sympathetic branch of the autonomic nervous system. In their review of stress detection, Giannakakis et al. (2022) describe HR as one of the most commonly reported physiological responses, which generally rises under stress. Its main limitation is that it is a basic measure. It reflects overall activation, but says little about how the autonomic nervous system regulates itself. HRV addresses this limitation. HRV is regulated by the autonomic nervous system, and lower HRV is generally associated with higher stress and reduced self-regulatory capacity. This link is explained by the neurovisceral integration model, which connects autonomic flexibility to emotional and cognitive regulation (Kennedy & Parker, 2019). The same review also mentions the Yerkes-Dodson curve, which describes stress as helpful up to a point (eustress) and harmful beyond it (distress). Kennedy

and Parker (2019) reviewed 17 studies published between 2000 and 2017 and found that biofeedback reliably reduces both physiological and subjective stress, while also enhancing performance.

Given the link between HRV and autonomic regulation, HRV biofeedback (HRVB) offers a way to act on the autonomic nervous system directly. Lehrer and Gevirtz (2014) explain how it works. During slow-paced breathing, heart rate oscillations align closely with respiration, a pattern known as respiratory sinus arrhythmia (RSA). HRVB trains users to maximise this pattern, producing a smooth, sine-wave-like oscillation in heart rate. This strengthens the baroreflex, the mechanism that regulates blood pressure through heart rate adjustments. The authors suggest that this increased baroreflex gain reflects improved autonomic homeostasis, meaning a better-regulated balance within the autonomic nervous system. This would explain why HRVB has been applied to a wide range of conditions, not only stress.

The effectiveness of HRVB is well documented. A review of 58 studies (Lehrer et al., 2020) and a meta-analysis of 24 studies with 484 participants (Goessl et al., 2017) both report consistent improvements in stress and anxiety outcomes. Goessl et al. (2017) found moderate to large effect sizes (Hedges' g between 0.81 and 0.83).

Individual randomised controlled trials show more mixed results. Van der Zwan et al. (2015) found that five weeks of HRVB were as effective as physical activity or mindfulness meditation, with no significant differences between the three. Brinkmann et al. (2020) randomly assigned 69 healthy adults to HRVB, a mindfulness-based intervention or a waitlist control. After six weeks, neither intervention differed significantly from the control group. The authors note that this may be due to the small sample. Other trials report stronger results. Chung et al. (2021) found that eight weeks of wearable-based HRVB significantly reduced anxiety and depressive symptoms in 14 participants, alongside an increase in RMSSD. Ratanasiripong et al. (2012) found that HRVB kept stress and anxiety stable in 60 nursing students over five weeks, while a control group's stress and anxiety increased significantly.

EEG can also be used to assess stress. One relevant marker is FAA, which reflects differences in activation between the left and right prefrontal cortex. Berretz et al. (2022) tested this in 51 healthy adults, recording EEG during the Trier Social Stress Test and a control condition. Under stress, participants showed stronger left-hemisphere activation over frontal electrodes and reduced left-hemisphere activation over occipital electrodes. This suggests that the left prefrontal cortex becomes more active during stress, helping to suppress negative or distracting information and keep emotional reactions under control. Akil et al. (2024) found a similar pattern in a broader adult sample, with a negative correlation between FAA and self-reported anxiety.

The two types of signal (cardiac and EEG) can be used together. Attar et al. (2021) found significant correlations between HRV and EEG features during a Stroop-induced stress condition, which suggests that the two provide complementary information about the stress response. Accordingly, the present study uses mean HR as its main physiological measure, with EEG as a complementary one, to capture both autonomic and cortical responses. HRV could not be used because of a sensor limitation, as explained in Section 7.4.

3 Design and Implementation

3.1 Design Requirements and Overview

The system developed for this project is a breathing exercise complemented by physiological biofeedback. Three requirements guided its design. First, the breathing pace had to be displayed clearly, so that participants could follow it without difficulty. Second, the biofeedback had to be easy to interpret with little explanation. HRV was selected as the underlying physiological measure, given the substantial body of literature supporting its use as an indicator of stress and relaxation (see Chapter 2). Third, the biofeedback had to be implemented in a comparable way in the computer application and in the robot, so that any differences between conditions could be attributed to the interaction modality and not to how the exercise was presented.

The system has two main parts: a web application and the robot Tyto. The web application runs the whole experiment in both conditions and stores the data in a database (Section 3.5). Tyto presents the breathing pace and the biofeedback in the robot condition. Figure 3.1 shows how these parts connect with the sensors, and the technical details of each component are given in Appendix A.

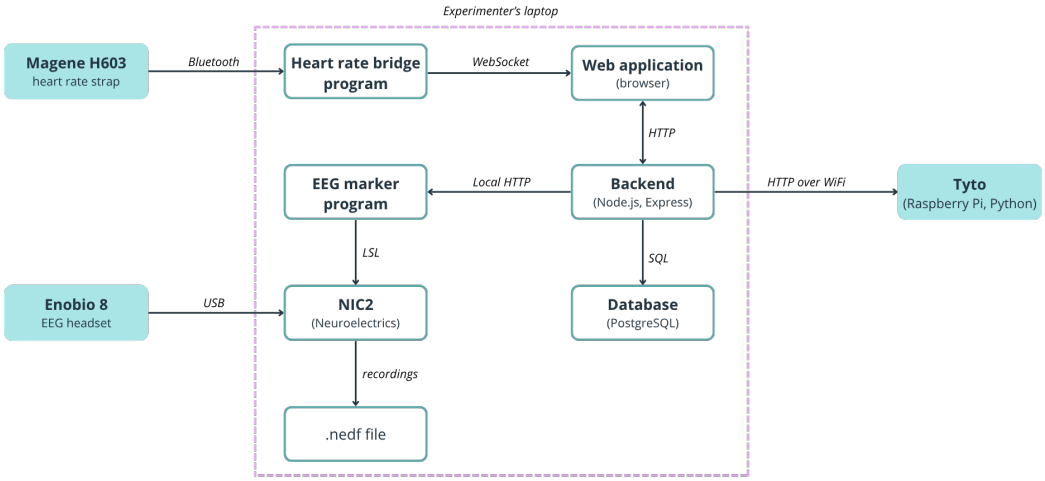


Figure 3.1: Overview of the system: web application, robot Tyto and sensors.

3.2 Breathing Exercise and Biofeedback Design

Different relaxation breathing techniques were considered before selecting one, including box breathing and the 4-7-8 technique. Although widely promoted, these techniques have limited empirical support for improving HRV. In contrast, breathing at approximately 6 breaths per minute has been shown to produce greater HRV improvements than both alternatives (Marchant et al., 2025). This rate falls within the resonance frequency range reported for adults, between 4.5 and 6.5 breaths per minute (Shaffer &

Meehan, 2020), although it was not calibrated for each participant. A fixed rate of 6 breaths per minute was used, for simplicity and reproducibility across conditions (see Chapter 4).

The breathing pace was shown with two complementary cues. Visually, a ball grew during the inhalation phase and shrank during the exhalation phase, a representation widely used in commercial breathing applications.[1] Audibly, a high-pitched tone (660 Hz) marked the beginning of the inhalation phase and a low-pitched tone (330 Hz) marked the beginning of the exhalation phase in the computer condition. In the robot condition, owl hoot sounds were used instead. A higher-pitched hoot marked the beginning of the inhalation phase and a lower-pitched hoot marked the beginning of the exhalation phase.

Two ways of showing the HRV information to the participant were considered. The first was a live graph of the HRV level over time, as in comparable systems.[2] It was discarded because a graph next to the breathing ball could be overwhelming, and participants might focus on the graph and lose the breathing pace, which is the central element of the exercise. Instead, the ball itself changed colour with the participant's HRV. The colour was calculated from the RMSSD of the last 60 readings received from the cardiac sensor, and it followed a gradient from red through amber to green, where greener meant higher HRV. Participants with no training in breathing exercises are likely to have low HRV at first, so a few fixed colours could keep the ball red during the whole exercise, which could be discouraging. A continuous gradient, from darker to brighter tones within each colour, let participants notice small increases or decreases. As explained in Section 7.4, the sensor did not provide true beat-to-beat intervals, so the colour was an approximation of the actual HRV.

3.3 Web Application

The web application was the core element of the experiment. It ran the whole experiment in both conditions: it administered the questionnaires, ran the three phases (the baseline phase, the stress-induction phase and the relaxation with biofeedback phase) and stored all the data in the database. The experimenter only used it at the start, to select the condition assigned to the participant. From then on, the experiment advanced automatically.

The interface was simple and minimalist, so that participants could stay focused on the task. The screens followed a fixed sequence. First, participants completed the pre-experimental questionnaires: demographics, the Affinity for Technology Interaction (ATI) scale, the Negative Attitudes towards Robots Scale (NARS) and the Generalised Anxiety Disorder scale (GAD-7). They also rated their perceived stress on a Visual Analogue Scale (VAS), from 0 (not at all stressed) to 10 (extremely stressed). The VAS was completed three times: before the baseline phase, after the stress-induction phase and after the relaxation with biofeedback phase. At the end, participants completed the post-experimental questionnaires: the Raw NASA Task Load Index (NASA-TLX), the System Usability Scale (SUS) and, in the robot condition only, the Godspeed questionnaire. The last screen announced the oral interview. The application did not allow participants to continue until all the mandatory questions of the questionnaires and the VAS were answered. Throughout the application, a progress bar showed how much of the experiment had been completed, as a percentage and without naming the current step, so that participants did not know what would come next.

During the baseline phase, a message asked participants to relax and breathe normally. No timer was shown, so that participants would not become nervous about how long the phase lasted.

In the stress-induction phase, participants performed the Stroop task. They had to click the button matching the ink colour of a colour word written in a different colour. The time to respond decreased

[1] Examples include HeartMath's emWave Pro (https://www.heartmath.com) and the Breathing Pacer web tool (https://everythingcalculators.com/health-calculators/breathing-pacer-web-breath-app/).

[2] For example, HeartMath's emWave Pro displays real-time coherence graphs during biofeedback.

from 3 seconds to a minimum of 0.8 seconds, which made the task progressively harder and more stressful. A running count of correct and incorrect answers was shown, and a different sound was played for each kind of answer.

In the relaxation with biofeedback phase, the screen depended on the condition. In the computer condition, it displayed the breathing ball described in Section 3.2, expanding and contracting at 6 breaths per minute and coloured according to the participant's approximate RMSSD. In the robot condition, the screen only showed a short instruction to follow the robot, since the breathing pace and the colour were shown on the rectangular screen on Tyto's belly. In this condition, the application sent the colour to the robot, as described in Section 3.6. The application also received the heart rate readings during the three phases and sent markers to the EEG recording at the start and end of each phase.

3.4 The Robot Tyto

Tyto was designed for this study, so its form was based on what it had to do. With limited time to design and build it, only the essentials were included: speakers to produce sound and a screen to show the breathing ball. Eyes were added to make the robot feel more alive, since eyes, especially when they blink, make a robot seem more engaging and likeable according to Yoshikawa et al. (2006).

It was decided to keep the robot static to save development time. This meant leaving out limbs, because limbs make users expect movement, and a robot that does not meet these expectations can be less satisfying to interact with (Dennler et al., 2023). A human-like look was also avoided. It raises expectations about what the robot can do (Dennler et al., 2023), and it has been linked to higher blood pressure and more negative emotions (Broadbent et al., 2011), which is the opposite of what a breathing exercise aims for. Finally, the appearance of a robot should match its task (Goetz et al., 2003), and Tyto only had to show the breathing pace and the biofeedback colour, so a simple design was enough.

Given this, an animal-like (zoomorphic) form was considered a better fit. Zoomorphic robots are associated with lower expectations and tend to be well received (Bharatharaj et al., 2016; Dennler et al., 2023; Šabanović et al., 2013), with successful examples in relaxation contexts such as Paro the seal (Geva et al., 2020; Wada & Shibata, 2007). They also look lifelike without looking human, which avoids the uncanny valley (de Graaf & Ben Allouch, 2017).

To choose the animal, several animals associated with calmness were considered: sloths, capybaras, pandas, koalas and seals. Sloths, capybaras, pandas and koalas were discarded because of their recognisable limbs, which might raise the same expectation of movement. A seal, following the example of Paro, was also considered, but it is better suited to being held, while Tyto had to stay fixed in place with a more vertical form. Birds fit this requirement better, since their vertical posture suits a stationary robot and their wings might make people expect less movement than limbs do. Among birds associated with calmness, the owl stood out as a form that users are likely to find endearing. This choice is also consistent with research on owl-shaped robots, which favours owls for their large, front-facing eyes, which facilitate eye contact, and for their small size compared with other companion animals (Mineshita & Lim, 2025).

A minimalist aesthetic was chosen over a cartoonish one, as it was considered more appropriate for a broad adult population. Several owl-shaped robots already exist commercially, such as the Luka Reading Robot and IXI-Play, which are designed as toys or companions for children, and TCL's Ai Me, a family companion robot with a playful, animated design. [3] Childlike proportions can make people react more positively, even young adults, but this has mostly been shown for companion robots aimed at a narrow young-adult group, and Tyto was meant for a wider range of adults. A simple design was also preferred because simple designs are better at sustaining attention (Lin et al., 2025).

[3] TCL's Ai Me: https://www.tcl.com; Luka Reading Robot: https://www.jojolearning.com/products/luka-the-ai-reading-companion; IXI-Play: https://spectrum.ieee.org/ixi-play-dancing-robot-bird.

With this in mind, the final design was based on minimalist owl sculptures. The head follows the heart-shaped silhouette of a barn owl, with a low-poly beak. Two circular screens on the head serve as eyes, modelled on the eyes of other owl species because the fully black eyes of the barn owl might look unsettling. A rectangular screen on the belly displays the ball during the breathing exercise. Wings are suggested through a pattern of small holes along the sides, which outline a wing-like silhouette without implying movement and also help the sound project from inside the robot. In total, Tyto has three screens, but only the belly screen displays the biofeedback.

The eyes blinked at random intervals between 3 and 10 seconds, using a sequence of hand-drawn frames. This frequency was chosen to avoid a repetitive animation while keeping the robot perceived as animated. Each blink was kept brief. In a similar owl-shaped robot, a physical blinking mechanism felt slow and unnatural, while a blink on a screen shorter than 0.5 seconds was perceived as natural (Mineshita & Lim, 2025). No other eye behaviour, such as pupil movement, was implemented, because of time limits and because participants were expected to look mainly at the belly screen. Even so, robotic eyes can act as ambient social cues, making a robot seem animate and observant even when they are not the focus of attention (Shinohara et al., 2025). A simple blink was therefore considered enough, and pupil movement was left for a future version.

As for the hardware, Tyto was controlled by a Raspberry Pi 4 Model B (8 GB).[4] The belly screen was a 2.4-inch Waveshare LCD module,[5] and the eyes were two 1.28-inch round LCD modules with a GC9A01 driver chip.[6] A Waveshare WM8960 Audio HAT and a pair of speakers reproduced the owl hoot sounds, and an RGB LED indicated a correct connection with the web application. The belly screen was the same model used in a previous Bachelor's thesis project, and the eye screens share their driver chip with the round displays of that project, so the same programming approach could be reused.[7] The body, of about $134 \times 155 \times 280$ mm, was 3D printed in white PLA in three parts (front shell, back cover and internal platform), in line with the minimalist aesthetic. A custom connection board, shaped as a HAT on the Raspberry Pi, kept the wiring stable during transportation. Tyto is also portable, and it does not depend on any commercial robot platform or software licence. Further details of the 3D design, wiring and internal architecture are given in Appendix A.

3.5 Database

A PostgreSQL database stores all the data collected by the web application and the sensors. It consists of eight tables, which are summarised in Table 3.1, and their relationships are shown in Figure 3.2. A more detailed description of every field of each table is provided in Appendix A.3.

[4] Raspberry Pi 4 Model B: https://www.raspberrypi.com/products/raspberry-pi-4-model-b/

[5] Waveshare 2.4inch LCD Module: https://www.waveshare.com/2.4inch-lcd-module.htm

[6] GC9A01 1.28-inch round LCD module (2-unit pack): https://www.amazon.es/Tiardey-Pantalla-Pulgadas-Arduino-Unidades/dp/B0DM569VDT

[7] Cadevall Soto, H. (2022). *Robòtica aplicada a teràpies i tractament de trastorns de l'espectre autista en infants* [Bachelor's thesis, Universitat Politècnica de Catalunya]. UPCommons. https://hdl.handle.net/2117/377449 (in Catalan)

Table	Content
`users`	Anonymous identifier of each participant and the condition they were assigned to (computer or robot)
`sessions`	One entry for each run of the experiment, with its date and time
`phases`	Start and end time of the baseline phase, the stress-induction phase and the relaxation with biofeedback phase
`stress_phase`	Number of attempts and errors in the Stroop task
`phase_readings`	Raw heart rate readings (BPM and RR interval) captured during each phase
`hrv_value`	Values calculated from the heart rate readings, used to set the colour of the ball
`eeg_recordings`	Location of the EEG recording file of each participant (the signal itself is stored on disk, not in the database)
`questionnaire_responses`	Answers to the questionnaires and VAS scales, one row per question

Table 3.1: Tables of the database and their content.

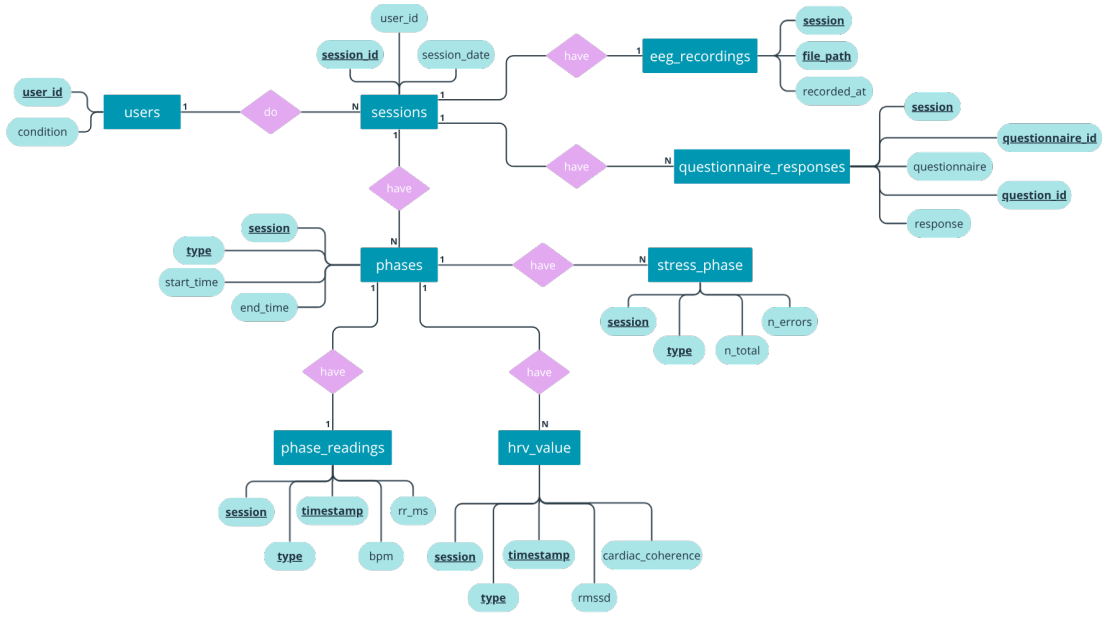


Figure 3.2: Entity-relationship diagram of the database.

3.6 System Architecture and Communication

Figure 3.1 shows how the components of the system are connected. Each connection uses the protocol that suits what it has to send, instead of a single protocol for the whole system.

The heart rate data follows this path. A small bridge program (`heart_rate_sensor_reader.py`) runs on the experimenter's laptop. It connects to the Magene H603 chest strap through Bluetooth and, at the same time, acts as a WebSocket server, so that the browser receives each new reading in real time. The web application then sends every reading to the backend, which stores it in the database.

The EEG headset (Enobio 8) is connected by USB to the same laptop. NIC2, the software of Neuroelectrics, is in charge of recording the EEG signal and saving it in a `.nedf` file. To keep this recording aligned with the rest of the experiment, the backend sends a request to a separate program (`enobio_markers.py`), which runs on the same laptop, whenever a phase starts or ends. This program sends a numeric marker to NIC2 through Lab Streaming Layer (LSL), and NIC2 saves it in the `.nedf` file together with the signal. In this way, the browser does not need to communicate with this program directly. The marker codes are described in Appendix A.7.1.

In the robot condition, whenever the browser calculates a new colour, it sends it to the backend, which forwards it to Tyto over the WiFi network with a simple HTTP request. The start and stop commands are sent in the same way. For this to work, the laptop and Tyto must be connected to the same WiFi network, and the network must allow devices to send requests to each other. The breathing pace, the sounds and the blinking of the robot do not depend on a message arriving each time, since they are controlled by timers running on Tyto itself. This was done on purpose because, in an earlier version, the breathing movement depended on each message arriving, so a lost WiFi message could freeze the animation. With the final design, if a colour update is lost, the ball keeps breathing with its last colour, and the participant can continue the exercise without interruption.

The backend does not wait for a response from the robot or from the marker program. If either cannot be reached, the request fails, a warning is logged and the experiment continues.

4 Methodology

4.1 Study Design

The study followed a randomised between-subjects design. Each participant was assigned to only one of the two conditions, the robot or the computer, by random assignment. The experiment had three phases: the baseline phase, the stress-induction phase (Stroop task) and the relaxation with biofeedback phase. In the rest of the report, they are also referred to as baseline, Stroop task and relaxation phase.

A between-subjects design was chosen over a within-subjects design for two reasons. First, if the same participant went through both conditions, what they had already experienced in the first could influence their results in the second. For example, they could get used to the stress-induction task, or compare the robot with the computer. Second, doing both conditions in a single session would have taken about two hours. Participants wore an EEG headset and a chest strap the whole time, so such a long duration could cause discomfort and fatigue, which could affect the signals recorded in the second condition.

Except for the interaction modality used in the relaxation phase, all participants followed the same protocol: questionnaires, physiological recording, the stress-induction phase and timing. In this way, any difference between conditions can be attributed to the interaction modality and not to the procedure. The only exception was the Godspeed questionnaire series, which was completed only by participants in the robot condition to evaluate their perception of Tyto, since participants in the computer condition did not interact with the robot.

4.2 Experimental Conditions and Variables

The study had one independent variable, the interaction modality, which refers to how the breathing exercise was delivered in the relaxation phase. It had two levels, which define the two conditions: robot and computer.

Computer condition. The guided breathing exercise with biofeedback was carried out through a web application on a computer screen. The interface showed a single circle that combined two sources of information. Its pulsing, growing and shrinking at a fixed pace, marked the breathing rhythm, and its fill colour gave biofeedback based on the heart rate (an approximate RMSSD, see Section 7.4). The breathing pace was also marked with generic tones, a high-pitched tone for the inhalation and a low-pitched tone for the exhalation.

Robot condition. The same exercise was carried out through Tyto (Chapter 3). The aim of this condition was to introduce a physically present social agent while keeping the interaction simple. A rectangular screen on the belly of the robot showed the same circle as in the computer condition, with the same pulsing and colour. The breathing pace was also marked with sound, but in this case with owl hoots: a high-pitched hoot for the inhalation and a low-pitched hoot for the exhalation, instead of the generic tones of the computer condition.

The dependent variables were the mean heart rate in each phase, the EEG signals, as indicators of cortical activation, and the scores of the VAS (perceived stress) and of the questionnaires. HRV could not be analysed because of a limitation of the heart rate sensor (see Section 7.4).

Several variables were kept constant in both conditions, so that any difference could be attributed to the interaction modality alone: the duration and order of the phases, the stress-induction task (the Stroop

task), the fixed breathing rate of 6 breaths per minute, and the type of feedback (simultaneous visual and auditory cues).

4.3 Participants

The study aimed to recruit between 12 and 20 participants. This sample size does not allow robust statistical inference, but it is enough to identify preliminary trends, which fits the proof-of-concept nature of the study (Chapter 1). The reduced sample was mainly due to the limited time available to carry out the experiment and analyse the data.

Participants had to be adults aged 18 or older, in good general health, without a clinical diagnosis related to stress or mental health, and able to understand and follow the instructions of the protocol. All of them signed an informed consent form before taking part in the experiment.

The exclusion criteria were:

- Severe cardiovascular conditions, or other medical conditions that could interfere with the heart rate measurements.
- A score of 10 or higher in the Generalised Anxiety Disorder-7 (GAD-7), which indicates moderate to severe anxiety (Spitzer et al., 2006). The threshold finally applied was 12, as explained in Section 5.1.
- Clinical psychological diagnoses that could affect the stress response or emotional regulation, such as major depressive disorder, bipolar disorder, post-traumatic stress disorder, panic disorder, obsessive-compulsive disorder, and psychotic disorders such as schizophrenia. These conditions have been associated with altered HRV (Wang et al., 2025).
- Use of medication that significantly alters the heart rate or its variability, such as beta-blockers or other drugs with autonomic effects (Maki et al., 2024).
- Neurological or sensory conditions that could prevent completing the protocol.
- A history of epilepsy or epileptic seizures, because of the use of EEG.
- Active wounds, injuries or scalp irritation.
- Known allergies to conductive gels or other materials of the sensors.

Participants were recruited by convenience sampling, mainly among university students, volunteers from the academic environment and personal contacts. An effort was made to keep a balanced number of men and women. The recruited sample and the participants excluded from the analysis are described in Section 5.1.

As agreed in the internship proposal, contact was also made with three organisations to explore the recruitment of participants with chronic stress or conditions for which guided breathing could be beneficial: Fundació AMPANS,[1] Fundació Althaia[2] and ACAF (Associació Catalana d'Afectades i Afectats de Fibromiàlgia i d'altres Síndromes de Sensibilització Central).[3] With Althaia, a couple of emails were exchanged, but no further response was received. Meetings were held with several people in charge of AMPANS (in person) and ACAF (online) to discuss a possible collaboration and the groups of participants that could fit the project. However, due to the limited time available during the internship, the collaboration could not be carried out with any of them, although they remained open to collaborating in the future.

[1] Fundació AMPANS is a Catalan foundation that provides social, educational and labour inclusion services for people with intellectual disabilities and mental health conditions. More information: https://ampans.cat/en/.

[2] Fundació Althaia is a healthcare institution based in Manresa, involved in research and innovation. More information (only in Catalan): https://www.althaia.cat/althaia/ca/corporatiu.

[3] ACAF is a patient association representing people affected by fibromyalgia and other central sensitization syndromes in Catalonia. More information (it can be translated to English inside the website): https://acaf.cat/.

4.4 Devices

Four devices were used in the study: a computer, the social robot Tyto, the Magene H603 heart rate monitor and the Enobio 8 EEG headset.

The computer was an MSI Katana 17 B13V laptop with a 17-inch display. It ran the web application, and in the computer condition it showed the breathing exercise to the participant. No specific hardware was needed beyond standard web browsing.

Tyto is the social robot described in Chapter 3 and Appendix A, and it was used only in the robot condition.

The Magene H603[4] is a chest strap that continuously records the heart rate in beats per minute (BPM). The sensor sends the heart rate at fixed time intervals, and not a value for each heartbeat, which is why HRV could not be analysed (see Section 7.4).

The Enobio 8[5], from Neuroelectrics, is a portable EEG system with 8 channels. The EEG signal was recorded continuously with the software NIC2, and the markers of each phase were saved in the recording (Section 3.6). The electrodes were placed at Fp1, Fp2, F7, F8, F3, F4, C3 and C4, following configurations commonly reported in the stress detection literature. F3 and F4 are the most frequently used pair to estimate FAA, as they lie above the dorsolateral prefrontal cortex, a region involved in the stress response. Anterior (Fp1 and Fp2), lateral (F7 and F8) and central (C3 and C4) pairs have also been used in psychophysiological studies of stress (Giannakakis et al., 2022).

All devices in direct contact with participants were cleaned and disinfected after each participant.

4.5 Measures and Questionnaires

Two physiological signals were recorded continuously: the heart rate, with the Magene H603, and the EEG, with the Enobio 8. The reasons for using these measures are discussed in Chapter 2.

Before the experiment, participants completed the NARS, the ATI scale and the GAD-7. The NARS (Nomura et al., 2006) measures attitudes towards robots, and the ATI scale (Franke et al., 2019) measures the affinity for technology. The GAD-7 was used as a screening instrument, to exclude participants with clinically relevant anxiety.

Perceived stress was measured three times with the VAS: before the baseline phase, after the Stroop task and after the relaxation phase. After the relaxation phase, participants completed the NASA-TLX (Hart, 2006; Lee et al., 2026) to assess perceived workload. The raw version was used, which averages the six subscales without the pairwise weighting of the original. Participants also completed the SUS (Brooke, 2013) to evaluate the usability of the system. In the robot condition, they completed in addition the Godspeed questionnaire series (Bartneck et al., 2009), which assesses the perception of Tyto in five dimensions: anthropomorphism, animacy, likeability, perceived intelligence and perceived safety.

All the questionnaires were administered in Catalan. Since no verified official Catalan translation was found for most of them, the items, instructions and scales were translated into Catalan for this study and then reviewed by the supervisors. The complete content of each questionnaire, with its background and scoring, is provided in Appendix B.

[4] https://www.magene.com/en/all-products/54-h603-heart-rate-monitor.html
[5] https://www.neuroelectrics.com/solutions/enobio/8

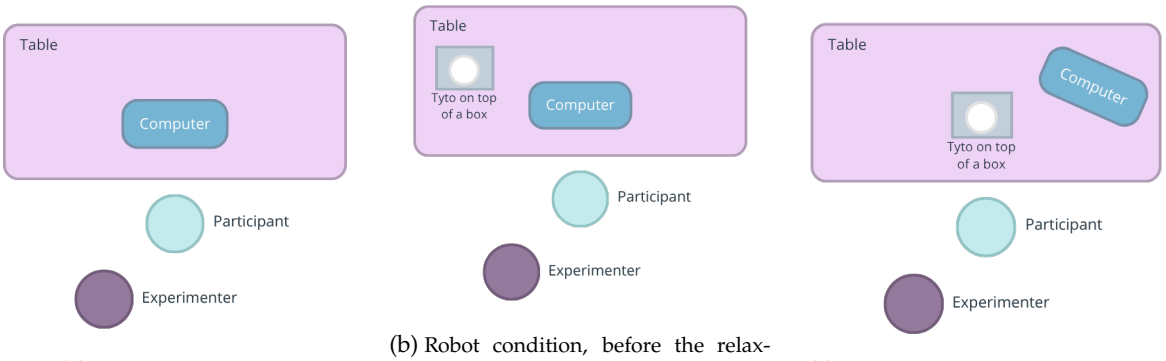
4.6 Setup

Participants sat on a chair in front of a table throughout the experiment. In both conditions, the computer was on the table in front of the participant, and it was plugged into the power supply the whole time so that it would not run out of battery. A mouse was available for participants who preferred to use one.

In the computer condition, the setup did not change during the experiment (Figure 4.1a).

In the robot condition, Tyto was placed next to the computer, on top of a small box, so that its belly screen was closer to the line of sight of the participant and at a more comfortable height (Figure 4.1b). During the relaxation phase, the computer was moved aside and the box with Tyto was placed in front of the participant, in the position of the computer (Figure 4.1c).

During the experiment, the experimenter stayed behind the participant and only intervened to give instructions before each step of the protocol.



(a) Computer condition.

(b) Robot condition, before the relaxation phase.

(c) Robot condition, relaxation phase.

Figure 4.1: Experimental setup, seen from above.

4.7 Experimental Procedure

The procedure consisted of eleven steps, summarised in Table 4.1. Steps 6, 7 and 8 are the three phases of the experiment.

Step	Duration	Main actions
1. Participant reception	3 min	Verify the inclusion and exclusion criteria; assign the condition (robot or computer)
2. Study explanation	5 min	Deliver and read the Participant Information Sheet (PIS); present the objectives; explain the tasks; answer questions
3. Informed consent	5 min	Sign the informed consent form; confirm voluntary participation; explain data handling and the right to withdraw
4. Sensor placement and setup	10–15 min	Fit the heart rate monitor and the EEG headset; verify the signal quality and the electrode impedance; ensure the comfort of the participant
5. Pre-experimental questionnaires	10 min	Collect demographic data; GAD-7 (screening); NARS and ATI; rate perceived stress on the VAS (T1)
6. Baseline phase	5 min	Record the heart rate and the EEG at rest
7. Stress-induction phase	6 min	Stroop task under time pressure; rate perceived stress on the VAS (T2)
8. Relaxation with biofeedback phase	10 min	Guided breathing at 6 breaths per minute, with the robot or the computer depending on the condition; biofeedback with the colour of the ball; rate perceived stress on the VAS (T3)
9. Post-experimental questionnaires	10 min	NASA-TLX and SUS; Godspeed questionnaire (robot condition only)
10. Qualitative interview	5 min	Audio-recorded interview about the overall experience, the acceptance of the system and suggestions
11. Debriefing	3 min	Explain the real purpose of the study and the stress-induction mechanism; answer remaining questions; confirm data handling

Table 4.1: Steps of the experimental procedure.

Three methodological choices in this procedure are worth highlighting. First, the whole experiment was conducted in Catalan, the native language of all the participants, to ensure that the instructions and the informed consent were fully understood. Second, the stress-induction phase used the Stroop task, a well-established cognitive task in which participants respond to word-colour conflicts under time pressure. It was chosen because it reliably produces a controlled and safe physiological stress response, without external or emotionally distressing stimuli (Renaud & Blondin, 1997). Third, the relaxation phase used a fixed rate of 6 breaths per minute in both conditions, for the reasons explained in Section 3.2 and to keep the protocol simple and reproducible.

The script used to guide the experimenter through each step, originally written in Catalan, is provided with its English translation in Appendix C.

4.8 Data Management and Ethical Considerations

The personal data of this study were processed in accordance with Regulation (EU) 2016/679 (General Data Protection Regulation, GDPR) and the Spanish Organic Law 3/2018 (LOPDGDD). All the digital data, including the heart rate and EEG recordings, the questionnaire answers and the audio of the interviews, were stored on a secure internal server of the Universitat Politècnica de Catalunya (UPC), based on Nextcloud and hosted in the European Union. Access was restricted to the experimenter and

the supervisors, with secure credentials. Each participant had an alphanumeric code, which was the only identifier in the data files. The table linking each code to the identity of the participant was stored separately, with restricted access.

The study was submitted to the Research Ethics Committee of the UPC (CEUPC) and approved before data collection began. A first version of the protocol was submitted and approved. It was then modified to simplify and improve it, and the modified version was also approved. This second version is the one used in the experiments and described in this report. The submission included three documents: the research protocol, the Participant Information Sheet (PIS) and the informed consent form, signed by each participant. The approval certificates of both versions and the documents of the second version, originally in Catalan, are provided with their English translation in Appendix D.

The study followed the ethical principles of the Declaration of Helsinki. The main risks foreseen were mild and temporary stress during the stress-induction phase, slight fatigue or discomfort from the cognitive tasks, and slight physical discomfort from the sensors. To reduce them, participation was voluntary and participants could withdraw at any time without giving a reason. The experimenter also watched for signs of discomfort during the whole experiment, and was ready to stop it if necessary.

The limitations of the study, including the prototype nature of the robot and the dual role of designer and experimenter, are discussed in Section 7.4.

5 Data Analysis

This chapter describes the final sample and how the physiological, perceived stress, questionnaire and interview data were processed and analysed.

5.1 Final Sample

In total, 19 participants took part in the study, 9 in the computer condition and 10 in the robot condition. Four of them were excluded from the analysis. Three were excluded because their GAD-7 score was 12 or higher (scores of 13, 12 and 14). The participant with a score of 13 also had no heart rate data. A fourth participant was excluded for a different reason, since the full EEG recording and the physiological data of the relaxation phase were missing.

The exclusion threshold applied here differs from the one in the research plan, which specified exclusion for a GAD-7 score of 10 or higher. Here, only participants with a score of 12 or higher were excluded, so Subject03 and Subject04 (with scores of 10 or 11) were kept. There were two reasons for this decision. First, a GAD-7 score of 10 is only a screening cutoff, not a strict limit. Second, excluding both participants would have reduced the computer condition to 5 participants, which would have unbalanced an already small sample even further.

After these exclusions, the final sample had 15 participants: 7 in the computer condition and 8 in the robot condition. The computer condition had a mean age of 31.1 years (range 21 to 60, median 23) and included 3 women and 4 men. The robot condition had a mean age of 42.5 years (range 29 to 62, median 37) and included 3 women and 5 men.

The only exception was the FAA, from which one participant in the robot condition (Subject14) was excluded. After the correction of the eye blinks with Signal-Space Projection (SSP), the signal of channel F3 was noticeably attenuated in this participant. Because the FAA is computed from F3 and F4 only, this attenuation distorted the asymmetry index. The participant was kept in the remaining EEG analyses, which use the mean of four channels (F3, F4, C3 and C4) and are therefore much less affected. As a result, the FAA analysis was based on 14 participants (7 per condition).

5.2 Physiological Measures

5.2.1 Heart Rate

The mean HR was calculated for each phase (baseline, Stroop task and relaxation phase) as the simple mean of the sensor readings recorded during that phase. The initial plan was to calculate HRV, but this was not possible because of a limitation of the heart rate sensor. The sensor does not send a value for each heartbeat, but the latest reading at fixed time intervals. Calculating HRV requires the time between consecutive heartbeats, and the sensor data could not provide it. For this reason, HRV was not analysed (see Section 7.4), and the mean HR of each phase was used instead, as an indication of how the heart reacted during each phase.

5.2.2 EEG Processing

The EEG signals recorded with the Enobio 8 headset were processed in Brainstorm (Tadel et al., 2011), an open-source application built on MATLAB (R2024b), in four steps. First, a band-pass filter of 1 to 40 Hz was applied. The 1 Hz limit removes slow drifts caused by sweat and small electrode movements, and the 40 Hz limit removes the power-line noise (50 Hz in Europe) and most of the gamma activity. Gamma was not analysed because it can be contaminated by muscle activity (Giannakakis et al., 2022), for example from forehead, jaw and neck movements. Second, an SSP correction was applied to reduce the effect of blinking, using one component per participant from channel Fp1, each reviewed visually. Third, the remaining artefacts were removed by visual inspection of the recording of each participant. This step was done without taking into account the condition of each participant. Fourth, the power spectral density (PSD) was calculated with Welch's method (2 s windows, 50% overlap) and averaged within each band: alpha (8 to 13 Hz) and beta (13 to 30 Hz). These ranges are the same as those reported by Giannakakis et al. (2022) and used by Vanhollebeke et al. (2023). Band power (μV^2) was obtained by multiplying the mean PSD ($\mu V^2/Hz$) by the bandwidth. Total power was calculated in the same way between 1 and 40 Hz, which is also the range of the band-pass filter. Relative power was defined as band power divided by total power, as in Vanhollebeke et al. (2023).

Four measures were derived from the spectral analysis: relative alpha, relative beta, the beta/alpha ratio and the FAA. Stress is generally considered to decrease alpha activity and increase beta activity, and relative power indices have been widely used to follow the intensity of the stress response (Giannakakis et al., 2022). The beta/alpha ratio was calculated as the mean beta power divided by the mean alpha power. This ratio is used as an indicator of cognitive load and arousal, and it is expected to increase under stress (Giannakakis et al., 2022). These three measures were computed from the band power averaged over F3, F4, C3 and C4. The FAA was calculated as $\ln\alpha_{F3} - \ln\alpha_{F4}$, using only F3 and F4. This is the definition used by Giannakakis et al. (2022), who report that this index decreases during stress. Lower values therefore mean relatively more alpha power on the right side than on the left. F3 and F4 are the electrode pair most commonly used for this index, because they lie above the dorsolateral prefrontal cortex (Giannakakis et al., 2022). Fp1 and Fp2 were discarded because their signals were heavily contaminated by blinking artefacts. F7 and F8 were also discarded because of contamination from jaw movement and pressure.

5.3 Perceived Stress

The VAS scores (0 to 10) were collected at three time points (T1, before the baseline phase; T2, after the Stroop task; and T3, after the relaxation phase) as a self-reported measure of stress, so that they could be compared with the physiological measures. For description, the mean and standard deviation of the VAS score of each condition were calculated at each time point.

5.4 Statistical Analysis

A mixed ANOVA was applied to each measure (mean HR, VAS, relative alpha, relative beta, beta/alpha ratio and FAA), with phase (baseline, Stroop task, relaxation phase) as the within-subject factor and condition (robot or computer) as the between-subject factor. Normality was checked with the Shapiro-Wilk test, the homogeneity of variance between conditions with Levene's test, and the sphericity of the within-subject factor with Mauchly's test. Post hoc comparisons used Holm's correction, for phase and for the phase $\times$ condition interaction. Effect sizes are reported as partial η^2. A significance level of $\alpha = .05$ was used in all the analyses.

A total of 30 Spearman correlations were computed between the change scores of the six measures: 15 for the stress response (baseline to Stroop task) and 15 for the recovery (Stroop task to relaxation phase). Change scores were calculated as the later phase minus the earlier phase. For the VAS, they were T2 minus T1 and T3 minus T2. Both conditions were pooled. The 10 correlations involving FAA were based on 14 participants, and the other 20 on 15 participants. The correlations were exploratory and two-sided, and no correction for multiple comparisons was applied.

All analyses were performed in JASP (version 0.98.1), an open-source statistical software supported by the University of Amsterdam.

5.5 Questionnaires

The questionnaires were scored following the procedure of each instrument, including the inversion of the reverse-worded items. The scoring of each questionnaire is described in Appendix B. The mean, median, standard deviation, minimum and maximum of each score were calculated for each condition, and for the full sample in the questionnaires completed by all the participants. Cronbach's alpha was also calculated for the NARS subscales, the ATI and the Godspeed dimensions, to assess their internal consistency. Each SUS score was classified into a grade, using the cutoffs shown in Table B.1.6 of Appendix B.

The relationships between questionnaires were explored with Spearman's rank correlation, which was chosen because of the small sample size and because normality could not be assumed. Three sets of correlations were computed: between the ATI and the SUS, between the ATI and each of the five Godspeed dimensions, and between each NARS subscale (S1, S2, S3) and each Godspeed dimension. The correlation between the ATI and the SUS used the full sample ($N = 15$). All the others were computed only within the robot condition ($N = 8$), since the Godspeed questionnaire was only administered in that condition. The NARS-Godspeed correlations were included because both questionnaires relate to perceptions of robots. The general attitudes toward robots of the participants, measured before the experiment with the NARS, could be related to how they later evaluated Tyto with the Godspeed questionnaire.

5.6 Qualitative Analysis

A simple thematic analysis was conducted on the interviews carried out at the end of the experiment. The comments of each participant were annotated with short tags, which were then grouped into the main themes. The resulting themes of this analysis were: time perception, breathing pace, experience during the Stroop task, state during the relaxation phase, visual and sound feedback, and robot perception.

6 Results

6.1 Physiological Measures

6.1.1 Heart Rate

Mean HR increased from baseline (computer M = 76.62, SD = 20.82; robot M = 70.64, SD = 10.03) to the Stroop task (computer M = 82.08, SD = 22.19; robot M = 73.91, SD = 10.74) and decreased during the relaxation phase (computer M = 79.00, SD = 19.21; robot M = 72.14, SD = 11.75) in both conditions. Mauchly's test indicated that the assumption of sphericity was met, W = 0.95, p = .754.

The mixed ANOVA showed a significant effect of phase on HR, $F(2, 26) = 5.98$, p = .007, η_p^2 = .32. In this and the following analyses, the mean differences (M_{diff}) of the post hoc comparisons were calculated as the earlier phase minus the later phase. Post hoc comparisons with Holm correction showed that HR increased from baseline to the Stroop task ($M_{\text{diff}} = -4.37$, p_{Holm} = .006, $d = -0.27$). The difference between the Stroop task and the relaxation phase was not significant (M_{diff} = 2.42, p_{Holm} = .159, d = 0.15). There was no significant effect of condition, $F(1, 13) = 0.72$, p = .412, η_p^2 = .052, and no significant phase × condition interaction, $F(2, 26) = 0.38$, p = .686, η_p^2 = .029.

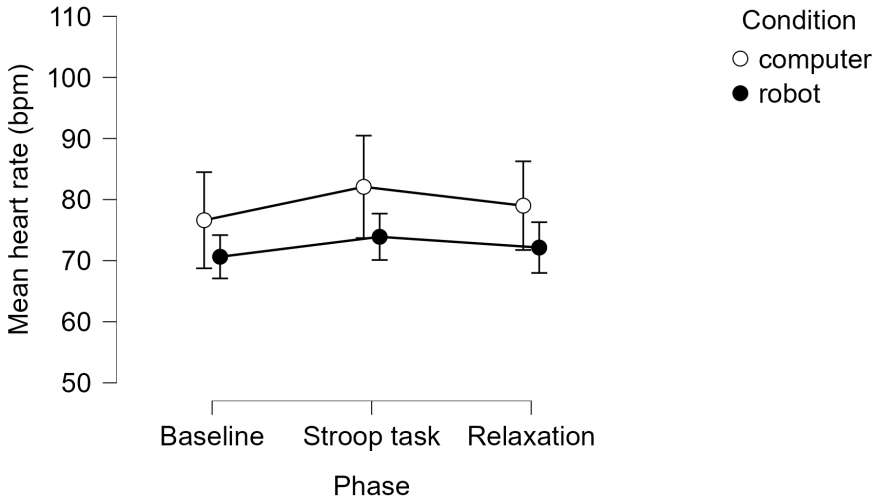


Figure 6.1: Mean HR (bpm) in each phase, by condition.

6.1.2 EEG

The descriptive statistics of the four EEG measures are shown in Table 6.1. Mauchly's test indicated that the assumption of sphericity was met for phase in all four measures (beta/alpha ratio: W = 0.85, p = .386; relative alpha: W = 0.73, p = .155; relative beta: W = 0.66, p = .086; FAA: W = 0.76, p = .218).

Measure	**Phase**	**Computer** (n = 7)	**Robot** (n = 8)
Relative alpha	Baseline	0.219 (0.100)	0.205 (0.116)
	Stroop	0.149 (0.048)	0.134 (0.034)
	Relaxation	0.293 (0.102)	0.217 (0.140)
Relative beta	Baseline	0.211 (0.062)	0.242 (0.097)
	Stroop	0.237 (0.054)	0.245 (0.075)
	Relaxation	0.184 (0.064)	0.248 (0.103)
Beta/alpha ratio	Baseline	1.131 (0.601)	1.477 (0.791)
	Stroop	1.680 (0.563)	1.922 (0.660)
	Relaxation	0.726 (0.429)	1.517 (0.956)
FAA[a]	Baseline	0.191 (0.333)	0.100 (0.234)
	Stroop	0.115 (0.385)	0.084 (0.323)
	Relaxation	0.142 (0.317)	0.134 (0.243)

Note. [a]For FAA, n = 7 in both conditions (14 participants in total).

Table 6.1: Descriptive statistics of the EEG measures, as mean (SD), by phase and condition.

For relative alpha, there was a significant effect of phase, $F(2, 26) = 13.71$, $p < .001$, $\eta_p^2 = .51$. Post hoc comparisons with Holm correction showed that relative alpha decreased from baseline to the Stroop task ($M_{\text{diff}} = 0.07$, $p_{\text{Holm}} = .014$, $d = 0.72$) and increased from the Stroop task to the relaxation phase ($M_{\text{diff}} = -0.11$, $p_{\text{Holm}} = .002$, $d = -1.16$). Relative alpha was also higher during the relaxation phase than at baseline ($M_{\text{diff}} = -0.04$, $p_{\text{Holm}} = .020$). There was no significant effect of condition, $F(1, 13) = 0.63$, $p = .443$, $\eta_p^2 = .046$, and no significant phase $\times$ condition interaction, $F(2, 26) = 1.32$, $p = .285$, $\eta_p^2 = .092$.

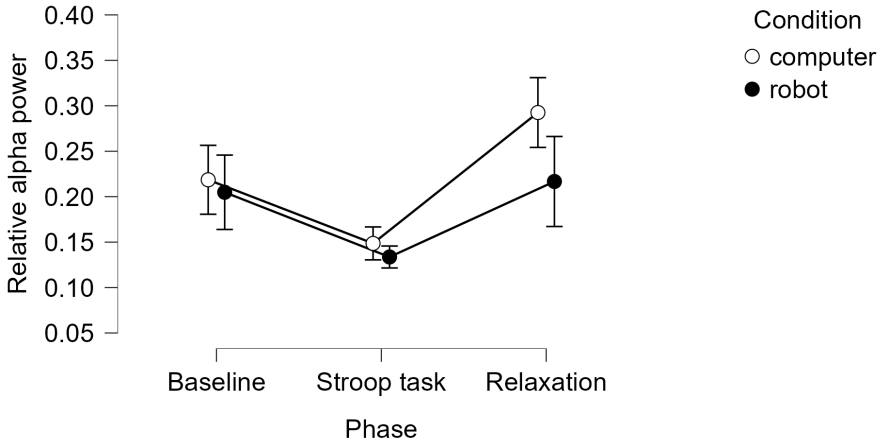


Figure 6.2: Relative alpha power in each phase, by condition.

For relative beta, there was no significant effect of phase, $F(2, 26) = 2.18$, $p = .133$, $\eta_p^2 = .14$, and no significant effect of condition, $F(1, 13) = 0.80$, $p = .388$, $\eta_p^2 = .058$. The phase $\times$ condition interaction was not significant, but a trend can be seen in the data, $F(2, 26) = 2.84$, $p = .077$, $\eta_p^2 = .18$. Relative beta was lower during the relaxation phase than at baseline in the computer condition, but not in the robot condition (Table 6.1).

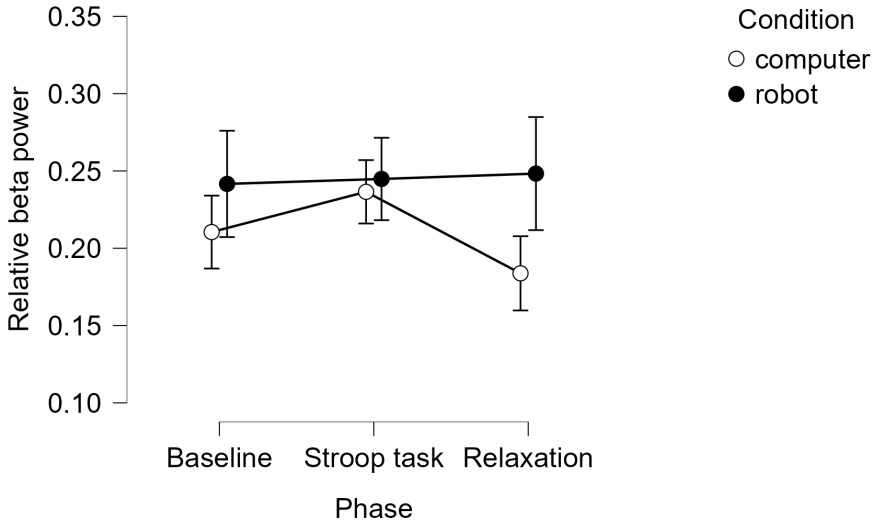


Figure 6.3: Relative beta power in each phase, by condition.

For the beta/alpha ratio, there was a significant effect of phase, $F(2, 26) = 12.18$, $p < .001$, $\eta_p^2 = .48$. Post hoc comparisons with Holm correction showed that the ratio increased from baseline to the Stroop task ($M_{\text{diff}} = -0.50$, $p_{\text{Holm}} = .019$, $d = -0.71$) and decreased from the Stroop task to the relaxation phase ($M_{\text{diff}} = 0.68$, $p_{\text{Holm}} = .001$, $d = 0.97$). There was no significant effect of condition, $F(1, 13) = 2.05$, $p = .176$, $\eta_p^2 = .136$, and no significant phase $\times$ condition interaction, $F(2, 26) = 2.09$, $p = .144$, $\eta_p^2 = .138$.

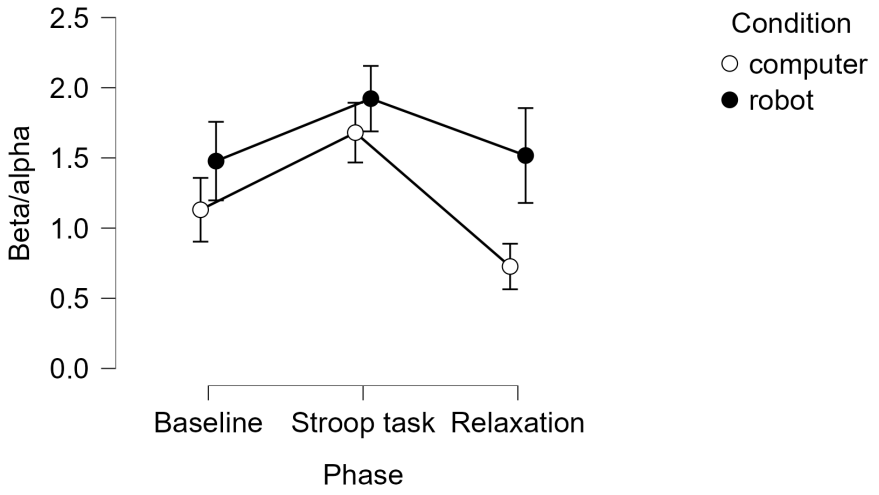


Figure 6.4: Beta/alpha ratio in each phase, by condition.

Finally, for the FAA no significant effect was found for phase, $F(2, 24) = 0.85$, $p = .441$, $\eta_p^2 = .066$, condition, $F(1, 12) = 0.07$, $p = .791$, $\eta_p^2 = .006$, or the phase $\times$ condition interaction, $F(2, 24) = 0.64$, $p = .538$, $\eta_p^2 = .050$.

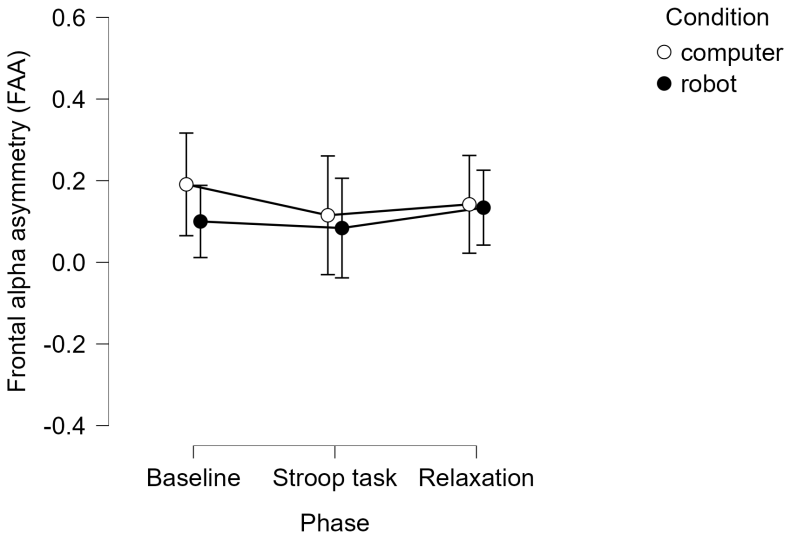


Figure 6.5: FAA in each phase, by condition.

6.2 Perceived Stress

Mean VAS scores increased from T1 (computer $M = 1.86$, $SD = 1.22$; robot $M = 2.50$, $SD = 1.60$) to T2, after the Stroop task (computer $M = 2.71$, $SD = 2.22$; robot $M = 4.88$, $SD = 2.48$), and decreased at T3, after the relaxation phase (computer $M = 2.43$, $SD = 2.88$; robot $M = 2.25$, $SD = 1.83$), in both conditions. Mauchly's test indicated that the assumption of sphericity was met, $W = 0.95$, $p = .749$.

The mixed ANOVA showed a significant effect of phase on the VAS, $F(2, 26) = 4.37$, $p = .023$, $\eta_p^2 = .25$. Post hoc comparisons with Holm correction did not show significant differences between time points, but a trend can be seen in both: an increase from T1 to T2, $M_{\text{diff}} = -1.62$, $p_{\text{Holm}} = .070$, $d = -0.77$, and a decrease from T2 to T3, $M_{\text{diff}} = 1.46$, $p_{\text{Holm}} = .081$, $d = 0.69$. There was no significant effect of condition, $F(1, 13) = 1.09$, $p = .315$, $\eta_p^2 = .078$, and no significant phase $\times$ condition interaction, $F(2, 26) = 1.94$, $p = .164$, $\eta_p^2 = .13$.

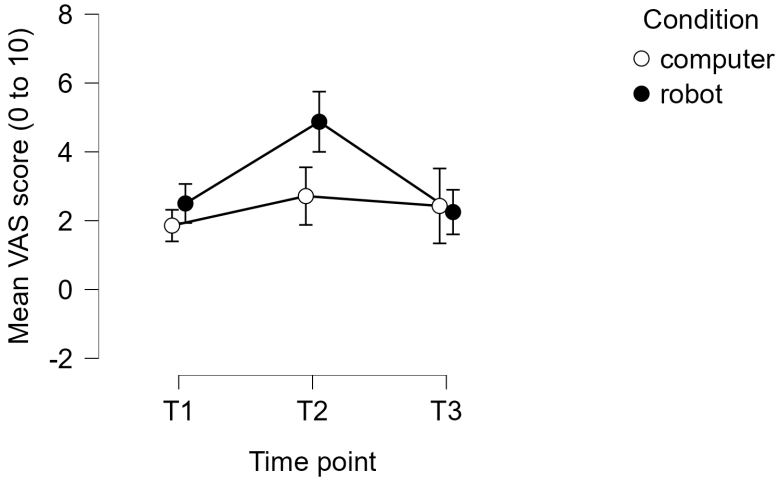
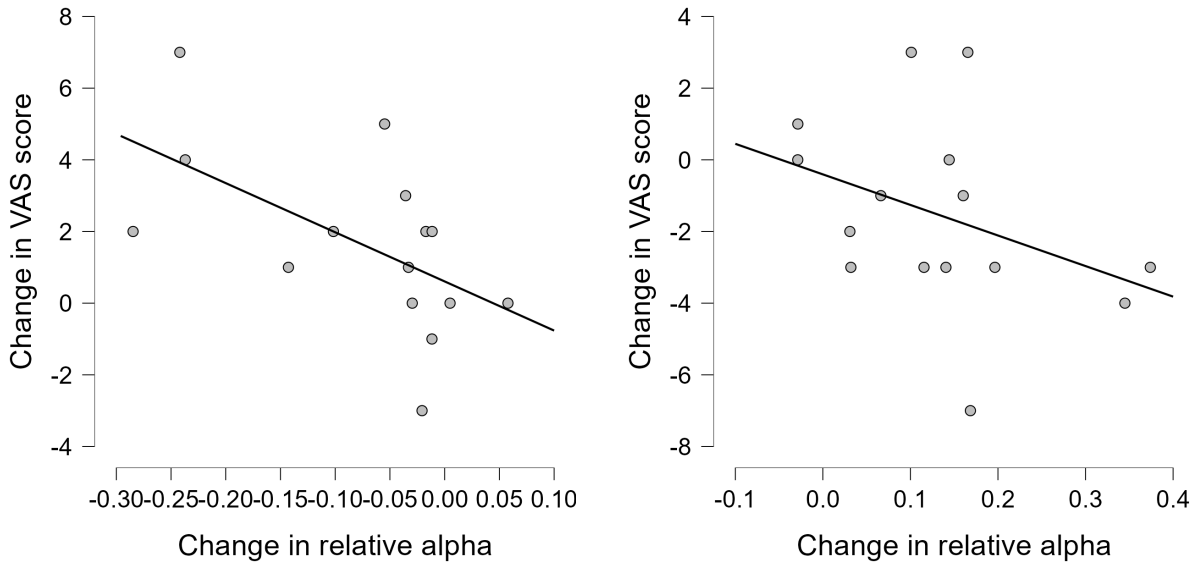


Figure 6.6: VAS scores at each time point, by condition.

6.3 Relationships Between Stress Measures

In the change from baseline to the Stroop task, the beta/alpha ratio and relative beta showed a strong positive correlation, $r_s(13) = .804$, $p < .001$. Two negative correlations were also significant: between relative alpha and the VAS, $r_s(13) = -.630$, $p = .012$, and between the beta/alpha ratio and the FAA, $r_s(12) = -.644$, $p = .015$. No other correlation was significant.

In the change from the Stroop task to the relaxation phase, the beta/alpha ratio and relative beta again showed a strong positive correlation, $r_s(13) = .911$, $p < .001$. Relative beta and the FAA showed a significant negative correlation, $r_s(12) = -.574$, $p = .035$. The correlation between relative alpha and the VAS was not significant, but a trend can be seen in the same direction as in the stress response, $r_s(13) = -.478$, $p = .072$. No other correlation was significant.



(a) Change in relative alpha against change in VAS from baseline to the Stroop task.

(b) Change in relative alpha against change in VAS from the Stroop task to the relaxation phase.

Figure 6.7: Relationship between the change in relative alpha and the change in VAS.

6.4 Questionnaires

The descriptive statistics of the NARS, the ATI and the NASA-TLX are reported in Appendix B.

NARS and ATI. All three NARS subscales had a mean below 3, the midpoint of the scale, in both conditions (all participants: S1 = 2.51, S2 = 2.03, S3 = 2.82). The largest difference between conditions was in S2, with a lower mean in the computer condition (1.69) than in the robot condition (2.33). Cronbach's alpha was below .70 for S1 (computer .57, robot .59) and S3 (computer .22, robot .33), and above it for S2 (computer .73, robot .80). For the ATI, the mean was 4.92 in the computer condition and 4.71 in the robot condition (4.81 in the full sample), and both conditions were above the midpoint of the scale (3.5). Cronbach's alpha was .67 in the computer condition, .78 in the robot condition and .72 in the full sample.

Godspeed. The results are shown in Table 6.2. All five dimensions had a mean above 3. Likeability had the highest values ($M = 4.60$, $Mdn = 4.9$) and animacy the lowest mean ($M = 3.27$). Anthropomorphism

had the largest standard deviation (0.94). Cronbach's alpha was .85 or higher for four dimensions, but it was negative for perceived safety (−0.42).

Dimension	*M*	*Mdn*	*SD*	Range	Cronbach's α
Anthropomorphism	3.35	3.50	0.94	1.60–4.60	.93
Animacy	3.27	3.33	0.62	2.00–4.00	.85
Likeability	4.60	4.90	0.51	3.80–5.00	.94
Perceived intelligence	3.58	3.30	0.71	3.00–5.00	.90
Perceived safety	3.92	3.67	0.58	3.33–5.00	−.42

Table 6.2: Godspeed results in the robot condition (n = 8).

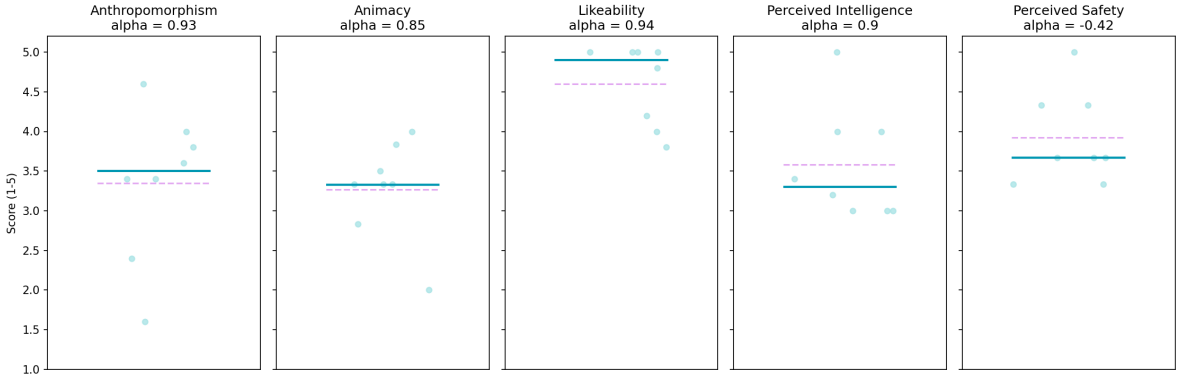


Figure 6.8: Scores of the Godspeed dimensions in the robot condition.

NASA-TLX. The overall score was slightly lower in the robot condition (*M* = 27.02, *SD* = 19.15, *Mdn* = 25.5) than in the computer condition (*M* = 29.55, *SD* = 17.78, *Mdn* = 28.5). Among the dimensions, performance had the highest mean in both conditions (computer 41.71, robot 38.12), and physical demand had the lowest median (computer 7.0, robot 0.0). The largest differences between conditions were in temporal demand, which was lower in the robot condition (25.75) than in the computer condition (39.71), and in frustration, which was higher in the robot condition (32.88) than in the computer condition (25.29). In the robot condition, one participant gave a physical demand score of 60, much higher than the rest.

SUS. The results are shown in Table 6.3. The mean SUS score was 86.07 (*SD* = 11.26, *Mdn* = 87.5) in the computer condition and 80.31 (*SD* = 13.33, *Mdn* = 83.75) in the robot condition. The mean and the median of both conditions corresponded to grade A. In both conditions, 5 participants obtained grade A and 2 obtained grade C. The remaining participant of the robot condition obtained grade D.

Condition	*n*	*M*	*SD*	*Mdn*	Min	Max	Participants per grade			
							A	B	C	D
Computer	7	86.07	11.26	87.50	70.0	100.0	5	0	2	0
Robot	8	80.31	13.33	83.75	62.5	100.0	5	0	2	1

Note. No participant obtained grade F.

Table 6.3: SUS scores and grades by condition.

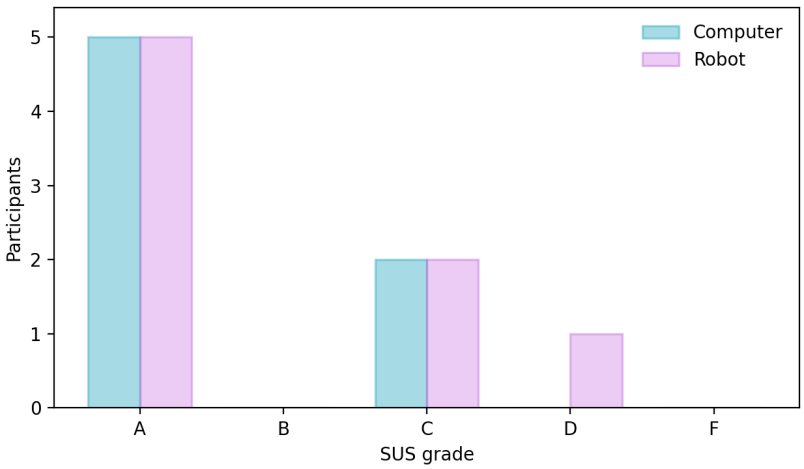


Figure 6.9: Distribution of SUS grades by condition.

Correlations. The correlation between the ATI and the SUS was weak and not significant, $r_s(13) = -.27$, $p = .33$. Within the robot condition ($N = 8$), the ATI had a significant negative correlation with anthropomorphism, $r_s(6) = -.77$, $p = .03$. The correlation with animacy was not significant, but a trend can be seen in the same direction, $r_s(6) = -.64$, $p = .09$. The ATI did not correlate with likeability ($r_s(6) = -.28$, $p = .51$), perceived intelligence ($r_s(6) = .11$, $p = .80$) or perceived safety ($r_s(6) = -.03$, $p = .95$).

No correlation between the NARS subscales and the Godspeed dimensions was significant. The strongest was between S1 and likeability, where a negative trend can be seen, $r_s(6) = -.70$, $p = .052$. It was followed by S1 and perceived safety, $r_s(6) = -.59$, $p = .12$, and S3 and animacy, $r_s(6) = -.51$, $p = .201$. The correlation between S3 and likeability was weaker ($r_s(6) = -.26$, $p = .529$). All the other correlations had $p > .40$. The complete results are reported in Appendix B.

6.5 Qualitative Analysis

The quotes in this section were translated from Catalan for this report.

Time perception. Six participants (4 computer, 2 robot) said that the whole session, or at least one of the phases, felt long. When asked about the general experience, Subject14 said, "Good, although a little long." Subject03, describing the relaxation phase, said, "It was a bit monotonous in the end. Spending 10 minutes looking at a screen feels long."

Breathing pace. Eight participants out of 15 (4 of 7 in the computer condition and 4 of 8 in the robot condition) said that the rhythm was hard to follow or did not suit them. Some found it too fast, such as Subject03: "I thought the ball was going too fast. I could still breathe more, and it was already switching to exhale." Others found it too slow, such as Subject07: "For me, breathing was going very slow. I find it hard to breathe so slowly."

Experience during the Stroop task. Six of the 15 participants (4 computer, 2 robot) said that they did not feel any stress. Three participants (1 computer, 2 robot) said that they felt a little stress, and three (all robot) said that they felt stressed or frustrated. Two participants, both in the computer condition, explained that they were not stressed because nothing would happen if they failed. Subject08 said, "I wasn't stressed because I knew nothing would happen if I got it wrong. If it had been an exam, then I would have been stressed." Another point that some participants highlighted about the Stroop task was its speed. Five participants, all in the robot condition, said that the task was too fast or that they

did not have enough time to click. Subject10 said, "In the end it was frustrating because it was very fast."

State during the relaxation phase. Seven participants (3 computer, 4 robot) said that they felt relaxed during this phase, and Subject13 said, "Very good. I got sleepy." However, four participants (2 computer, 2 robot) reported some stress or nervousness. Subject06 said, "It was a bit more stressful during the breathing part, because I was trying to follow the ball's rhythm, but it was hard."

Visual and sound feedback. Three participants (1 computer, 2 robot) said that the colour of the ball was frustrating or made them nervous, for example because it did not seem to change. Two participants (1 computer, 1 robot) said that they were guided more by the sound than by the ball. One of them said, "At some point I even closed my eyes and only listened to the beep." In contrast, Subject13 followed the ball and used the sound mostly to feel calmer.

Robot perception. No participant in the robot condition made a negative comment about Tyto. Five of the eight described it as cute, pleasant, funny or sweet. Two participants highlighted the robot's eyes, and one said that its little sounds were funny. The sounds were not perceived positively by everyone. Subject09 said, "The high-pitched sound sounds like my alarm clock, and it made me feel activated. It didn't help me relax."

7 Discussion

This pilot study compared the owl robot Tyto with a computer interface during a guided breathing exercise at 6 breaths per minute with biofeedback, with 15 healthy participants (7 in the computer condition and 8 in the robot condition). No evidence was found that the robot condition led to better stress regulation than the computer condition (RQ). Neither the effect of condition nor the phase $\times$ condition interaction was significant in any measure (RQ1 and H0). With 7 or 8 participants per condition, however, these results are not conclusive, and some of the patterns described below are an interesting starting point for a larger study.

7.1 Physiological and Subjective Stress Measures

The measures did change across phases. From baseline to the Stroop task, mean HR and the beta/alpha ratio increased and relative alpha decreased significantly, which is the direction expected under stress. This agrees with the literature, where stress is associated with a higher heart rate, lower alpha activity and higher beta activity (Giannakakis et al., 2022). Jun and Smitha (2016) also reported lower alpha power and higher beta power during a Stroop task than at rest, in a sample of 10 participants. From the Stroop task to the relaxation phase, relative alpha increased and the beta/alpha ratio decreased significantly. Alpha activity predominates during relaxation, whereas higher-frequency activity is linked to alertness and processing demands (Giannakakis et al., 2022), so these changes are consistent with a more relaxed state. HR also decreased during the relaxation phase, but not significantly.

The VAS showed a significant effect of phase, but no post hoc comparison was significant, although a trend can be seen in both transitions. This is consistent with the interviews, in which six of the 15 participants said that they felt no stress during the Stroop task, some of them because there were no consequences for errors, and only three felt stressed or frustrated. The Stroop task, therefore, did not induce as much stress as intended. Pande et al. (2025) also found that most participants reported little stress in a pilot study with the NAO robot and a Stroop-based game. Even so, the physiological measures did show a stress response, and a more stressful task might have produced larger differences between phases. Relative beta and the FAA did not change significantly across phases. Vanhollebeke et al. (2023) also found no significant change in frontal beta power under psychosocial stress, and suggested that band power measured at sensor level may not be sensitive enough. For relative beta, a trend can be seen in the descriptive data: it increased during the Stroop task and decreased during the relaxation phase in the computer condition, but stayed almost constant in the robot condition. This difference was not significant (phase $\times$ condition interaction, $p = .077$), so it should not be interpreted as an effect.

The EEG measures followed the expected changes more clearly than HR and the VAS. Relative alpha and the beta/alpha ratio had the largest effects of phase ($\eta_p^2 = .51$ and .48) and medium to large effect sizes in both transitions (relative alpha, $d = 0.72$ and 1.16; beta/alpha ratio, $d = 0.71$ and 0.97). In contrast, HR changed significantly only in the stress response, with a small effect size ($d = -0.27$). Relative alpha was also higher in the relaxation phase than at baseline, and it was the only measure that correlated significantly with the VAS. Alpha and beta activity are already used as indicators of stress (Giannakakis et al., 2022), and these results suggest that, in this guided breathing protocol, they were also sensitive to the recovery, and more so than HR, even with a small sample. This interpretation has limits, since alpha activity is not specific to relaxation. It increases when visual and cognitive demands decrease and when the eyes are closed (Barry et al., 2007), so the increase after the Stroop task could partly reflect the end of a demanding task. Without a control phase with no guided breathing, these explanations cannot be separated. A related question, although the difference was not significant, is why relative beta did not decrease in the robot condition during the relaxation phase while HR

decreased slightly. HR depends on the autonomic nervous system and, at 6 breaths per minute, also directly on the breathing itself (Lehrer & Gevirtz, 2014), whereas beta activity is linked to alertness. Following the ball, the sound and the robot at a pace that was difficult for some participants may have kept cognitive engagement high even when HR decreased. This was not tested.

In both conditions, the baseline and the Stroop task were presented in the same way, on the computer in front of the participant, and the robot was only used in the relaxation phase of the robot condition. A difference between conditions would therefore mainly appear in how participants changed from the Stroop task to the relaxation phase, that is, as a phase $\times$ condition interaction, and this interaction was not significant for any measure. Klęczek et al. (2024) also found no significant differences between a Pepper robot group and a laptop group in self-reported stress, galvanic skin response or FAA. Descriptively, during recovery, HR, relative alpha, relative beta and the beta/alpha ratio changed more in the computer condition, and only the FAA and the VAS changed more in the robot condition. The larger decrease of the VAS in the robot condition should be read with care, because this group also had a higher VAS at T2 (4.88 compared with 2.71) and therefore more room to decrease. Several interactions had medium to large effect sizes (η_p^2 = .18 for relative beta, .138 for the beta/alpha ratio and .13 for the VAS, where .06 is considered medium and .14 large; Cohen, 1988; Richardson, 2011). With only 7 or 8 participants per condition, the study had little power to detect them, so a larger sample would be needed to see whether these differences are real. The absence of significant differences does not show that the two conditions are equal.

The VAS and the physiological measures changed in the same direction in each transition, in both conditions, taking into account that relative alpha is expected to decrease with stress (RQ2). The only exception was relative beta in the robot condition, which did not decrease during recovery. Only 5 of the 30 correlations between change scores were significant. The beta/alpha ratio and relative beta correlated in both transitions, which is understandable since both are based on beta power. The FAA correlated with the beta/alpha ratio (stress response) and with relative beta (recovery), but since the FAA showed no effect on its own, these correlations are difficult to interpret. The most interesting result was the negative correlation between relative alpha and the VAS in the stress response ($r_s(13) = -.630$, p = .012), with a similar trend in recovery ($r_s(13) = -.478$, p = .072). This suggests that participants whose alpha changed more also reported larger changes in perceived stress (RQ3). However, the correlations were exploratory, with both conditions pooled and without correction for multiple comparisons, so this result needs to be confirmed in future work.

7.2 Perception of the System

Both groups had similar attitudes toward robots (NARS) and affinity for technology interaction (ATI). The only notable difference was in NARS S2, which was lower in the computer condition (1.69 compared with 2.33). Since the NARS was answered before the interaction with the robot, Tyto cannot explain this difference. The usability of both systems was rated positively, with a mean SUS grade A in both conditions (86.07 in the computer condition and 80.31 in the robot condition) (Sauro, 2018).

The NASA-TLX showed a low workload in both conditions, with similar values in most dimensions. The exceptions were temporal demand, which was almost 14 points lower in the robot condition, and frustration, which was about 7.5 points higher in the robot condition. The difference in temporal demand could be related to the perception of time, since six participants said that the session felt long (four in the computer condition and two in the robot condition). It cannot be explained by the breathing pace alone, because the share of participants who had difficulties with it was similar in both conditions (4 of 7 and 4 of 8). For frustration, a possible explanation is that a physically embodied robot produces more social presence than a computer (Powers et al., 2007), which can increase the feeling of being observed and judged. This was not measured, so it remains a hypothesis, and with 15 participants it cannot be determined whether these differences are real or due to chance.

Regarding the perception of the robot, all Godspeed dimensions had a mean above 3, the neutral point, and likeability reached 4.6 out of 5. The interviews support this, since none of the eight participants in the robot condition made a negative comment about Tyto, and five described it with positive adjectives such as cute, pleasant, funny or sweet. Animacy had the lowest mean (3.27), so there is room to improve how lively the robot looks. The Godspeed questionnaire was only answered in the robot condition, so these results cannot be compared with the computer condition. Perceived safety had a negative Cronbach's alpha, which may be related to a pair of items that was unclear in the Catalan translation.

The interviews also showed problems in the relaxation phase. Eight of the 15 participants (four in each condition) reported some discomfort with the breathing pace. A rate of 6 breaths per minute has been proposed as helpful for relaxation (Lehrer & Gevirtz, 2014; Marchant et al., 2025), but in people without breathing training it can cause moderate hyperventilation (Szulczewski & Rynkiewicz, 2018). This could be one reason why some participants felt more stressed during the exercise, although it was not tested here.

Three participants said that the colour of the ball was frustrating or made them nervous, because it did not seem to change. As explained in Section 7.4, the colour was calculated from an approximation of HRV that was not reliable. The colour was meant to show progress, and when it showed none, it gave no sign that the participant was relaxing, which may have led participants to try harder. Trying to relax under load can have the opposite effect (Wegner et al., 1997). Two other participants said that they were guided more by the sound than by the ball, and one of them closed their eyes during the exercise. Alpha power is higher with eyes closed than with eyes open (Barry et al., 2007), so for that participant the alpha increase cannot be attributed to the relaxation phase alone. Finally, one participant said that the high-pitched sound reminded them of their alarm clock and made them feel activated. Together, these reports suggest that for some participants the biofeedback did not work as intended, although with so few participants nothing can be said about how often this happened.

7.3 Research Questions and Hypotheses

Table 7.1 summarises the answer to each research question and hypothesis.

	Question or hypothesis	**Answer**
RQ	Does interaction with a social robot lead to greater physiological and subjective stress regulation than interaction with a conventional computer interface, during guided biofeedback-based breathing exercises?	No evidence was found.
RQ1	Are there differences in stress-related physiological indicators, such as HRV and EEG-derived measures, between the robot condition and the computer condition?	No significant differences in mean HR or in any EEG measure. HRV could not be analysed.
RQ2	How consistent are participants' subjective stress perception and the objective physiological measures obtained during each condition?	They changed in the same direction in each transition, except relative beta in the robot condition.
RQ3	Is there a measurable correlation between physiological indicators and perceived-stress scores, and does this relationship differ between conditions?	The VAS correlated significantly only with relative alpha in the stress response. Differences between conditions could not be assessed.

	Question or hypothesis	Answer
RQ4	Does interacting with a socially present physical agent lead to a greater sense of emotional support and engagement compared to a computer interface?	It could not be answered, because no measure compared these aspects between conditions (see H5).
H0	There will be no significant differences between the robot condition and the computer condition in any of the stress measures.	There is no evidence to reject H0. This does not show that the conditions are equal.
H1	Participants interacting with the social robot will show a greater reduction in perceived stress compared to the computer condition.	Not supported. The VAS decreased more in the robot condition during recovery, but the phase $\times$ condition interaction was not significant (p = .164).
H2	The robot condition will show higher HRV values, indicating better physiological stress regulation.	Could not be tested, because reliable HRV could not be computed. Mean HR did not differ significantly between conditions.
H3	A partial dissociation will occur between subjective and physiological measures, such that the reduction in stress reported by the participants will be proportionally greater than the improvement observed in physiological measures.	Not supported. The VAS change (d = 0.69) was smaller than those of relative alpha (d = 1.16) and the beta/alpha ratio (d = 0.97) and larger than that of mean HR (d = 0.15).
H4	There will be a significant correlation between HRV values and perceived-stress scores.	Could not be tested as written. The VAS correlated significantly only with relative alpha in the stress response.
H5	The robot condition will generate a greater sense of emotional support and engagement during the exercise.	Could not be evaluated. The Godspeed questionnaire gave positive results in the robot condition, but the same information was not collected in the computer condition, so the two could not be compared.

Table 7.1: Answers to the research questions and hypotheses.

7.4 Changes and Limitations

Several things changed with respect to the internship proposal. First, the target population changed from people with chronic stress to a healthy population. As explained in Section 4.3, contact was made with AMPANS, Althaia and ACAF, but no collaboration could be reached. In addition, the study had to be approved by an ethics committee, which can take several weeks, and longer if sensitive topics are involved. The study already involved sensitive data, since it recorded biological data from the participants, and adding a vulnerable population could have made the approval process longer. As a result, the study induced short-term stress with the Stroop task in healthy participants, instead of studying chronic stress. Second, the study was originally conceived as a larger one, but because of time limits, the aim of simplifying the system and the difficulty of finding enough volunteers without offering any reward, it was decided to carry out a proof of concept, which requires fewer participants. Third, the plan was to use the Qui-Bot robot, but it was not available when it was needed. A robot was therefore designed and built specifically for this experiment. The proposal mentioned humanoid robots because those were the ones available at the university, and building the robot made it possible to choose its form, so a zoomorphic form was considered more suitable (see Section 3.4). The result was Tyto, a zoomorphic owl. This delayed the user tests, since time had to be spent building the robot instead of writing a simple program for Qui-Bot. Fourth, the protocol was approved by the CEUPC in two versions. After the first approval, changes were made to the protocol and it was submitted again,

and the second version was also approved. Finally, HRV was planned as one of the measures, but it could not be used because the sensor was not suitable for it, and mean HR was analysed instead.

The main limitation was the small sample size ($N = 15$), which did not give enough statistical power to draw firm conclusions. The age also differed between conditions, with a mean of 31.1 years in the computer condition and 42.5 years in the robot condition. The groups also seemed to differ in how they experienced the Stroop task, even though it was identical in both conditions. All the participants who found it too fast, and all those who said they felt stressed or frustrated during it, were in the robot condition (Section 6.5), and the VAS at T2 was higher in this group. With 7 or 8 participants per condition, differences between groups unrelated to the condition can easily appear by chance.

The second limitation was the cardiac sensor. The Magene H603 does not send a value at each heartbeat, but the latest reading at fixed time intervals, and calculating HRV requires the time between consecutive heartbeats. This was not stated in the specifications and was discovered after several tests, in which the HRV values calculated were not consistent. For this reason, HRV could not be analysed, and mean HR was the only cardiac measure. The colour of the ball was still calculated in real time from the RMSSD of the last 60 values received, as described in Section 3.2. Since these values were not true beat-to-beat intervals, the colour was not reliable, which may also explain why the ball tended to show reddish colours. As a result, the feedback shown to participants may not have reflected their real physiological state.

Third, no measure of breathing was recorded. It was therefore not possible to verify that participants followed the 6 breaths per minute pace, or to test whether the pace caused hyperventilation. The pace was also the same for all participants. The first version of the protocol included a step before the baseline to find the resonance frequency of each participant, based on Shaffer and Meehan (2020). Participants breathed for a few minutes at each rate between 6.5 and 4.5 breaths per minute, in steps of 0.5 breaths per minute, and a calculation was used to find which rate gave the best breathing performance for each participant. Then, during the relaxation phase, the participant would be guided gradually towards their ideal resonance frequency by slowly decreasing the rate. In the end, this was discarded to simplify the protocol and reduce its length, since it would have added about 30 minutes, resulting in a test of about 90 minutes. A test this long could discourage volunteers, and it could cause tiredness, which could affect the emotional state and the measures.

Fourth, the study did not include a phase without guided breathing, so the recovery cannot be attributed to the breathing exercise alone (see Section 7.1). The Stroop task also caused less stress than planned, according to the interviews (Section 6.5).

Fifth, Tyto was built specifically for this study. It has not been validated and there are no previous results to compare with, and being new, it may have caused a novelty effect. The questionnaires were also translated into Catalan for this study, and only the original English versions have been validated, which may explain some of the low Cronbach's alpha values. Finally, the system was developed and the experiments were run by the same person, so the experimenter was not blind to the condition.

8 Conclusions

This study compared a zoomorphic social robot, Tyto, with a computer interface during a guided breathing exercise with biofeedback. It was a pilot study with 15 healthy participants, 7 in the computer condition and 8 in the robot condition. Stress was induced with a Stroop task, and the response was measured with HR, EEG and the VAS. Usability, workload, attitudes toward robots and the perception of the robot were measured with questionnaires and interviews.

No evidence was found that the robot condition led to better stress regulation than the computer condition. No significant differences between conditions were found in any physiological measure or in the VAS. With a small sample of 15 participants, this does not show that the two conditions are equal. It only shows that no difference could be detected.

The measures did change over the session. HR, relative alpha and the beta/alpha ratio changed in the direction expected for an increase in stress during the Stroop task. Relative alpha and the beta/alpha ratio then changed in the direction of recovery during the relaxation phase. They were the most sensitive measures, even with a small sample, which suggests that alpha activity may be a useful indicator to follow relaxation in a larger study. However, as the study did not include a phase without guided breathing, this recovery cannot be attributed to the biofeedback or to the guided breathing, since it could also be due to the passing of time. The VAS correlated with relative alpha in the stress response, but this correlation is exploratory and needs to be confirmed.

Both systems, the computer web app used in the computer condition and the robot Tyto used in the robot condition, obtained a high usability score (mean grade A in the SUS), and the robot was perceived positively (likeability of 4.6 out of 5 in the Godspeed questionnaire).

Regarding the hypotheses, H0 could not be rejected, and H1 and H3 were not supported. H2 and H4 could not be tested because reliable HRV could not be computed with the cardiac sensor used, and H5 could not be evaluated because no measure compared emotional support or engagement between conditions. The main limitations were the small sample, the cardiac sensor, which also made the feedback shown to the participants unreliable, the lack of a control phase, the lack of previous validation of the robot, and the low level of stress caused by the Stroop task.

Despite these limitations, this pilot study achieved its main purpose. It produced a working system, formed by the robot Tyto, the computer web app, the integration of the EEG and cardiac sensors, and the experimental protocol. It also gave preliminary data that can be used to design a larger study, and it made it possible to identify which parts of the protocol worked and which did not. Although no significant differences between conditions were found, the changes in the physiological measures over the session, the differences between conditions that were not significant but had medium to large effect sizes, and the positive perception of the robot suggest that this line of research is worth following in a larger study, applying the lessons learned from this pilot. The improvements proposed to continue this work are described in Chapter 9.

On a personal level, this internship made it possible to go through all the steps of a real research project, from the initial idea to the analysis of the data. Some of the tasks had already been experienced in the master's courses, such as writing protocols, running user tests, analysing questionnaires and interviews with a thematic analysis, and programming medium-complexity systems. Here, they were applied to a real study with real participants. Other aspects were completely new, such as the use of EEG, how to analyse it and the neuroscience concepts behind it, as well as HRV. Statistical analysis was also new, including significance testing, mixed ANOVA and power analysis. Finally, it was the first time preparing the documentation for an ethics committee and going through the approval process. The project also helped to understand what to take into account when designing an experiment, such as validating the sensors more thoroughly, including a control condition, and testing the stress task with more people before the experiments. It also showed the real time that a project of this kind requires, and what to expect in case of deciding to follow a research career in the future.

9 Future Work

Several lines of future work are proposed, ordered from the short to the long term.

In the short term, the anonymised EEG data obtained could be made available in an open data repository such as OpenNeuro or Zenodo, so that other researchers can use it. This is covered by the informed consent form signed by the participants and by the protocol approved by the CEUPC. The interview recordings would not be shared. In addition, following an open-source philosophy, a Git repository could be created with the documentation, the code and the files needed to print and reproduce the robot Tyto, including the list of components used, under an open licence. This would allow others to use it or improve it in future research.

In the medium term, a poster or article about this research could be prepared with the support of the project supervisors, with the aim of publishing it in a journal. The goal is that the work may contribute to the field of HRI and motivate new research.

In the long term, the study could be repeated with a larger sample and some modifications to the protocol. A larger sample is needed to confirm whether the trends observed in this pilot are real. Regarding the protocol, first, a control group would be added. This group would follow the same protocol, but instead of doing the breathing exercise after the Stroop task, participants would rest for the same duration (10 minutes). This would make it possible to tell whether the recovery is due to the breathing exercise or to the passing of time. Second, the Polar H10 cardiac sensor would be used, since it has been validated against ECG for RR intervals and linear HRV measures (Schaffarczyk et al., 2022). It is about twice as expensive as the sensor used in this study, but it would allow a reliable HRV analysis. Third, a measure of breathing would be added, to check that participants follow the pace. Fourth, the biofeedback would probably be dropped, and guided breathing alone would be used. Since the feedback in this pilot was not reliable, the study cannot say whether it had an effect, and removing it would also remove one variable that could explain the recovery. If it were kept, the colour range would be changed to avoid red, which has negative associations. Finally, alternatives to the Stroop task would be explored for the stress induction, because it did not cause the expected level of stress during this pilot study.

The sample size should be defined with a power analysis before the study. As a reference, a similar study had 18 participants per group (Klęczek et al., 2024). With a control group, there would be three conditions. A medium effect of the phase $\times$ condition interaction ($f = 0.25$) is a reasonable value to plan for. A large effect ($f = 0.40$) is unlikely, since the present pilot, with 15 participants, would probably have detected it (power of about 0.90, assuming a correlation of 0.5 between phases). In contrast, the pilot had low power for a medium effect (about 0.50), so a medium effect cannot be ruled out. A mixed ANOVA with three conditions and three phases needs about 36 participants in total (12 in each condition) to detect a medium interaction with 80% power, according to a power analysis made with G*Power (Faul et al., 2007).[1] A small effect ($f = 0.10$) would need about 200 participants in total, which is not feasible for a study of this kind, so it is ruled out as a design target. Therefore, 15 to 20 participants per condition would be enough for medium effects, but not for small ones. This also leaves a margin over the minimum of 12 per condition, in case some participants have to be excluded, as happened in this study.

Tyto could also be improved. Animacy was the lowest dimension of the Godspeed questionnaire, so the robot could be given more life, for example with pupil movement in the eyes and a light effect in the belly area that glows in time with the breathing exercise, to give the sensation of the lungs moving. The light effect was not made because it was technically complicated. The body is 3D-printed, so the plastic would need to be thin or translucent enough to let the light through, while still hiding the electronic components.

[1] G*Power version 3.1.9.7 is a free program for calculating the sample size needed for a study. The analysis was an a priori power analysis for the interaction of a repeated measures ANOVA (within-between), assuming a medium effect ($f = 0.25$) and a correlation of 0.5 between repeated measures.

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A System Implementation

A.1 Frontend

The frontend was built with React, chosen for its extensive documentation and widespread use, which made it easy to find help during development. Its state management was also one of the reasons for choosing it, as it updates the relevant parts of the interface whenever a value changes, such as the colour of the ball or the Stroop task counters, without having to control manually when the screen refreshes. Vite was used as the build tool, as it is the standard choice for React projects. The interface was styled with the Bulma CSS framework, because of prior experience with it and a preference for its clean, simple look. Navigation between screens is handled by react-router-dom, the standard routing library for React.

The application runs entirely in the browser. Once loaded, it updates its content as the participant progresses through the experiment, without requesting a new page from the server at each step. All communication with the backend uses standard HTTP requests to the Express API.

The first screen is the only one intended for the experimenter, who selects the condition assigned to the participant. Starting the experiment from this screen, creates the corresponding records in the `users` and `sessions` tables, and stores the selected condition in the browser's local storage, so that all the following screens can use it without querying it again.

The pre-experimental and post-experimental questionnaires are each shown on a single scrollable page, with all their questions at once. A single "Next" button at the bottom submits the whole questionnaire and moves on to the next one. If a mandatory question is missing, it is marked on screen and the participant cannot continue until it is answered. The format of the questions depends on the questionnaire:

- ATI, NARS, GAD-7 and SUS use a Likert scale matrix, a grid of discrete response options.
- The VAS is a row of eleven options from 0 to 10 along a line, with only the two endpoints labelled ("not at all stressed" and "extremely stressed").
- The Godspeed questionnaire (robot condition only) uses a semantic differential scale, where each item has its own pair of opposite adjectives at the two ends of the line.
- The Raw NASA Task Load Index (RTLX) uses a continuous slider from 0 to 100 for each item, following the format of the original paper-based questionnaire. Since a slider always starts at a default midpoint value, an additional mechanism checks that the participant has moved it before the question is considered answered.
- The demographics questionnaire collects basic information, including the participant's age and sex.

During the three phases, the application connects to the heart rate bridge program through a WebSocket and sends each reading to the backend (Section 3.6). The backend also sends the markers that flag the start and end of each phase in the EEG recording.

A.2 Backend

The backend was built with Node.js, based on prior experience with it and its ease of use. Express.js is used as the web framework for the REST API, as it is a simple and widely used option for lightweight APIs in Node.

The backend receives the data from the frontend through the REST API and stores it in the PostgreSQL database (Section 3.5). It also forwards information to two external systems, through two dedicated modules. `robotBridge.js` forwards the colour, start and stop commands to Tyto, and `markerBridge.js` notifies the EEG marker program whenever a phase is opened or closed (Section 3.6). In both cases, the backend does not wait for a response. If the robot or the marker program cannot be reached, the request fails, a warning is logged and the experiment continues.

The API is organised by resource, with a separate route file for each one: users, sessions, phases, Stroop task results, raw heart rate readings, HRV values, questionnaire responses and robot commands. This structure mirrors the database tables. All endpoints are available under the `/api` path, around 20 in total.

Among them, the phases endpoints have a central role, since opening or closing a phase also triggers the EEG markers described above. The questionnaire endpoint also works differently from the rest. Instead of storing a whole questionnaire as a single entry, it inserts each answer as a separate row, one per question.

A.3 Database Tables

This section describes every field of each table of the database (Section 3.5). In the code, each run of the experiment for a participant is stored in a table named `sessions`, and the other tables refer to it through the `session` field. In this report, participants are referred to as SubjectX. PK marks a primary key and FK a foreign key.

Field	Type	Description
`user_id` (PK)	`varchar(255)`	Identifier of the participant. To preserve anonymity, it is a randomly generated alphanumeric value.
`condition`	`varchar(255)`	Condition assigned to the participant: robot or computer.

Table A.3.1: `users`: participants of the experiment.

Field	Type	Description
`session_id` (PK)	`integer`	Auto-incremented identifier.
`user_id` (FK to `users`)	`varchar(255)`	Participant who carried out the experiment.
`session_date`	`timestamp`	Date and time at which the entry was created.

Table A.3.2: `sessions`: one entry for each run of the experiment.

Field	Type	Description
`session` (PK, FK to `sessions`)	`integer`	Experiment to which the phase belongs.
`type` (PK)	`varchar(255)`	Name of the phase: `baseline`, `stroop_task` or `biofeedback`.
`start_time`	`timestamp`	Moment at which the phase begins.
`end_time`	`timestamp`	Moment at which the phase ends (empty while the phase is ongoing).

Table A.3.3: `phases`: the three phases of the experiment and their start and end time.

The value `biofeedback` of the field `type` corresponds to the relaxation with biofeedback phase. The same value is used in the code of the EEG markers (Section A.7.1).

Field	Type	Description
`session` (PK, FK to `sessions`)	`integer`	Experiment to which the results belong.
`type` (PK)	`varchar(255)`	Phase of the results (always `stroop_task`).
`n_total`	`integer`	Total number of attempts, correct and incorrect.
`n_errors`	`integer`	Number of incorrect attempts.

Table A.3.4: `stress_phase`: results of the Stroop task.

Field	Type	Description
`session` (PK, FK to `sessions`)	`integer`	Experiment to which the reading belongs.
`type` (PK)	`varchar(255)`	Phase during which the reading was captured.
`timestamp` (PK)	`timestamp`	Moment of the reading.
`bpm`	`integer`	Heart rate, in beats per minute.
`rr_ms`	`numeric(6,2)`	Interval between heartbeats in milliseconds, calculated from the heart rate. It does not correspond to true beat-to-beat intervals (see Section 7.4).

Table A.3.5: `phase_readings`: raw readings of the heart rate sensor, one row per reading.

Field	Type	Description
session (PK, FK to sessions)	integer	Experiment to which the value belongs.
type (PK)	varchar(255)	Phase during which the value was calculated.
timestamp (PK)	timestamp	Moment of the calculation.
rmssd	decimal(4,2)	Approximate RMSSD, calculated from the last 60 readings. It is not a true measure of HRV (see Section 7.4).
cardiac_coherence	decimal(4,2)	Value derived from the RMSSD, used to set the colour of the ball.

Table A.3.6: hrv_value: values calculated from the heart rate readings during a phase.

Field	Type	Description
session (PK, FK to sessions)	integer	Experiment to which the recording belongs.
file_path (PK)	varchar(255)	Path of the .nedf file inside backend/ecg/.
recorded_at	timestamp	Moment at which the recording was made.

Table A.3.7: eeg_recordings: location of the EEG recording of each participant. The signal itself is stored on disk (backend/ecg/*.nedf), not in the database.

Field	Type	Description
session (PK, FK to sessions)	integer	Experiment to which the answer belongs.
questionnaire_id (PK)	integer	Numeric identifier of the questionnaire.
questionnaire	varchar(255)	Name of the questionnaire or scale: demographics, ati, nars, gad7, vas1, vas2, vas3, sus, godspeed or nasa.
question_id (PK)	varchar(255)	Identifier of the question inside the questionnaire.
response	varchar(255)	Answer given by the participant.

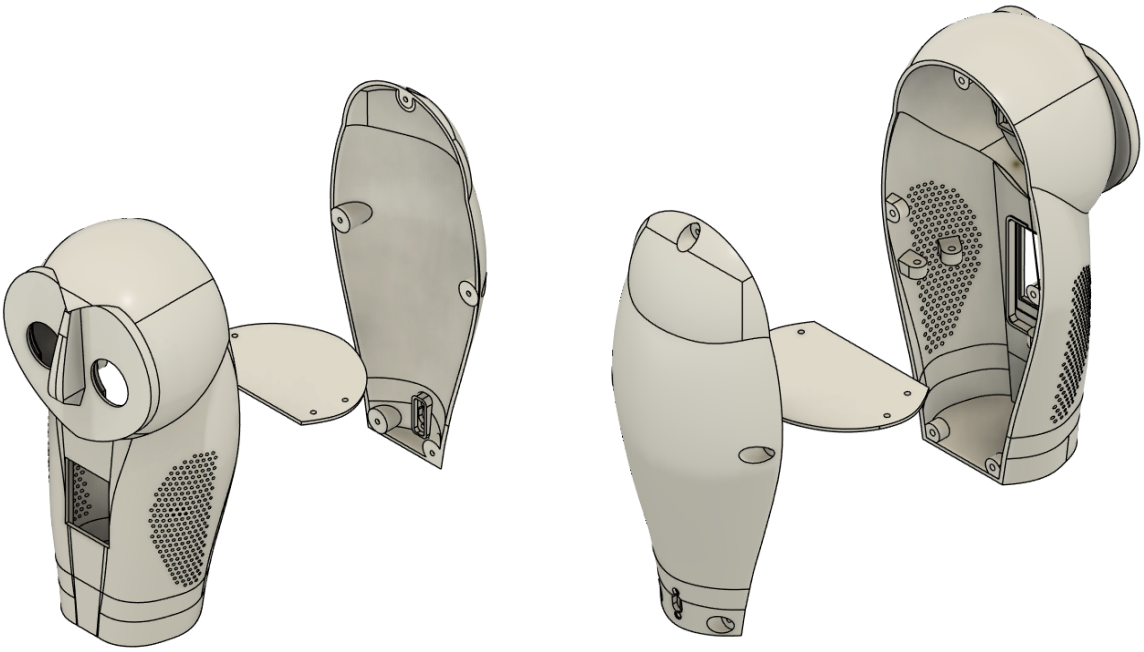
Table A.3.8: questionnaire_responses: answers to all the questionnaires and VAS scales, one row per question.

A.4 3D Design and Fabrication

The 3D model of Tyto was designed with Autodesk Fusion. There was no prior experience with this software, and it was chosen because it is comprehensive and free for personal use. Fusion also allows past operations in the design history to be modified easily, which was useful given the iterative nature of the design process.

The final design measures approximately 134 mm in width, 155 mm in depth and 280 mm in height. It consists of three main parts: the front shell, the back cover and an internal platform that divides the interior of the robot and provides a surface to mount the Raspberry Pi. To hold the internal components

in place, small protrusions were added to the design. Those holding the screens are in the front shell, and those holding the USB-C extension cable are in the back cover, so that each part fits snugly into position (Figures A.4.1 and A.4.2).



(a) Exploded view from the front.

(b) Exploded view from the back, showing the interior and the protrusions.

Figure A.4.1: 3D design of Tyto: front shell, back cover and internal platform.

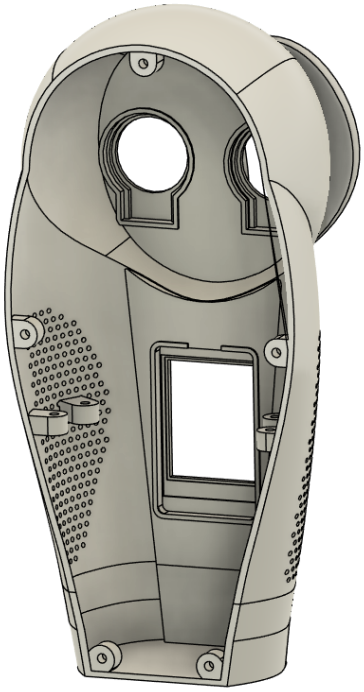


Figure A.4.2: Interior of the front shell, showing the protrusions that hold the screens.

The back cover is attached to the front shell with five M4 screws, and the internal platform is fixed with another four M4 screws. Since screwing directly into 3D-printed plastic is not reliable and can strip the threads, metal threaded inserts were embedded in the corresponding holes before assembly.

All parts were printed with an Anycubic Kobra X printer, using matte white PLA filament. This is consistent with the minimalist aesthetic described in Section 3.4, as it avoids a finish that draws unnecessary attention.



Figure A.4.3: Tyto printed and assembled.

One of the most challenging aspects of the design was achieving correct tolerances for the openings and protrusions holding each component. To obtain a precise fit, every part was measured with a caliper before being modelled.

A.5 Hardware and Wiring

Table A.5.1 lists the components of Tyto with the model used for each one.

Component	**Model**	**Function**
Processing unit	Raspberry Pi 4 Model B (8 GB)[1]	Controls all the other components. It has heatsinks and a fan above the processor to prevent overheating during long use.
Belly screen	2.4-inch LCD module (ILI9341) from Waveshare[2]	Shows the breathing ball.
Eye screens	Two 1.28-inch round LCD modules (GC9A01)[3]	Show the eyes and the blinking animation.
Audio output	Waveshare WM8960 Audio HAT, with speakers	Plays the owl hoot sounds that mark the inhalation and the exhalation.
Connection indicator	Generic RGB LED	Indicates a correct connection between the robot and the web application.

Table A.5.1: Components of Tyto.

The belly screen is the same model used in the Bachelor's thesis of the author, and the eye screens use the same GC9A01 controller as those of that project.[4]

To prevent the wiring from disconnecting while the robot was transported, all components were connected to a custom board, built as a soldered prototyping board shaped as a HAT (Hardware Attached on Top) for the Raspberry Pi. The components are connected to the board with Dupont wires, and resistors are included in the circuit of the RGB LED. This gives a stable and compact connection without loose wiring. Figure A.5.1 shows the wiring diagram of the board.

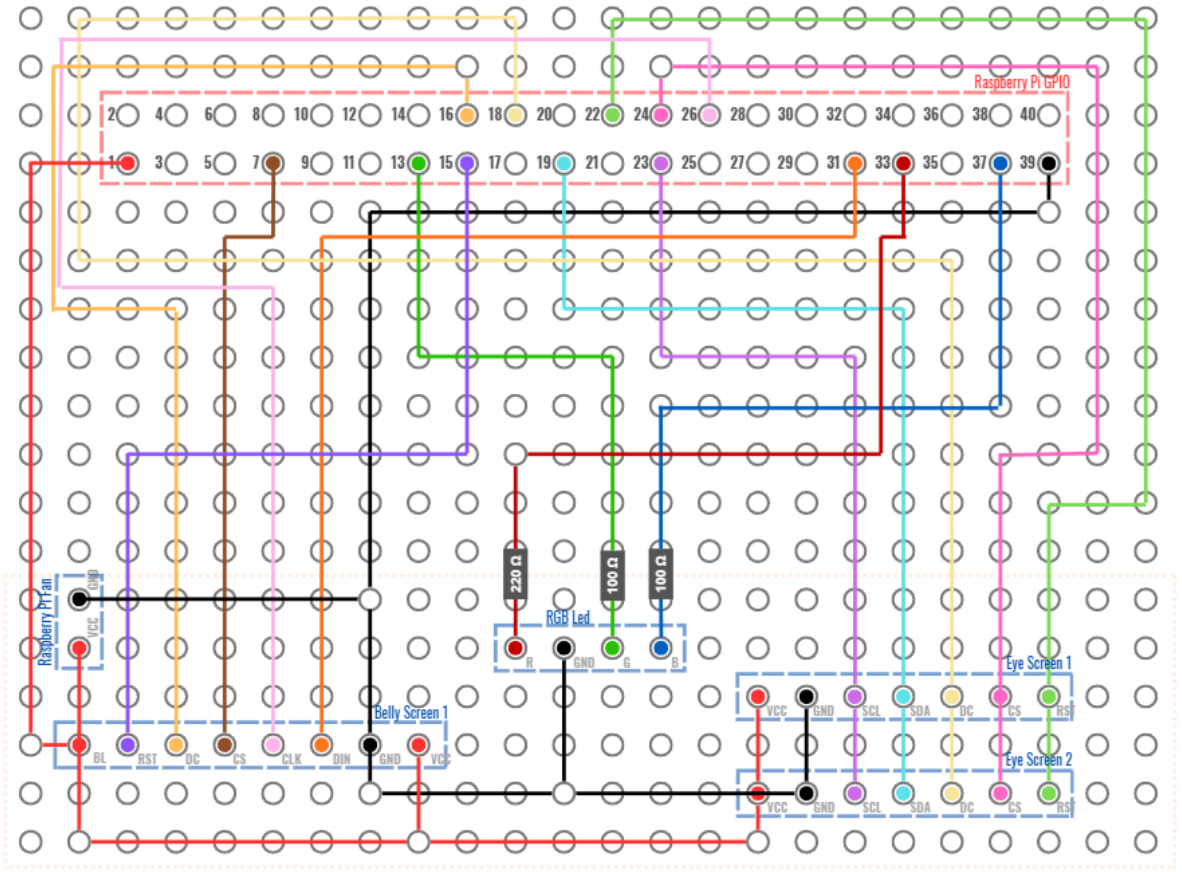


Figure A.5.1: Wiring diagram of the custom board. Numbers on the Raspberry Pi GPIO header are physical pin numbers. Red wires carry power and black wires ground, and the other colours distinguish the signals. The pins of each screen are named as printed on its module. On the eye screens, SCL and SDA are the clock and data pins of the SPI interface.

A.6 Internal Architecture of Tyto

Inside Tyto, the components communicate through two independent SPI buses, plus the I2C and I2S buses of the audio HAT.

[1]https://www.raspberrypi.com/products/raspberry-pi-4-model-b/
[2]https://www.waveshare.com/2.4inch-lcd-module.htm
[3]https://www.amazon.es/Tiardey-Pantalla-Pulgadas-Arduino-Unidades/dp/B0DM569VDT
[4]Cadevall Soto, H. (2022). *Robòtica aplicada a teràpies i tractament de trastorns de l'espectre autista en infants* [Bachelor's thesis, Universitat Politècnica de Catalunya]. https://hdl.handle.net/2117/377449 (in Catalan).

The two eye screens are wired in parallel on a shared SPI bus, with every pin connected in common, including chip select, data/command and reset. Since both screens always show the same content, this wiring simplified the design and left another SPI bus available for other components. For the same reason, the software treats them as a single component.

The belly screen uses its own SPI bus. Unlike the eyes, it shows different content, the ball of the biofeedback (Section 3.2), so it cannot share a bus with them.

The audio HAT (WM8960) uses two further buses: an I2C bus to control the playback settings, such as the volume, and an I2S bus to transmit the audio signal. These settings are configured through the operating system of the Raspberry Pi, and the software of the robot does not interact with either bus directly. The software only triggers the playback through the audio system of the operating system.

Finally, the status LED is controlled with three digital output pins (red, green and blue), each with a current-limiting resistor in series. Colours are not mixed with PWM.

A.7 Communication Protocol

Table A.7.1 lists all the connections between the components of the system (Figure 3.1), with the protocol and the information that travels through each one. The reasons for each choice are explained in Section 3.6.

From	To	Protocol	Information
Magene H603	Bridge program	Bluetooth	Heart rate readings
Bridge program	Browser	WebSocket	Each new reading, in real time
Browser	Backend	HTTP (REST API, `/api`)	Readings, answers, phases and robot colour
Backend	PostgreSQL	SQL	Storage of all the data
Backend	Marker program	HTTP GET (local)	Start and end of each phase
Marker program	NIC2	Lab Streaming Layer (LSL)	Numeric markers
Enobio 8	NIC2	USB	EEG signal
Backend	Tyto	HTTP over WiFi	Colour, start and stop commands

Table A.7.1: Connections between the components of the system.

A.7.1 EEG Markers

Whenever a phase starts or ends, the backend sends a local HTTP GET request to the marker program (`enobio_markers.py`), with the name of the phase and the type of event (start or end) as parameters. The marker program sends the corresponding numeric code to NIC2 through an LSL stream named `PythonMarkers` (type `Markers`, integer values), and NIC2 saves it in the `.nedf` file together with the EEG signal. Table A.7.2 shows the codes. The code of the end of a phase is the code of its start plus 100.

Phase value	Start	End	Phase
`baseline`	1	101	Baseline phase
`stroop_task`	2	102	Stress-induction phase (Stroop task)
`biofeedback`	3	103	Relaxation with biofeedback phase

Table A.7.2: EEG marker codes.

The value `biofeedback` is the same one used in the database (Table A.3.3). If the marker program cannot be reached, the backend logs a warning and the experiment continues, so a missing marker would only be noticed when processing the EEG recording.

A.7.2 Connection with Tyto

In the robot condition, the backend sends the colour of the ball, and the start and stop commands, to Tyto through HTTP requests over the WiFi network. The laptop and Tyto have to be connected to the same network, and the network must allow devices to send requests to each other. The breathing pace, the sounds and the blinking of the robot run on timers inside Tyto, so they do not depend on a message arriving each time (Section 3.6).

A.8 Robot Budget

Table A.8.1 lists the components used to build Tyto, with the quantity and the price of each one. The total cost of the components is about 316 €. The Raspberry Pi is more than half of this amount. Prices are in euros, with VAT included, and each one is the total for the quantity indicated. They are the prices paid at the time of purchase, except those marked as estimated, which are reference prices of the same stores or approximate values for small parts that were already available. The cost of the printing, the tools (such as the soldering iron and the 3D printer) and the working time are not included.

Component	Model or notes	Qty.	Price (€)
Raspberry Pi 4 Model B (8 GB)	Main board	1	175.95
microSD card, 64 GB	With NOOBS preinstalled	1	17.95
Heatsinks	Set of 4	1	1.95 (est.)
Fan	30×30×7 mm, 3.3 V/5 V	1	3.95
Belly screen	Waveshare 2.4-inch LCD module (ILI9341)	1	20.99
Eye screens	1.28-inch round LCD modules (GC9A01), pack of 2	1	11.99
Audio HAT with speakers	Waveshare WM8960, with two speakers	1	22.35
Power supply	Official Raspberry Pi 4, USB-C, 5 V, 3 A	1	9.95
USB-C extension cable	AAOTOKK	1	11.89
USB-C right-angle adapter	RUXELY, pack of 3	1	8.99
Prototyping board	Double-sided PCB, 4×6 cm	1	0.95
Stackable header	Samtec ESQ-120-14-G-D, 2×20 pins, 2.54 mm	1	11.43
Pin headers	2.54 mm, 29 pins used from a 40-pin strip	1	0.85 (est.)
Dupont wires	Assorted pack, 11 female-female wires used	1	7.98
RGB LED	5 mm	1	0.50 (est.)
Resistors	220 Ω, 100 Ω and 100 Ω	3	0.10 (est.)
PLA filament	Matte white, about 400 g of a 1 kg spool (13.99 €)	1	5.60
M4 screws	5 for the back cover and 4 for the internal platform	9	1.00 (est.)
M4 threaded inserts	Metal, embedded in the printed holes	9	2.00 (est.)
Total			**316.37**

Table A.8.1: Cost of the components of Tyto.

The Raspberry Pi is the most expensive component, and Tyto probably does not need all its capacity. With some modifications to the software and the wiring, it could probably be adapted to cheaper processors, such as a Raspberry Pi with less memory or a microcontroller board like an ESP32. This was not tested in this project.

B Questionnaires

This appendix gives the items, instructions and scoring of each questionnaire (Section B.1), the complete descriptive statistics that were summarised in Chapter 6 (Section B.2) and the correlations between the NARS subscales and the Godspeed dimensions (Section B.3).

B.1 Content and Scoring

All questionnaires were presented to the participants in Catalan. Most of the instruments did not have a previously verified Catalan version, so the items, instructions and scales were translated for this study and reviewed by the supervisors. The Godspeed items are the Catalan translation published by Bartneck (2023). The English column of each table gives the wording of the original instrument. The instructions are given in Catalan, followed by the original English instructions when the instrument has them. The questionnaires are listed in the order in which they were administered.

B.1.1 Demographic Questions

The participants were first asked for some basic data. Instructions: *"Abans de començar, indica'ns algunes dades bàsiques sobre tu."*

No.	Question	Answer
1	Edat	Number (0–120)
2	Sexe	Home / Dona / Altre / Prefereixo no dir-ho

Table B.1.1: Demographic questions.

B.1.2 GAD-7

The instructions were *"Durant les últimes 2 setmanes, amb quina freqüència us han afectat o molestat els problemes següents?"* (original: *"Over the last 2 weeks, how often have you been bothered by the following problems?"*). Each item is answered on a 4-point scale: *Gens* (0), *Diversos dies* (1), *Més de la meitat dels dies* (2) and *Gairebé cada dia* (3), corresponding to *not at all*, *several days*, *over half the days* and *nearly every day* in the original. The total score (0 to 21) is the sum of the seven items. Scores of 10 or higher indicate moderate to severe anxiety (Spitzer et al., 2006). No item is reverse-worded.

An additional question about the functional impact (item 8 in Table B.1.2) was also presented. It is not part of the total score.

No.	Catalan	English original
1	Sentir-se nerviós(-osa), ansiós(-osa) o tens(-a).	Feeling nervous, anxious, or on edge
2	No poder deixar de preocupar-se o controlar les preocupacions.	Not being able to stop or control worrying
3	Preocupar-se massa per diferents coses.	Worrying too much about different things

No.	Catalan	English original
4	Dificultat per relaxar-se.	Trouble relaxing
5	Estar tan inquiet(-a) que costa estar-se quiet(-a).	Being so restless that it's hard to sit still
6	Irritar-se o enfadar-se amb facilitat.	Becoming easily annoyed or irritable
7	Sentir por com si pogués passar alguna cosa terrible.	Feeling afraid as if something awful might happen
8	Si heu marcat algun problema, fins a quin punt aquests us han dificultat fer la vostra feina, ocupar-vos de les coses a casa o relacionar-vos amb altres persones? (No ha estat gens difícil / Una mica difícil / Molt difícil / Extremadament difícil)	If you checked off any problems, how difficult have these made it for you to do your work, take care of things at home, or get along with other people? (Not difficult at all / Somewhat difficult / Very difficult / Extremely difficult)

Table B.1.2: GAD-7 items.

B.1.3 NARS

The instructions were *"A continuació, us presentem 14 afirmacions que descriuen diferents reaccions i punts de vista envers els robots. Valoreu cadascuna de les línies segons el vostre criteri mitjançant una escala Likert de 5 punts. Responeu amb sinceritat i de manera fluida, sense reflexionar en excés sobre cap afirmació."* The scale has five labelled points: *Totalment en desacord* (1), *Bastant en desacord* (2), *Neutral* (3), *Bastant d'acord* (4) and *Totalment d'acord* (5).

Items 3, 5 and 6 are reverse-worded and were inverted before calculating the score of each subscale as the mean of its items (Nomura et al., 2006). The items were assigned to the subscales as defined in the research protocol submitted to the ethics committee (Appendix D): S1 includes items 1, 2, 4, 5, 8 and 14, S2 items 3, 6, 7, 10 and 12, and S3 items 9, 11 and 13 (Table B.1.3). This assignment is different from the one published for the original scale by Nomura et al. (2006), where S1 (negative attitudes toward situations of interaction with robots) includes items 4, 7, 8, 9, 10 and 12, S2 (negative attitudes toward the social influence of robots) items 1, 2, 11, 13 and 14, and S3 (negative attitudes toward emotions in interaction with robots) items 3, 5 and 6. For this reason, the subscale scores of this study are not directly comparable with those of the original validation.

No.	Catalan	English original	Reversed	Subscale
1	Em sentiria incòmode(-a) si els robots tinguessin emocions.	I would feel uneasy if robots really had emotions.	No	S1
2	Alguna cosa dolenta podria passar si els robots es convertissin en éssers vivents.	Something bad might happen if robots developed into living beings.	No	S1
3	Em sentiria tranquil(-la) conversant amb un robot.	I would feel relaxed talking with robots.	Yes	S2
4	Em sentiria incòmode(-a) si obtinc una feina on hagués de fer servir robots.	I would feel uneasy if I was given a job where I had to use robots.	No	S1

No.	Catalan	English original	Reversed	Subscale
5	Si els robots tinguessin emocions jo podria fer amistat amb ells.	If robots had emotions, I would be able to make friends with them.	Yes	S1
6	Em sento reconfortat en presència de robots.	I feel comforted being with robots that have emotions.	Yes	S2
7	La paraula "robot" no significa res per a mi.	The word "robot" means nothing to me.	No	S2
8	Em sentiria nerviós(-osa) d'haver de fer anar un robot davant d'altres persones.	I would feel nervous operating a robot in front of other people.	No	S1
9	Odiaria la idea que robots i intel·ligència artificial fessin judicis sobre coses.	I would hate the idea that robots or artificial intelligences were making judgments about things.	No	S3
10	Em sentiria molt nerviós(-osa) amb el sol fet d'estar davant d'un robot.	I would feel very nervous just standing in front of a robot.	No	S2
11	Sento que si depenc molt dels robots alguna cosa dolenta podria passar.	I feel that if I depend on robots too much, something bad might happen.	No	S3
12	Em sentiria paranoic(-a) parlant amb un robot.	I would feel paranoid talking with a robot.	No	S2
13	Estic preocupat(-da) que els robots podrien exercir una mala influència en els nens.	I am concerned that robots would be a bad influence on children.	No	S3
14	Sento que en el futur la societat estarà dominada per robots.	I feel that in the future society will be dominated by robots.	No	S1

Table B.1.3: NARS items, in Catalan and in the English original of Nomura et al. (2006), with the subscale used in the analysis.

The Catalan version of item 6 does not include the words "that have emotions" of the original.

B.1.4 ATI

The instructions were *"A l'enquesta següent, et preguntarem sobre la teva interacció amb sistemes tècnics. El terme 'sistemes tècnics' fa referència a aplicacions mòbils, programari informàtic, així com a dispositius digitals complexos (telèfons intel·ligents, ordinadors, televisors, navegadors del cotxe, etc.). Si us plau, indiqueu el grau en què esteu d'acord/en desacord amb les següents afirmacions."* The original instructions are: *"In the following questionnaire, we will ask you about your interaction with technical systems. The term 'technical systems' refers to apps and other software applications, as well as entire digital devices (e.g., mobile phone, computer, TV, car navigation). Please indicate the degree to which you agree/disagree with the following statements."* The scale has six labelled points: *Totalment en desacord* (1, completely disagree), *Bastant en desacord* (2, largely disagree), *Lleugerament en desacord* (3, slightly disagree), *Lleugerament d'acord* (4, slightly agree), *Bastant d'acord* (5, largely agree) and *Totalment d'acord* (6, completely agree).

The score is the mean of the 9 items (Franke et al., 2019). Items 3, 6 and 8 are reverse-worded and were inverted before averaging.

No.	Catalan	English original	Reversed
1	M'agrada ocupar-me amb més detall dels sistemes tècnics.	I like to occupy myself in greater detail with technical systems.	No
2	M'agrada provar les funcions de nous sistemes tècnics.	I like testing the functions of new technical systems.	No
3	Em relaciono amb els sistemes tècnics principalment perquè m'hi veig obligat(-da).	I predominantly deal with technical systems because I have to.	Yes
4	Quan tinc un nou sistema tècnic al davant, el provo intensivament.	When I have a new technical system in front of me, I try it out intensively.	No
5	Gaudeixo passant temps familiaritzant-me amb un nou sistema tècnic.	I enjoy spending time becoming acquainted with a new technical system.	No
6	En tinc prou que un sistema tècnic funcioni; no m'importa com ni per què.	It is enough for me that a technical system works; I don't care how or why.	Yes
7	Intento entendre com funciona exactament un sistema tècnic.	I try to understand how a technical system exactly works.	No
8	En tinc prou de conèixer les funcions bàsiques d'un sistema tècnic.	It is enough for me to know the basic functions of a technical system.	Yes
9	Intento aprofitar al màxim les capacitats d'un sistema tècnic.	I try to make full use of the capabilities of a technical system.	No

Table B.1.4: ATI items, in Catalan and in the English original of Franke et al. (2019).

B.1.5 Visual Analogue Scale of Perceived Stress

The perceived stress was measured with a single question, *"En aquest moment, com d'estressat/da et sents? Indica, en una escala del 0 al 10."*. The scale has 11 points, each of them showing its number. Only the two ends have a text label: 0, *Gens estressat/da*, and 10, *Extremadament estressat/da*. The question was identical in the three moments in which it was presented: just before the baseline phase, just after the Stroop task and just after the relaxation phase. The score is the number selected, and no item is reversed.

B.1.6 NASA-TLX

The instructions were *"Per a cada una de les escales següents, indica amb quina intensitat has experimentat cada factor durant la tasca. Mou el marcador de cada barra fins al punt (entre els dos extrems indicats) que millor representi la teva experiència."* Each dimension was answered with a continuous slider from 0 to 100, with marks every 5 units and the marker initially at 50. Only the two ends have a text label, shown in the last column of Table B.1.5.

The raw version was used, without the pairwise weighting of the original questionnaire, following Hart (2006), who found that the unweighted version gives results practically equivalent to the weighted one.

Unlike the standard NASA-TLX, where the performance scale runs in the opposite direction to the other five, all six dimensions were labelled so that a higher score meant a worse outcome. No inversion was therefore needed, and the overall score is the mean of the six dimensions.

Dimension	**Catalan**	**English original**	**Ends (low / high)**
Mental demand	Quanta activitat mental i perceptiva ha calgut (per exemple, pensar, decidir, calcular, recordar, mirar, cercar, etc.)? Ha estat la tasca fàcil o exigent, senzilla o complexa, precisa o indulgent?	How much mental and perceptual activity was required (e.g., thinking, deciding, calculating, remembering, looking, searching, etc.)? Was the task easy or demanding, simple or complex, exacting or forgiving?	Molt baixa / Molt alta
Physical demand	Quanta activitat física ha calgut (per exemple, empènyer, estirar, girar, controlar, activar, etc.)? Ha estat la tasca fàcil o exigent, lenta o ràpida, relaxada o extenuant, descansada o laboriosa?	How much physical activity was required (e.g., pushing, pulling, turning, controlling, activating, etc.)? Was the task easy or demanding, slow or brisk, slack or strenuous, restful or laborious?	Molt baixa / Molt alta
Temporal demand	Quanta pressió de temps has sentit a causa del ritme al qual s'han succeït les tasques o els elements de la tasca? Ha estat el ritme lent i pausat o ràpid i frenètic?	How much time pressure did you feel due to the rate or pace at which the tasks or task elements occurred? Was the pace slow and leisurely or rapid and frantic?	Molt baixa / Molt alta
Performance	Com d'exitós/osa creus que has estat a l'hora d'aconseguir els objectius de la tasca establerts (per l'experimentador o per tu mateix/a)? Com de satisfet/a estàs amb el teu rendiment a l'hora d'assolir aquests objectius?	How successful do you think you were in accomplishing the goals of the task set by the experimenter (or yourself)? How satisfied were you with your performance in accomplishing these goals?	Bo / Dolent
Effort	Amb quina duresa has hagut de treballar (mentalment i físicament) per assolir el teu nivell de rendiment?	How hard did you have to work (mentally and physically) to accomplish your level of performance?	Molt baix / Molt alt

Dimension	Catalan	English original	Ends (low / high)
Frustration	Com d'insegur/a, descoratjat/da, irritat/da, estressat/da i molest/a -o bé segur/a, satisfet/a, content/a, relaxat/da i tranquil/il·la- t'has sentit durant la tasca?	How insecure, discouraged, irritated, stressed and annoyed versus secure, gratified, content, relaxed and complacent did you feel during the task?	Molt baix / Molt alt

Table B.1.5: NASA-TLX items. The English text is the original wording of the questionnaire.

B.1.7 SUS

The instructions were *"Indiqueu el vostre grau d'acord o desacord amb les afirmacions següents sobre el sistema amb el qual acabeu d'interactuar (l'aplicació en pantalla o Tyto (Robot), segons la condició), utilitzant una escala Likert de 5 punts d'1 (Totalment en desacord) a 5 (Totalment d'acord)."* Each item shows the five numbers, and only the two ends have a text label.

For odd-numbered items (positively worded) the contribution is the response minus 1, and for even-numbered items (negatively worded) it is 5 minus the response. The ten contributions are summed and multiplied by 2.5, which gives a score from 0 to 100 (Brooke, 2013). Each score was classified into a grade with the ranges of Sauro (2018), collapsed into whole letters, without the plus and minus subdivisions of the original source (Table B.1.6).

Grade	SUS score range
A	78.9–100
B	72.6–78.8
C	62.7–72.5
D	51.7–62.6
F	0–51.6

Table B.1.6: Ranges of SUS scores used for each grade.

No.	Catalan	English original
1	Crec que m'agradaria utilitzar aquest sistema sovint.	I think that I would like to use this system frequently.
2	He trobat que el sistema és altament complex de manera innecessària.	I found the system unnecessarily complex.
3	He pensat que aquest sistema era fàcil d'utilitzar.	I thought this system was easy to use.
4	Crec que necessitaria el suport d'una persona tècnica per poder utilitzar aquest sistema.	I think that I would need the support of a technical person to be able to use this system.
5	He trobat que les diverses funcions d'aquest sistema estan ben integrades.	I found the various functions in this system were well integrated.

No.	Catalan	English original
6	He pensat que hi havia massa inconsistència en aquest sistema.	I thought there was too much inconsistency in this system.
7	Imagino que la majoria de la gent aprendria a utilitzar aquest sistema molt ràpidament.	I would imagine that most people would learn to use this system very quickly.
8	He trobat que aquest sistema és molt feixuc d'utilitzar.	I found this system very cumbersome to use.
9	M'he sentit molt segur(-a) utilitzant aquest sistema.	I felt very confident using this system.
10	He necessitat aprendre moltes coses abans de poder posar-me en marxa amb aquest sistema.	I needed to learn a lot of things before I could get going with this system.

Table B.1.7: SUS items, in Catalan and in the English version used as reference (Brooke, 2013). The even-numbered items are reverse-worded.

B.1.8 Godspeed

The Godspeed questionnaire was presented only in the robot condition. The instructions were *"Per a cada parella d'adjectius, indiqueu amb un punt de l'escala l'opció que millor reflecteixi la vostra impressió sobre el sistema amb el qual acabeu d'interactuar."* The original instructions of the questionnaire are: *"Please rate your impression of the robot on these scales."* Each item is a semantic differential scale of 5 points between two opposite adjectives, without visible numbers.

The score of each dimension is the mean of its items, rated from 1 to 5 (Bartneck et al., 2009). The dimensions have 5 (anthropomorphism), 6 (animacy), 5 (likeability), 5 (perceived intelligence) and 3 (perceived safety) items. In perceived safety, items 2 and 3 are reverse-worded and were inverted before averaging.

No.	Catalan	English original	Reversed
Anthropomorphism			
1	Fals / Natural	Fake / Natural	No
2	Màquina / Amb aspecte humà	Machinelike / Humanlike	No
3	Inconscient / Conscient	Unconscious / Conscious	No
4	Artificial / Realista	Artificial / Lifelike	No
5	Es mou rígidament / Es mou amb fluïdesa	Moving rigidly / Moving elegantly	No
Animacy			
1	Mort / Amb vida	Dead / Alive	No
2	Aturat / Enèrgic	Stagnant / Lively	No
3	Mecànic / Orgànic	Mechanical / Organic	No
4	Artificial / Sembla viu	Artificial / Lifelike	No
5	Estàtic / Interactiu	Inert / Interactive	No
6	Indiferent / Atent	Apathetic / Responsive	No

No.	Catalan	English original	Reversed
Likeability			
1	No m'agrada / M'agrada	Dislike / Like	No
2	Antipàtic / Simpàtic	Unfriendly / Friendly	No
3	Descortès / Amable	Unkind / Kind	No
4	Desagradable / Agradable	Unpleasant / Pleasant	No
5	Horrible / Maco	Awful / Nice	No
Perceived intelligence			
1	Incompetent / Competent	Incompetent / Competent	No
2	Ignorant / Entés	Ignorant / Knowledgeable	No
3	Irresponsable / Responsable	Irresponsible / Responsible	No
4	Sense intel·ligència / Intel·ligent	Unintelligent / Intelligent	No
5	Insensat / Assenyat	Foolish / Sensible	No
Perceived safety			
1	Nerviós / Relaxat	Anxious / Relaxed	No
2	Tranquil / Agitat	Agitated / Calm	Yes
3	No sorprès / Sorprès	Quiescent / Surprised	Yes

Table B.1.8: Godspeed items (pole of value 1 / pole of value 5), in Catalan and in the English original of Bartneck et al. (2009).

In item 2 of perceived safety, the Catalan version lists the two poles in the opposite order to the original.

B.2 Descriptive Statistics

Table B.2.1 gives the descriptive statistics of the NARS subscales and the ATI, and Table B.2.2 those of the NASA-TLX. The descriptive statistics of the SUS and the Godspeed questionnaire are in Chapter 6.

Score	Group	*n*	*M*	*Mdn*	*SD*	Range	Cronbach's α
NARS S1	Computer	7	2.40	2.50	0.58	1.33–3.00	.57
	Robot	8	2.60	2.42	0.48	2.17–3.67	.59
	All	15	2.51	2.50	0.52	1.33–3.67	.52
NARS S2	Computer	7	1.69	2.00	0.47	1.00–2.20	.73
	Robot	8	2.33	2.50	0.54	1.40–3.00	.80
	All	15	2.03	2.00	0.59	1.00–3.00	.80
NARS S3	Computer	7	2.90	3.00	0.69	1.67–3.67	.22
	Robot	8	2.75	2.67	0.56	2.00–3.67	.33
	All	15	2.82	3.00	0.60	1.67–3.67	.22
ATI	Computer	7	4.92	4.89	0.51	4.00–5.44	.67
	Robot	8	4.71	4.94	0.54	4.00–5.33	.78
	All	15	4.81	4.89	0.52	4.00–5.44	.72

Table B.2.1: Descriptive statistics of the NARS subscales and the ATI.

Dimension	**Computer** (n = 7)				**Robot** (n = 8)			
	M	***SD***	***Mdn***	**Range**	***M***	***SD***	***Mdn***	**Range**
Mental demand	28.86	32.04	15.0	5–94	24.00	22.12	20.0	0–61
Physical demand	12.00	11.76	7.0	0–35	14.38	25.28	0.0	0–60
Temporal demand	39.71	25.16	40.0	10–71	25.75	25.39	18.0	0–75
Performance	41.71	19.47	40.0	10–70	38.12	29.40	35.0	5–85
Effort	29.71	29.27	20.0	0–83	27.00	25.50	22.5	0–66
Frustration	25.29	23.91	15.0	0–60	32.88	29.85	28.5	0–70
Overall	29.55	17.78	28.5	10.5–59.5	27.02	19.15	25.5	5.0–59.83

Table B.2.2: Descriptive statistics of the NASA-TLX, by condition (scores from 0 to 100).

B.3 Correlations Between the NARS and the Godspeed Dimensions

Table B.3.1 gives the Spearman correlations between each NARS subscale and each Godspeed dimension. They were computed only in the robot condition (n = 8), since the Godspeed questionnaire was not presented in the computer condition, and no correction for multiple comparisons was applied because they are exploratory.

NARS subscale	Godspeed dimension	r_s	p
S1	Anthropomorphism	.07	.862
	Animacy	.18	.679
	Likeability	$-.70$	.052
	Perceived intelligence	.19	.655
	Perceived safety	$-.59$	.120
S2	Anthropomorphism	.25	.545
	Animacy	$-.04$	.931
	Likeability	$-.34$	.413
	Perceived intelligence	$-.19$	.650
	Perceived safety	$-.30$	.473
S3	Anthropomorphism	$-.32$	.447
	Animacy	$-.51$	.201
	Likeability	$-.26$	.529
	Perceived intelligence	$-.14$	.747
	Perceived safety	.13	.768

Table B.3.1: Spearman correlations between the NARS subscales and the Godspeed dimensions in the robot condition ($n = 8$).

C Experimenter Script

This appendix contains the script used to guide the experimenter through each step of the experiment. The script was written in Catalan and is provided in Section C.1. Section C.2 gives its English translation. The parts spoken to the participant are shown in quotation blocks, and the actions of the experimenter are shown in italics. The minutes indicate the approximate planned duration of each step.

C.1 Script in Catalan

Recepció del participant (3 minuts)

> Hola, sóc l'Helena i seré la persona que durà a terme aquest estudi.
>
> La sessió tindrà una durada aproximada de 50 minuts i estarà dividida en diverses fases. Si en qualsevol moment tens algun dubte o et sents incòmode/a, m'ho pots comunicar.

Comprovar els criteris d'inclusió i exclusió, assignar el participant a la condició experimental corresponent i assignar el codi d'identificació.

Explicació de l'estudi (5 minuts)

> Abans de començar, t'explicaré en què consisteix l'estudi i què farem durant la sessió.
>
> Aquest estudi s'emmarca dins de les meves pràctiques acadèmiques del Màster en Interaction, Graphics and Design, i els resultats obtinguts formaran part del meu Treball Final de Màster (TFM).
>
> De manera general, l'estudi té com a objectiu analitzar l'experiència de les persones durant una breu tasca de respiració guiada i comparar dues formes de presentar aquesta interacció: mitjançant un robot social o mitjançant una interfície digital en pantalla.
>
> En un moment de la sessió realitzaràs una tasca cognitiva amb pressió de temps. Aquesta tasca està dissenyada per provocar un nivell moderat i controlat d'activació o estrès. Posteriorment, faràs uns exercicis de respiració guiada amb el sistema que t'hagi estat assignat.
>
> La teva participació és voluntària i les dades recollides s'utilitzaran amb finalitats acadèmiques i de recerca, d'acord amb la informació proporcionada en el Full d'Informació al Participant i el consentiment informat.

Lliurar el Full d'Informació al Participant (FIP).

> Durant la sessió enregistrarem algunes mesures fisiològiques, com la freqüència cardíaca, la respiració i l'activitat elèctrica cerebral mitjançant EEG. També et demanarem que responguis alguns qüestionaris abans i després de les tasques.
>
> Ara et deixaré uns minuts perquè puguis llegir el Full d'Informació al Participant. Si tens qualsevol dubte sobre el procediment o sobre la informació que hi apareix, m'ho pots preguntar abans de continuar.

Un cop llegit el FIP:

> Has pogut entendre el procediment general? Tens algun dubte abans de continuar?

Resoldre els dubtes del participant.

Consentiment informat (5 minuts)

Si has entès la informació i estàs d'acord a participar voluntàriament en l'estudi, et demanaré que signis el document de consentiment informat.

Lliurar el document de consentiment informat.

Tens alguna pregunta sobre el consentiment, el tractament de les dades o qualsevol altre aspecte de l'estudi abans de començar?

Recorda que pots decidir abandonar l'estudi en qualsevol moment.

Col · locació de sensors i preparació del sistema (10–15 minuts)

Ara procediré a col·locar els sensors que utilitzarem per registrar les variables fisiològiques durant l'experiment.

Col·locar el monitor de freqüència cardíaca i el casc d'EEG.

Durant la col·locació:

Si en algun moment el sensor de pit o el casc et resulta incòmode, avisa'm i intentarem ajustar-lo perquè estiguis el més còmode/a possible.

Comprovar també que els sistemes de registre funcionen correctament.

Qüestionaris preexperimentals (10–15 minuts)

Perfecte. Abans de començar les tasques, farem uns qüestionaris inicials. Aquests qüestionaris ens permetran conèixer algunes característiques inicials i ens ajudaran posteriorment a analitzar millor les dades obtingudes.

No hi ha respostes correctes o incorrectes. Respon segons la teva situació o opinió personal.

Els qüestionaris es faran mitjançant formularis a l'ordinador. Per passar al qüestionari següent hauràs de clicar el botó de "Següent".

El participant completa els qüestionaris.

En finalitzar:

Perfecte. Un cop finalitzats els qüestionaris, passarem a la següent part.

Fase baseline (5 minuts)

Començar el protocol al NIC2.

> En aquesta part de l'experiment necessitaré que reposis durant cinc minuts. Durant aquest temps, simplement et demanaré que estiguis assegut/da i tranquil/a.
>
> Intenta mantenir una postura còmoda i respirar de manera natural. No cal que facis cap tasca concreta. Jo t'indicaré quan comenci i quan finalitzi el període.

Iniciar el registre fisiològic i el període de repòs.

Fase d'inducció d'estrès (6 minuts)

> Ara realitzaràs una tasca de Stroop.
>
> Durant 6 minuts, se't mostraran paraules que fan referència a diferents colors, però el color amb què està escrita la paraula pot no coincidir amb el seu significat. La teva tasca serà indicar el color de la tinta amb què està escrita la paraula, i no el que diu la paraula.
>
> Intenta respondre tan ràpidament i correctament com puguis durant tota la tasca.

Abans de començar:

> Tens alguna pregunta sobre com funciona la tasca?

Si no hi ha dubtes:

> D'acord. Comencem.

Iniciar la tasca de Stroop.

En finalitzar:

> La tasca ha finalitzat. Ara, abans de continuar, indica el teu nivell d'estrès percebut en aquest moment mitjançant l'escala que es mostra a la pantalla.

Fase de relaxació i biofeedback (10 minuts)

Introducció comuna:

> Ara passarem a la fase de respiració guiada. L'objectiu és seguir un ritme de respiració lent i regular durant els pròxims deu minuts.
>
> El sistema et mostrarà el ritme de respiració que has de seguir i també proporcionarà informació sobre la teva resposta fisiològica en temps real.
>
> No cal que facis res més que seguir la guia de respiració de la manera més còmoda possible.

Condició robot (Tyto):

> En el teu cas, la guia de respiració es presentarà mitjançant el robot Tyto.
>
> A la pantalla rectangular del robot apareixerà una bola que s'anirà fent gran i petita. Aquest moviment marcarà el ritme que has de seguir per inhalar i exhalar.
>
> Si no tens cap pregunta, prem el botó de començar que es mostra a la pantalla.

Iniciar la fase de respiració guiada.

Condició ordinador:

> En el teu cas, la guia de respiració es presentarà mitjançant la interfície que veus a la pantalla.
>
> Hi veuràs una bola que es farà gran i petita, marcant el ritme que has de seguir per inhalar i exhalar.
>
> Si no tens cap pregunta, prem el botó de començar que es mostra a la pantalla.

Iniciar la fase de respiració guiada.

En finalitzar:

> La fase de respiració ha finalitzat. Ara indica, de nou, el teu nivell d'estrès percebut en aquest moment mitjançant l'escala que veus a la pantalla.

Qüestionaris postexperimentals (10–12 minuts)

> Ara et demanaré que responguis uns últims qüestionaris sobre l'experiència que acabes de tenir. Els qüestionaris, igual que els de l'inici de l'estudi, es realitzaran a l'ordinador.
>
> Ens interessa especialment conèixer la teva percepció del sistema i de la tasca. No hi ha respostes correctes o incorrectes; respon segons la teva experiència personal.

El participant completa els qüestionaris postexperimentals.

En finalitzar:

> Perfecte. Ara farem una breu entrevista sobre la teva experiència durant la sessió.

Entrevista qualitativa (5–8 minuts)

> Per acabar, m'agradaria conèixer una mica millor la teva experiència durant la sessió. Per aquest motiu, et faré unes quantes preguntes al respecte. No hi ha respostes correctes o incorrectes i ens interessa especialment la teva opinió personal.

Abans de començar l'enregistrament:

> Amb el teu permís, enregistrarem aquesta part de la sessió en àudio per poder transcriure-la i analitzar-la posteriorment. L'enregistrament estarà associat al teu codi de participant i no inclourà el teu nom ni informació identificativa directa.

Confirmar el consentiment per a l'enregistrament i iniciar l'enregistrament d'àudio.

Preguntes:

1. **Experiència general:** Com descriuries la teva experiència general durant la sessió?
2. **Tasca d'estrès:** Com t'has sentit durant la tasca de Stroop?
3. **Fase de relaxació:** Com t'has sentit durant la fase de respiració/relaxació?
4. **Interacció:** Com descriuries la interacció amb el sistema?
5. **Confort:** T'ha resultat còmode utilitzar-lo?

6. **Aspectes positius o negatius:** Hi ha hagut algun moment que t'hagi resultat especialment útil o que t'hagi agradat? O algun moment que t'hagi incomodat o hagi sigut difícil?
7. **Millores:** Hi ha alguna cosa que canviaries del sistema o de la manera com s'ha desenvolupat la sessió?
8. **Comentaris finals:** Hi ha alguna altra cosa que vulguis comentar sobre la teva experiència?

En finalitzar:

Moltes gràcies. Amb això finalitza l'entrevista.

Aturar l'enregistrament.

Debriefing (3–5 minuts)

Amb això hem finalitzat l'experiment. Abans d'acabar, t'explicaré amb més detall l'objectiu específic de l'estudi i el motiu d'algunes de les tasques que has realitzat.

Explicar de manera transparent:

L'objectiu principal de l'estudi és comparar l'experiència de realitzar una fase de respiració guiada amb un robot social respecte d'una interfície digital en pantalla.

La tasca de Stroop que has realitzat s'ha utilitzat com a mecanisme d'inducció d'estrès. La seva funció era generar una activació o resposta d'estrès de manera controlada per poder observar posteriorment la resposta durant la fase de relaxació.

Durant la fase de respiració, el sistema ha proporcionat una guia a un ritme de sis respiracions per minut i informació fisiològica en temps real. Això ens permet estudiar la resposta fisiològica i la percepció de l'experiència durant aquesta fase.

Les mesures d'EEG i les altres mesures fisiològiques registrades s'utilitzen únicament per obtenir les variables definides en l'estudi.

Resolució de dubtes:

Ara que coneixes l'objectiu complet de l'estudi, tens algun dubte o alguna preocupació sobre l'experiment o sobre les dades que hem recollit?

Recordatori final:

Moltes gràcies per la teva participació i pel teu temps. Ho agraeixo molt.

Si vols obtenir accés als resultats de l'estudi un cop hagi finalitzat, fes-m'ho saber.

C.2 Script in English

Welcoming the participant (3 minutes)

Hello, I'm Helena and I will be the person carrying out this study.

The session will last about 50 minutes and is divided into several phases. If at any moment you have a question or feel uncomfortable, you can let me know.

Check the inclusion and exclusion criteria, assign the participant to the corresponding experimental condition and assign the identification code.

Explanation of the study (5 minutes)

Before we start, I will explain what the study is about and what we will do during the session.

This study is part of my academic internship of the Master in Interaction, Graphics and Design, and the results will be part of my Master's thesis.

In general, the aim of the study is to analyse people's experience during a short guided breathing task and to compare two ways of presenting this interaction: with a social robot or with a digital interface on a screen.

At one point in the session you will do a cognitive task under time pressure. This task is designed to cause a moderate and controlled level of activation or stress. Afterwards, you will do some guided breathing exercises with the system assigned to you.

Your participation is voluntary, and the data collected will be used for academic and research purposes, according to the information given in the Participant Information Sheet and the informed consent.

Hand over the Participant Information Sheet (PIS).

During the session we will record some physiological measures, such as heart rate, breathing and electrical brain activity using EEG. We will also ask you to answer some questionnaires before and after the tasks.

Now I will give you a few minutes to read the Participant Information Sheet. If you have any question about the procedure or the information in it, you can ask me before we continue.

Once the PIS has been read:

Have you understood the general procedure? Do you have any question before we continue?

Answer the questions of the participant.

Informed consent (5 minutes)

If you have understood the information and agree to take part voluntarily in the study, I will ask you to sign the informed consent form.

Hand over the informed consent form.

Do you have any question about the consent, the processing of the data or any other aspect of the study before we start?

Remember that you can decide to leave the study at any moment.

Placement of the sensors and preparation of the system (10–15 minutes)

> Now I will place the sensors that we will use to record the physiological variables during the experiment.

Place the heart rate monitor and the EEG cap.

During the placement:

> If at any moment the chest sensor or the cap is uncomfortable, let me know and we will try to adjust it so that you are as comfortable as possible.

Also check that the recording systems work correctly.

Pre-experiment questionnaires (10–15 minutes)

> Perfect. Before starting the tasks, we will do some initial questionnaires. They will let us know some initial characteristics and will help us analyse the data better later.
>
> There are no right or wrong answers. Answer according to your own situation or opinion.
>
> The questionnaires are done with forms on the computer. To go to the next questionnaire you have to click the "Next" button.

The participant completes the questionnaires.

When they finish:

> Perfect. Now that the questionnaires are finished, we will move on to the next part.

Baseline phase (5 minutes)

Start the protocol in NIC2.

> In this part of the experiment I need you to rest for five minutes. During this time, I will simply ask you to sit calmly.
>
> Try to keep a comfortable posture and breathe naturally. You do not need to do any specific task. I will tell you when the period begins and ends.

Start the physiological recording and the resting period.

Stress-induction phase (6 minutes)

Now you will do a Stroop task.

For 6 minutes, you will be shown words that refer to different colours, but the colour in which the word is written may not match its meaning. Your task is to indicate the ink colour in which the word is written, and not what the word says.

Try to respond as quickly and as correctly as you can during the whole task.

Before starting:

Do you have any question about how the task works?

If there are no questions:

All right. Let's start.

Start the Stroop task.

When it finishes:

The task is finished. Now, before we continue, indicate your perceived stress level at this moment using the scale shown on the screen.

Relaxation with biofeedback phase (10 minutes)

Common introduction:

Now we will move on to the guided breathing phase. The aim is to follow a slow and regular breathing rhythm for the next ten minutes.

The system will show you the breathing rhythm to follow and will also give information about your physiological response in real time.

You do not need to do anything else than follow the breathing guide in the most comfortable way possible.

Robot condition (Tyto):

In your case, the breathing guide will be presented by the robot Tyto.

On the rectangular screen of the robot, a ball will appear that gets bigger and smaller. This movement marks the rhythm you have to follow to breathe in and out.

If you have no questions, press the start button shown on the screen.

Start the guided breathing phase.

Computer condition:

In your case, the breathing guide will be presented by the interface you see on the screen.

You will see a ball that gets bigger and smaller, marking the rhythm you have to follow to breathe in and out.

If you have no questions, press the start button shown on the screen.

Start the guided breathing phase.

When it finishes:

> The breathing phase is finished. Now indicate again your perceived stress level at this moment using the scale you see on the screen.

Post-experiment questionnaires (10–12 minutes)

> Now I will ask you to answer some last questionnaires about the experience you have just had. Like those at the beginning of the study, they are done on the computer.
>
> We are especially interested in your perception of the system and of the task. There are no right or wrong answers; answer according to your own experience.

The participant completes the post-experiment questionnaires.

When they finish:

> Perfect. Now we will do a short interview about your experience during the session.

Qualitative interview (5–8 minutes)

> To finish, I would like to know your experience during the session a little better. For this reason, I will ask you a few questions about it. There are no right or wrong answers, and we are especially interested in your personal opinion.

Before starting the recording:

> With your permission, we will record this part of the session in audio so that we can transcribe and analyse it later. The recording will be linked to your participant code and will not include your name or any direct identifying information.

Confirm the consent for the recording and start the audio recording.

Questions:

1. **General experience:** How would you describe your general experience during the session?
2. **Stress task:** How did you feel during the Stroop task?
3. **Relaxation phase:** How did you feel during the breathing and relaxation phase?
4. **Interaction:** How would you describe the interaction with the system?
5. **Comfort:** Was it comfortable to use?
6. **Positive or negative aspects:** Was there any moment that was especially useful or that you liked? Or any moment that made you uncomfortable or was difficult?
7. **Improvements:** Is there anything you would change about the system or about the way the session was carried out?
8. **Final comments:** Is there anything else you would like to say about your experience?

When it finishes:

> Thank you very much. This is the end of the interview.

Stop the recording.

Debriefing (3–5 minutes)

This is the end of the experiment. Before finishing, I will explain in more detail the specific aim of the study and the reason for some of the tasks you have done.

Explain in a transparent way:

The main aim of the study is to compare the experience of doing a guided breathing phase with a social robot with that of a digital interface on a screen.

The Stroop task you did was used as a stress-induction mechanism. Its function was to generate an activation or stress response in a controlled way, so that the response during the relaxation phase could be observed afterwards.

During the breathing phase, the system gave a guide at a rhythm of six breaths per minute and physiological information in real time. This lets us study the physiological response and the perception of the experience during this phase.

The EEG measures and the other physiological measures recorded are used only to obtain the variables defined in the study.

Resolving questions:

Now that you know the complete aim of the study, do you have any question or concern about the experiment or about the data we have collected?

Final reminder:

Thank you very much for your participation and your time. I really appreciate it.

If you want access to the results of the study once it is finished, let me know.

D Ethics Committee Documents

This appendix contains the documents submitted to the Research Ethics Committee of the UPC (CEUPC): the approval certificates of the two versions of the protocol (Section D.1), the research protocol of the second version (Section D.2), the Participant Information Sheet (Section D.3) and the informed consent form (Section D.4). The certificates are provided in their original Catalan version. The protocol, the Participant Information Sheet and the informed consent form are given in Catalan, followed by their English translation.

D.1 Approval Certificates

The first version of the protocol was approved by the CEUPC on 13 July 2026 (reference 2026-085). The modified version, which is the one used in the experiments, was approved on 17 September 2026 (reference 2026-088). In both cases the opinion was favourable.

UNIVERSITAT POLITÈCNICA DE CATALUNYA BARCELONATECH

Comitè d'Ètica de la UPC

DICTAMEN DEL COMITÈ D'ÈTICA DE LA UPC

El Comitè d'Ètica de la Universitat Politècnica de Catalunya (CEUPC)

FA CONSTAR QUE:

el CEUPC reunit el dia **13 de juliol de 2026**, ha avaluat la documentació adjuntada en relació al projecte de recerca:

Títol:	*Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback*
Sol·licitant:	*Helena Cadevall Soto*
Directors:	*Marta Isabel Tarrés Puertas, Linus Manubens-Gil*
Departament:	*Enginyeria Minera, Industrial i TIC*
Codi d'identificació:	*2026-085*

El CEUPC emet un **DICTAMEN FAVORABLE** tot i recordant que totes les activitats descrites en el projecte han de tenir present:

- Els principis ètics de la Declaració de Helsinki i de l'informe Belmont
- Els principis del Codi Ètic de la UPC i el Codi d'integritat de la recerca de la UPC
- Els objectius, els protocols de la recerca i l'anàlisi de beneficis i riscos de l'estudi

El comitè vincula aquest dictamen favorable a que:

- en cas de realitzar una recerca col·laborativa amb una institució externa, aquesta també emeti el corresponent dictamen;
- que qualsevol prova que requereixi l'ús de dispositius mèdics estigui supervisada per un tècnic amb els corresponents coneixements;
- igualment, en la recerca, en cas d'apreciacions psicològiques sobre els participants, que aquestes vinguin avalades per alguna persona amb la formació adient.

Aquest dictamen també recorda:

- que el CEUPC podrà demanar fer un seguiment de l'estudi en qualsevol moment de la seva execució;
- que qualsevol ampliació de l'estudi, temporal o de contingut, requerirà la formalització d'una nova sol·licitud de revisió;
- l'obligatorietat de fer públics els resultats de la recerca, tant si són positius com negatius;
- que tant els equips de recerca, com el reclutament de participants en els experiments, han d'incorporar la perspectiva de gènere, intentant garantir aquesta paritat;
- que tota publicació que faci referència a dades de persones inclogui, en la mesura que sigui possible, la variable sexe; a més, que les dades publicades han d'estar convenientment segregades.

Secretari del CEUPC

Alejandro León

Barcelona, a data de la signatura electrònica.

NOTA FINAL. Les futures referències a l'aprovació d'aquesta sol·licitud favorable per part del CEUPC hauran d'incloure, com a mínim, el text següent:
El Comitè d'Ètica de la UPC ha emès un dictamen favorable sobre els aspectes ètics relacionats amb la recerca realitzada en aquest projecte/article.

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Jordi Girona, 31
08034 Barcelona - SPAIN

https://comite-etica.upc.edu

UNIVERSITAT POLITÈCNICA DE CATALUNYA BARCELONATECH

Comitè d'Ètica de la UPC

DICTAMEN DEL COMITÈ D'ÈTICA DE LA UPC

El Comitè d'Ètica de la Universitat Politècnica de Catalunya (CEUPC)

FA CONSTAR QUE:

el CEUPC reunit el dia **17 de setembre de 2026**, ha avaluat la documentació adjuntada en relació al projecte de recerca:

Títol:	*Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback*
Sol·licitant:	*Helena Cadevall Soto*
Tutors:	*Marta Isabel Tarrés Puertas, Linus Manubens-Gil*
Departament:	*Enginyeria Minera, Industrial i TIC*
Codi d'identificació:	*2026-088*

El comitè ha recollit les modificacions operacionals i metodològiques al respecte de la sol·licitud original (referència ***2026-085***). El CEUPC emet un **DICTAMEN FAVORABLE** tot i recordant que totes les activitats descrites en el projecte han de tenir present:

- Els principis ètics de la Declaració de Helsinki i de l'informe Belmont
- Els principis del Codi Ètic de la UPC i el Codi d'integritat de la recerca de la UPC
- Si escau, ajustar-se a la "Guia de bones pràctiques sobre l'ús de la Intel·ligència Artificial a la UPC"
- Els objectius, els protocols de la recerca i l'anàlisi de beneficis i riscos de l'estudi

Aquest dictamen també recorda:

- que el CEUPC podrà demanar fer un seguiment de l'estudi en qualsevol moment de la seva execució;
- que qualsevol ampliació de l'estudi, temporal o de contingut, requerirà la formalització d'una nova sol·licitud de revisió;
- l'obligatorietat de fer públics els resultats de la recerca, tant si són positius com negatius;
- que tant els equips de recerca, com el reclutament de participants en els experiments, han d'incorporar la perspectiva de gènere, intentant garantir aquesta paritat;
- que tota publicació que faci referència a dades de persones inclogui, en la mesura que sigui possible, la variable sexe; a més, que les dades publicades han d'estar convenientment segregades.

Secretari del CEUPC

Alejandro León

Barcelona, a data de la signatura electrònica.

NOTA FINAL. Les futures referències a l'aprovació d'aquesta sol·licitud favorable per part del CEUPC hauran d'incloure, com a mínim, el text següent:
El Comitè d'Ètica de la UPC ha emès un dictamen favorable sobre els aspectes ètics relacionats amb la recerca realitzada en aquest projecte/article.

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D.2 Research Protocol

The research protocol of the second version describes the study as it was planned, and two aspects differ from what was finally done. First, the protocol includes the heart rate variability and the respiratory rate among the measures, but they were not analysed because of the limitations of the heart rate sensor (Section 5.2.1). Second, the protocol excludes the participants with a GAD-7 score of 10 or higher, while the final analysis excluded those with a score of 12 or higher (Section 5.1).

The protocol was submitted together with an annex with the Catalan version of the questionnaires. The content of this annex is provided in Appendix B.

D.2.1 Catalan Version

Helena Cadevall Soto

Pla de recerca

1 Títol del projecte

Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback

2 Informació general del projecte

2.1 Investigador/a principal

- Nom: Helena Cadevall Soto
- Programme / màster / TFM: Master in Interaction, Graphics and Design

2.2 Supervisor/a

Supervisor 1:

- Nom: Marta Isabel Tarrés Puertas
- Position: Associate Professor
- Departament: Smart Sustainable Resources
- Email institucional: marta.isabel.tarres@upc.edu

Supervisor 2:

- Nom: Linus Manubens-Gil
- Position: Distinguished Investigator
- Departament: Smart Sustainable Resources
- Email institucional: linus.manubens@upc.edu

2.3 Institució

- Home institution: Institut Polytechnique de Paris (IP Paris)
- Host institution: Universitat Politècnica de Catalunya (UPC)

3 Resum del projecte

Aquest projecte investiga l'eficàcia de la Interacció Humà-Robot (Human–Robot Interaction, HRI) com a eina per a la regulació fisiològica de l'estrès. En concret, es planteja com un estudi pilot de tipus proof of concept, orientat a explorar la viabilitat d'un sistema basat en biofeedback fisiològic mitjançant interacció amb un robot social, i comparar-lo amb una solució digital convencional basada en pantalla.

L'estudi compararà dues condicions experimentals: una basada en la interacció amb un robot social de forma zoomòrfica i una altra basada en una interfície digital convencional. Durant les sessions experimentals, els participants realitzaran exercicis de respiració guiada basats en biofeedback fisiològic, després d'una fase d'inducció d'estrès controlada mitjançant tasques cognitives.

Aquest estudi preliminar estarà orientat principalment a persones sanes, amb l'objectiu d'obtenir una primera aproximació sobre l'impacte de diferents tecnologies interactives en la regulació de l'estrès i en l'experiència d'usuari. El perfil de participants podrà incloure adults sense patologies clíniques diagnosticades relacionades amb l'estrès o la salut mental.

Per avaluar l'efecte de les diferents condicions experimentals, es combinaran mesures fisiològiques objectives i subjectives. Les principals mesures fisiològiques inclouran la freqüència cardíaca (BPM) i la freqüència respiratòria (RR), registrades directament mitjançant els sensors utilitzats, a partir de les quals es derivaran els indicadors de variabilitat de la freqüència cardíaca (HRV), com el RMSSD. Així mateix, es consideraran possibles mesures d'activitat cerebral mitjançant electroencefalografia (EEG), per explorar indicadors d'activació cortical i càrrega cognitiva. Paral·lelament, es recolliran qüestionaris estandarditzats relacionats amb l'estrès percebut, l'experiència d'usuari i la percepció de la tecnologia i dels robots.

L'impacte esperat del projecte és aportar evidència preliminar sobre la viabilitat de la integració de biofeedback fisiològic en sistemes d'interacció humà-robot, així com identificar tendències inicials sobre el potencial dels robots socials com a alternatives o complements a interfícies digitals convencionals.

4 Context i motivació

4.1 Estrès i interacció humà-computador

L'estrès és una resposta fisiològica i psicològica adaptativa que pot esdevenir problemàtica quan es manté de manera sostinguda en el temps. En contextos contemporanis, l'augment de les demandes cognitives, laborals i digitals ha contribuït a una major prevalença d'estrès percebut en la població general, amb impactes potencials sobre el benestar físic i mental.

En aquest context, la recerca en Interacció Humà-Computador (HCI) ha explorat com el disseny de sistemes interactius pot influir en els estats emocionals i fisiològics dels usuaris, així com en la seva capacitat de regulació de l'estrès mitjançant intervencions digitals com la respiració guiada o sistemes de biofeedback [1].

No obstant això, moltes d'aquestes solucions es basen en interfícies purament digitals, fet que pot limitar la implicació emocional i la sensació de presència social durant la interacció.

4.2 Robòtica social i benestar

En els darrers anys, la robòtica social s'ha consolidat com un camp emergent dins de la Interacció Humà-Robot (HRI), amb aplicacions creixents en salut i benestar emocional. Els robots socials poden interactuar amb humans mitjançant canals multimodals, augmentant la sensació de presència social i la implicació durant la interacció [2].

Diversos estudis han explorat l'ús de robots socials per a la regulació de l'estrès i l'ansietat, especialment en el context d'exercicis de respiració guiada. Per exemple, s'ha observat que la interacció amb robots pot reduir respostes psicofisiològiques associades a l'estrès i millorar l'experiència emocional en comparació amb interfícies tradicionals [3, 4].

Més recentment, treballs en aquest àmbit han explorat sistemes de respiració guiada basats en robots socials, així com mecanismes de *physiological entrainment*, en què els ritmes del robot poden ajudar a sincronitzar la respiració de l'usuari i facilitar la regulació fisiològica [5, 6, 7, 8, 9]. Aquestes aproximacions suggereixen que la presència física d'un agent interactiu pot tenir un impacte diferencial respecte a interfícies basades en pantalla.

4.3 Ús de HRV com a indicator fisiològic

Per a l'estudi de l'estrès, las mesures subjectives han estat tradicionalment la principal font d'avaluació. Tanmateix, en els darrers anys s'ha incrementat l'interès per l'ús de mesures fisiològiques objectives que permetin caracteritzar

de manera més precisa la resposta a l'estrès.

En aquest context, la variabilitat de la freqüència cardíaca (*Heart Rate Variability*, HRV) s'ha establert com un dels indicadors més utilitzats per a l'avaluació de l'activitat del sistema nerviós autònom. En particular, paràmetres com el RMSSD permeten estimar l'equilibri entre els sistemes simpàtic i parasimpàtic, associat a estats d'estrès i relaxació respectivament [10].

L'ús de HRV, juntament amb altres mesures fisiològiques com l'activitat electrodermal o el patró respiratori, permet obtenir una visió complementària a les mesures subjectives d'estrès percebut [11].

5 Objectius i hipòtesis

5.1 Objectiu principal

L'objectiu principal d'aquest estudi és comparar l'eficàcia de dues modalitats d'interacció, una basada en un robot social i una altra basada en una interfície digital convencional, en la regulació fisiològica de l'estrès.

En concret, es pretén obtenir una primera aproximació sobre quina de les dues condicions resulta més efectiva per reduir la resposta fisiològica d'estrès i afavorir estats de relaxació durant la realització d'exercicis de respiració guiada.

5.2 Objectius secundaris

- Analitzar diferències preliminars en variables fisiològiques relacionades amb l'estrès entre la condició amb robot i la condició amb pantalla, especialment la variabilitat de la freqüència cardíaca (HRV) i mesures derivades de l'electroencefalografia (EEG), com indicadors d'activació cortical i càrrega cognitiva.
- Comparar la percepció subjectiva d'estrès reportada pels participants mitjançant qüestionaris amb les mesures objectives obtingudes a partir dels senyals fisiològics (HRV i EEG), com l'objectiu d'explorar la coherència entre mesures subjectives i objectives en un context exploratori.
- Explorar la relació i possible correlació entre els indicadors fisiològics (HRV i EEG) i les puntuacions dels qüestionaris d'estrès percebut i experiència d'usuari, per identificar tendències preliminars segons la modalitat d'interacció.

5.3 Hipòtesis

5.3.1 Hipòtesi nul·la (H0)

H0: No hi haurà diferències significatives entre la condició amb robot social i la condició amb interfície digital en cap de les mesures d'estrès, ni fisiològiques (HRV) ni subjectives (qüestionaris).

5.3.2 Hipòtesis alternatives

- **H1:** Els participants que interactuïn amb el robot social presentaran una major reducció de l'estrès percebut en comparació amb la condició basada en pantalla.
- **H2:** La condició amb robot social mostrarà valors de HRV més elevats, indicant una millor regulació fisiològica de l'estrès, respecte a la condició amb interfície digital.
- **H3:** Es produirà una dissociació parcial entre les mesures subjectives i fisiològiques, de manera que la reducció d'estrès percebuda en els qüestionaris serà proporcionalment major que la millora observada en les mesures fisiològiques.
- **H4:** Existirà una correlació significativa entre els valors de HRV i les puntuacions d'estrès percebut, indicant coherència entre mesures objectives i subjectives.
- **H5:** La condició robot generarà una major sensació de suport emocional i implicació durant l'exercici.

6 Disseny experimental

6.1 Tipus d'estudi

L'estudi segueix un disseny experimental aleatoritzat entre subjectes (randomized between-subjects design), en el qual cada participant és assignat a una única condició experimental: robot social o interfície en ordinador.

Tots els participants segueixen exactament el mateix protocol experimental (qüestionaris, mesura de senyals fisiològics, inducció d'estrès i fase de relaxació amb biofeedback). L'única diferència entre condicions és la modalitat utilitzada durant la fase de biofeedback (robot o ordinador).

Aquest disseny permet atribuir les diferències observades exclusivament al tipus d'interfície utilitzada, reduint la influència d'altres variables experimentals.

6.2 Condicions experimentals

L'estudi contempla dues condicions experimentals principals, que difereixen únicament en la modalitat d'interacció utilitzada durant la fase de relaxació i biofeedback.

6.2.1 Condició A – Ordinador

En aquesta condició, els exercicis de respiració guiada i biofeedback es realitzen mitjançant una aplicació web visualitzada en un ordinador. La interfície mostra de manera simultània el ritme respiratori objectiu i la respiració real del participant, permetent-li ajustar-la en temps real.

6.2.2 Condició B – Robot

En aquesta condició, els mateixos exercicis de respiració guiada i biofeedback són conduïts mitjançant Tyto, un robot social de forma zoomòrfica (inspirat en una òliba/mussol) dissenyat específicament per a aquest estudi, que reprodueix el mateix mecanisme de biofeedback visual i sonor que la condició d'ordinador.

Aquesta condició busca introduir un agent físic interactiu amb capacitat de presència social, mantenint una interacció simple i de baixa càrrega cognitiva.

6.2.3 Tasca i durada

En ambdues condicions, els participants realitzen el mateix protocol experimental, que inclou:

- Fase baseline (repòs)
- Fase d'inducció d'estrès
- Fase de relaxació amb biofeedback

La durada total de la sessió és aproximadament de 50 a 60 minuts, depenent del temps de preparació dels sensors.

6.3 Variables

- **Variables independents**
  - Modalitat d'interacció:
    - Robot (*Tyto*)
    - Ordinador (aplicació web)
- **Variables dependents**

- Variabilitat de la freqüència cardíaca (HRV), amb especial èmfasi en indicadors temporals com RMSSD
- Freqüència cardíaca i respiratòria
- Senyals d'electroencefalografia (EEG), com indicadors d'activació cortical i càrrega cognitiva
- Qüestionaris d'estrès i experiència subjectiva

7 Participants

7.1 Nombre estimat

Es preveu la participació d'entre 12 i 20 participants.

Aquest estudi es planteja com un pilot (proof of concept) amb l'objectiu d'explorar la viabilitat del sistema experimental i obtenir una primera aproximació als efectes de la interacció amb robot social versus interfície digital en la regulació de l'estrès.

Donat el caràcter exploratori del treball, la mostra no pretén permetre inferències estadístiques robustes a gran escala, sinó identificar tendències preliminars, validar el protocol experimental i avaluar la viabilitat de les mesures fisiològiques (HRV i EEG) i subjectives.

7.2 Criteris d'inclusió

Els criteris d'inclusió per a la participació en l'estudi són:

- Adults majors de 18 anys.
- Participants sans, sense diagnòstics clínics rellevants associats a trastorns greus d'estrès, ansietat o depressió.
- Capacitat per comprendre i seguir les instruccions del protocol experimental.
- Consentiment informat signat abans de la participació.

7.3 Criteris d'exclusió

S'exclouran de l'estudi els participants que compleixin algun dels següents criteris:

- Trastorns cardiovasculars greus o condicions mèdiques que puguin interferir amb les mesures de HRV.
- Puntuació al qüestionari GAD-7 igual o superior a 10, indicativa d'ansietat generalitzada moderada o greu.
- Diagnòstics psicològics clínics rellevants que puguin afectar la resposta a l'estrès o la regulació emocional.
- Ús de medicació que alteri significativament la variabilitat de la freqüència cardíaca (HRV), com betabloquejants o altres fàrmacs amb impacte autonòmic.
- Condicions neurològiques o sensorials que dificultin la realització del protocol experimental.
- Història d'epilèpsia o crisis epilèptiques (degut a l'ús d'EEG).
- Ferides, lesions o irritacions actives al cuir cabellut.
- Al·lèrgies conegudes a gels conductors o materials utilitzats en els sensors.

7.4 Reclutament

Els participants seran reclutats mitjançant mostreig de conveniència a través de:

- Estudiants universitaris.
- Voluntaris de l'entorn acadèmic.
- Contactes personals dels investigadors.

8 Procediment experimental

El procediment experimental es desenvolupa en fases seqüencials per garantir la correcta recollida de dades fisiològiques i subjectives, així com el benestar del participant.

8.1 Recepció del participant (3 minuts)

- Verificació dels criteris d'inclusió i exclusió.
- Assignació a la condició experimental (robot o ordinador).

8.2 Explicació de l'estudi (5 minuts)

- Lliurament i lectura dirigida del *Full d'Informació al Participant (FIP)*.
- Presentació dels objectius generals del projecte.
- Explicació de les tasques experimentals i resolució de qualsevol dubte que pugui sorgir.
- Informació detallada sobre la durada i el procediment general de la sessió.

8.3 Consentiment informat (5 minuts)

- Signatura del document de consentiment informat un cop resolts els dubtes del FIP.
- Confirmació de la participació voluntària del subjecte.
- Informació sobre el tractament de dades i el dret a abandonar l'estudi.

8.4 Col·locació de sensors i preparació del sistema (10–15 minuts)

- Col·locació del monitor de freqüència cardíaca i el casc d'EEG.
- Verificació de la qualitat del senyal i la impedància dels elèctrodes.
- Ajust de confort del participant per garantir que els sensors no causin molèsties.

8.5 Qüestionaris preexperimentals (10–15 minuts)

- Recollida de dades demogràfiques bàsiques.
- Avaluació de l'ansietat generalitzada basal (GAD-7), amb finalitat de cribratge i exclusió.
- Mesura de l'actitud prèvia i acceptació envers la tecnologia i la robòtica social (NARS, ATI).

8.6 Fase baseline (5 minuts)

- Registre en estat de repòs de les següents variables fisiològiques:
  - Freqüència cardíaca (BPM).
  - Freqüència respiratòria (RR).
  - Activitat elèctrica cerebral (EEG).
- Establiment dels valors de referència fisiològica individuals del participant.
- Registre de l'estrès percebut momentani (VAS, punt T1).

8.7 Fase d'inducció d'estrès (6 minuts)

- Execució de la tasca d'Stroop (Stroop Task), amb pressió de temps i penalització per errors.
- Objectiu: Provocar una activació fisiològica d'estrès de forma controlada i segura.
- Registre de l'estrès percebut momentani (VAS, punt T2), immediatament després de la tasca i abans de la fase de relaxació.

8.8 Fase de relaxació i biofeedback (10 minuts)

- Exercicis de respiració guiada a un ritme fix de 6 respiracions per minut (6 bpm), adaptats a la condició experimental assignada:
  - Robot social (*Tyto*).
  - Interfície digital en pantalla (ordinador).
- Provisió de feedback fisiològic en temps real, mostrant tant el ritme respiratori objectiu com la respiració real del participant, basat en la variabilitat de la freqüència cardíaca (HRV).
- Registre de l'estrès percebut momentani (VAS, punt T3), en finalitzar la fase.

Nota metodològica: El ritme fix de 6 bpm s'ha triat per garantir la simplicitat i reproductibilitat del protocol entre condicions [17, 18].

8.9 Qüestionaris postexperimentals (10–12 minuts)

- Mesura de la càrrega de treball mental i de la usabilitat general del sistema.
- Percepció de l'experiència interactiva (específica del robot o de la pantalla segons la condició).
- Valoració subjectiva del confort global de la sessió.

8.10 Entrevista qualitativa (5–8 minuts)

- Recollida de l'experiència general de la sessió de viva veu del participant.
- Valoració sobre l'acceptació del sistema i la comoditat de la interacció.
- Recollida de comentaris lliures i possibles millores de la interfície.

L'entrevista serà enregistrada en àudio per a la seva posterior transcripció i anàlisi qualitativa. Les dades d'àudio seran completament anonimitzades mitjançant codificació algorísmica (ID de participant) i no inclouran noms ni cap mena d'informació identificativa directa.

8.11 Debriefing (3–5 minuts)

- Explicació final i transparent de l'objectiu real de l'estudi i del mecanisme d'estrès.
- Resolució de qualsevol dubte o neguit residual del participant.
- Recordatori del protocol d'ús de dades, emmagatzematge segur i estricta confidencialitat.

9 Dispositius utilitzats

Durant l'estudi s'utilitzaran diferents dispositivs i eines experimentals per a la mesura de senyals fisiològics i per a la interacció home–robot.

9.1 Tyto

Tyto és un robot social de forma zoomòrfica, inspirat en el disseny d'una òliba/mussol, dissenyat i desenvolupat per la investigadora principal específicament per a aquest projecte, prenent com a base un robot desenvolupat prèviament en el marc del seu Treball de Fi de Grau [24].

El robot és de mida petita i completament estàtic, sense cap element mecànic ni motor que en permeti el moviment. El seu maquinari es compon de dues pantalles circulars que fan la funció d'ulls, una pantalla rectangular situada a la zona ventral (on es representa visualment el biofeedback), un parell d'altaveus (utilitzats per emetre senyals sonors indicatius dels moments d'inhalació i exhalació), un LED indicador de connexió correcta amb l'aplicació web de control, i una unitat de processament Raspberry Pi 4.

El sistema de biofeedback del robot reprodueix, amb la màxima fidelitat possible, el mateix mecanisme utilitzat a la condició d'interfície digital en ordinador, amb l'objectiu d'aïllar la modalitat d'interacció (robot vs. pantalla) com a única variable diferencial entre condicions.

Referències:

- Codi font i memòria del TFG previ: `https://hdl.handle.net/2117/377449`

9.2 Monitor de freqüència cardíaca Magene H603

El Magene H603 és un cinturó de freqüència cardíaca (*chest strap*) que permet el registre continu de BPM (*beats per minute*) i la derivació de la freqüència respiratòria (RR) a partir de la variabilitat del senyal cardíac.

Referència del producte:

- `https://es.magene.com/es/all-products/150-h603-heart-rate-monitor.html`

9.3 Casc EEG Enobio 8 (Neuroelectrics)

El Enobio 8 és un sistema EEG portàtil de 8 canals utilitzat per a la monitorització de l'activitat cerebral en entorns de recerca. Permet el registre en temps real i la sincronització amb altres senyals fisiològiques.

Referència del producte:

- `https://www.neuroelectrics.com/products/research/enobio/enobio8`

9.4 Protocol d'higiene i desinfecció dels dispositius

Atès que els dispositius biomètrics descrits anteriorment són de caràcter multiusuari i requereixen un contacte físic directe i prolongat amb la pell i el cuir cabellut dels participants, s'estableix un protocol d'higiene estricte. Després de la finalització de cada sessió experimental i abans de rebre el següent participant, es duran a terme les següents accions de neteja i desinfecció per tal de garantir la màxima seguretat dermatològica i higiènica:

- **Monitor de freqüència cardíaca (Magene H603):** La banda elàstica de pit es desmuntarà del sensor electrònic i es rentarà a mà amb aigua tèbia i sabó neutre dermatològic. El sensor i els elèctrodes de contacte plàstics es netejaran i desinfectaran minuciosament fent servir tovalloles humides impregnades amb alcohol isopropílic al 70%, garantint l'eliminació de restes de suor o greix cutani.
- **Casc d'EEG (Enobio 8):** Els elèctrodes i la banda de neoprè es netejaran immediatament després de l'ús per retirar completament qualsevol residu de gel conductor. Els elèctrodes es rentaran amb aigua tèbia i es desinfectaran amb una solució d'alcohol isopropílic específica per a material mèdic, assegurant-ne el secat complet a l'aire abans de la següent col·locació.
- **Superfícies i entorn de treball:** Tant el mobiliari (cadira i taula on s'asseu el participant) com els elements d'interacció física directa (teclat i ratolí de l'ordinador utilitzat per a la tasca d'Stroop) es desinfectaran entre sessions amb productes netejadors d'evaporació ràpida.

10 Mesures i qüestionaris

Aquest apartat descriu les mesures fisiològiques i els qüestionaris utilitzats en el protocol experimental, així com el moment d'administració de cada instrument.

10.1 Mesures fisiològiques

Durant tot l'experiment es registraran de manera contínua les següents senyals fisiològiques:

- **BPM (Beats Per Minute)**: freqüència cardíaca expressada en batecs per minut com a indicador general d'activació fisiològica.
- **HRV (Heart Rate Variability)**: variabilitat de la freqüència cardíaca, amb especial atenció a indicadors com **RMSSD** i **SDNN**, utilitzats com a marcadors d'activació del sistema nerviós autònom i regulació emocional.
- **RR (Respiratory Rate)**: freqüència respiratòria, especialment rellevant en la fase de biofeedback.
- **EEG**: activitat electroencefalogràfica per a l'anàlisi d'estats d'atenció, càrrega cognitiva i activació cortical.

10.2 Qüestionaris de l'estudi

Els qüestionaris s'administraran en diferents moments del protocol experimental per capturar tant l'estat basal com els canvis pre i post intervenció.

Nota sobre les versions lingüístiques: Atès que no s'ha trobat una traducció oficial prèviament verificada en llengua catalana per a la majoria de qüestionaris utilitzats, l'adaptació i traducció formal dels ítems, instruccions i escales ha estat realitzada íntegrament per la investigadora principal **Helena Cadevall Soto**, i revisada posteriorment pels supervisors del projecte, **Linus Manubens-Gil** i **Marta Isabel Tarrés Puertas**, per tal d'assegurar-ne el rigor i la fidelitat de la traducció.

Qüestionaris preexperimentals

- **NARS (Negative Attitudes toward Robots Scale)** [12]
- **ATI Scale (Affinity for Technology Interaction)** [13]
- **GAD-7 (Generalized Anxiety Disorder Scale)** [14]

Mesura d'estrès percebut momentani

- **VAS (Visual Analogue Scale) d'estrès percebut, escala 0-10**: Administrada en tres moments: baseline (T1), post-inducció d'estrès (T2), post-biofeedback (T3). [15, 16]

Qüestionaris postexperimentals

- **RTLX (Raw NASA-TLX)** [21, 22]
- **SUS (System Usability Scale)** [19]
- **Godspeed Questionnaire Series:** Administrat únicament a la condició de robot [20].

11 Gestió i protecció de dades

El tractament de les dades de caràcter personal recollides durant aquest estudi s'ajusta al Reglament (UE) 2016/679 General de Protecció de Dades (RGPD) i a la Llei Orgànica 3/2018 de Protecció de Dades Personals i garantia dels drets digitals (LOPDGDD).

11.1 Allotjament, custòdia i seguretat de la informació

- **Allotjament i infraestructura:** Totes les dades digitals recollides (registres fisiològics d'EEG i HRV, respostes dels qüestionaris i fitxers d'àudio de les entrevistes) s'emmagatzemaran en un servidor segur intern de la Universitat Politècnica de Catalunya (UPC), basat en la plataforma Nextcloud. La infraestructura d'aquest servidor està allotjada dins de la Unió Europea (Alemanya), garantint el compliment del Reglament (UE) 2016/679 (Reglament General de Protecció de Dades, RGPD) i les polítiques de seguretat de la informació de la UPC.
- **Gestió de l'accés:** L'accés al directori de dades estarà estrictament restringit a la investigadora responsable, Helena Cadevall Soto, i als supervisors del projecte, la Dra. Marta Isabel Tarés Puertas i el Dr. Linus Manubens-Gil, mitjançant credencials d'autenticació segures.
- **Anonimització i codificació:** S'utilitzarà un sistema de codificació alfanumèrica per a cada participant. Aquest codi serà l'únic identificador present en tots els fitxers de dades experimentals. La taula que relaciona la identitat real dels participants amb el seu codi alfanumèric es conservarà en un fitxer separat, amb accés restringit i fora del conjunt de dades analítiques.
- **Termini de retenció i eliminació:** Les dades identificatives seran eliminades un cop finalitzat el Treball Final de Màster i el seu procés d'avaluació, amb un termini màxim de dos anys des de la finalització de la recerca. Després d'aquest període, només es conservaran les dades degudament anonimitzades necessàries per a la verificació científica.
- **Publicació i reutilització:** Es preveu la publicació de les dades experimentals completament anonimitzades en repositoris científics de dades obertes (com ara *OpenNeuro*), sempre que s'hagi garantit tècnicament que no existeix un risc raonable de reidentificació dels participants, seguint la normativa vigent de protecció de dades.

11.2 Clàusula Informativa de Protecció de Dades (UPC)

Responsable del tractament	Universitat Politècnica de Catalunya (UPC).
Finalitat del tractament	Gestió de les dades de caràcter personal per a la realització del projecte de recerca acadèmica basat en biofeedback respiratori i robòtica social.
Legitimació	Consentiment exprés i lliure atorgat pel participant mitjançant la signatura del document de Consentiment Informat (Art. 6.1.a del RGPD).
Destinataris de les dades	Les dades no es cediran a tercers aliens a la universitat, excepte per obligació legal o en cas de publicacions científiques de forma agregada i anonimitzada.
Drets dels participants	Accés, rectificació, supressió, oposició, limitació del tractament i portabilitat de les seves dades.
Termini de conservació	Les dades se mantindran emmagatzemades durant un termini màxim de 2 anys des de la cloenda del projecte.
Delegat de Protecció de Dades	Servei de Protecció de Dades de la UPC. Adreça electrònica: `proteccio.dades@upc.edu`.

Table 1: Clàusula informativa corporativa de protecció de dades.

12 Consideracions ètiques i riscos

La Universitat Politècnica de Catalunya (UPC) disposa d'un codi d'integritat en la recerca que estableix les mesures necessàries per a la realització d'investigació experimental amb participants humans. L'equip investigador seguirà els principis establerts per les directrius internacionals aplicables a la recerca amb persones participants, incloent la Declaració d'Hèlsinki.

12.1 Riscos esperats

- Estrès lleu i temporal durant la fase d'inducció d'estrès.

- Fatiga o incomoditat menor derivada de les tasques cognitives.
- Lleu incomoditat física associada a l'ús dels dispositius fisiològics.

12.2 Mesures de mitigació

- **Voluntarietat:** Participació completament voluntària i possibilitat d'abandonar l'estudi en qualsevol moment, sense necessitat de justificar-ne el motiu i sense cap conseqüència.
- **Supervisió:** Supervisió contínua de l'estat del participant durant totes les fases de l'experiment.
- **Protocol d'interrupció:** Interrupció immediata de la sessió experimental si el participant manifesta incomoditat física o emocional.
- **Interacció amb el robot:** La interacció amb el robot social no comporta riscos físics rellevants; durant l'experiment, el robot romandrà completament estàtic. No hi haurà desplaçaments, moviments mecànics ni cap tipus de contacte físic amb els participants per part del robot. La interacció entre el participant i el robot es produeix exclusivament a través de canals visuals (pantalles) i auditius (altaveus), sense que en cap moment sigui necessari el contacte físic directe entre el participant i el dispositiu.
- **Privacitat i naturalesa de les dades EEG:** L'estudi utilitza un sistema d'EEG per obtenir indicadors fisiològics globals (com ara canvis en l'atenció o la relaxació). El protocol no aplica paradigmes de neurociència cognitiva per a la descodificació de continguts mentals, ni tècniques d'aprenentatge automàtic per a la inferència de pensaments, records o intencions. Les dades registrades no permeten identificar el participant biomètricament ni extreure informació sobre el seu contingut cognitiu específic.

13 Resultats esperats i impacte

Es preveu que aquest estudi aporti evidència preliminar sobre l'efectivitat de la interacció humà-robot com a eina per a la regulació fisiològica de l'estrès, aportant informació rellevant sobre com el tipus d'agent interactiu (robot vs interfície digital) pot influir en la resposta fisiològica i en l'experiència d'usuari durant tasques de regulació emocional.

Atès que el sistema robòtic emprat és un prototip dissenyat específicament per a aquest estudi, els resultats obtinguts s'interpretaran com a evidència preliminar associada a aquest dispositiu concret, sense pretendre generalitzar-los directament a robots socials com a categoria general.

13.1 Limitacions relacionades amb el dispositiu robòtic

Cal assenyalar que el robot social utilitzat en aquest estudi és un prototip d'autoria pròpia, sense historial previ de validació en investigació d'interacció humà-robot. Això implica dues limitacions rellevants: (1) els resultats obtinguts no són directament generalitzables a robots socials en general, sinó específics d'aquest dispositiu; i (2) no es pot descartar un possible efecte de novetat associat al caràcter únic i no estandarditzat del prototip. Per mitigar aquesta limitació, prèviament a l'inici de la recollida de dades es duran a terme proves pilot de funcionament tècnic del sistema, per assegurar l'estabilitat del dispositiu durant les sessions experimentals.

Addicionalment, es reconeix que la investigadora principal ha estat responsable tant del disseny del robot com de la recollida i anàlisi de les dades, cosa que introdueix un potencial conflicte de rol. Per mitigar-ho, l'anàlisi de dades fisiològiques (HRV, EEG) es basarà en indicadors quantitatius i protocols de processament estandarditzats, minimitzant el marge d'interpretació subjectiva.

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D.2.2 English Version

Helena Cadevall Soto

Research Plan

1 Project title

Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback

2 General information about the project

2.1 Principal investigator

- Name: Helena Cadevall Soto
- Programme / master's / thesis: Master in Interaction, Graphics and Design

2.2 Supervisors

Supervisor 1:

- Name: Marta Isabel Tarrés Puertas
- Position: Associate Professor
- Department: Smart Sustainable Resources
- Institutional email: marta.isabel.tarres@upc.edu

Supervisor 2:

- Name: Linus Manubens-Gil
- Position: Distinguished Investigator
- Department: Smart Sustainable Resources
- Institutional email: linus.manubens@upc.edu

2.3 Institution

- Home institution: Institut Polytechnique de Paris (IP Paris)
- Host institution: Universitat Politècnica de Catalunya (UPC)

3 Project summary

This project investigates the effectiveness of Human–Robot Interaction (HRI) as a tool for the physiological regulation of stress. Specifically, it is proposed as a pilot study of the proof of concept type, aimed at exploring the feasibility of a system based on physiological biofeedback through interaction with a social robot, and at comparing it with a conventional screen-based digital solution.

The study will compare two experimental conditions: one based on the interaction with a zoomorphic social robot and the other based on a conventional digital interface. During the experimental sessions, the participants will

do guided breathing exercises based on physiological biofeedback, after a controlled stress-induction phase using cognitive tasks.

This preliminary study will be aimed mainly at healthy people, with the objective of obtaining a first approximation of the impact of different interactive technologies on stress regulation and on the user experience. The profile of the participants may include adults without clinically diagnosed conditions related to stress or mental health.

To evaluate the effect of the different experimental conditions, objective physiological measures and subjective measures will be combined. The main physiological measures will include heart rate (BPM) and respiratory rate (RR), recorded directly with the sensors used, from which heart rate variability (HRV) indicators such as the RMSSD will be derived. Possible measures of brain activity using electroencephalography (EEG) will also be considered, to explore indicators of cortical activation and cognitive load. In parallel, standardised questionnaires related to perceived stress, user experience and the perception of technology and robots will be collected.

The expected impact of the project is to provide preliminary evidence about the feasibility of integrating physiological biofeedback in human–robot interaction systems, and to identify initial trends about the potential of social robots as alternatives or complements to conventional digital interfaces.

4 Context and motivation

4.1 Stress and human–computer interaction

Stress is an adaptive physiological and psychological response that can become problematic when it is sustained over time. In contemporary contexts, the increase in cognitive, work-related and digital demands has contributed to a higher prevalence of perceived stress in the general population, with potential impacts on physical and mental wellbeing.

In this context, research in Human–Computer Interaction (HCI) has explored how the design of interactive systems can influence the emotional and physiological states of users, as well as their capacity to regulate stress through digital interventions such as guided breathing or biofeedback systems [1].

However, many of these solutions are based on purely digital interfaces, which may limit emotional involvement and the sense of social presence during the interaction.

4.2 Social robotics and wellbeing

In recent years, social robotics has become established as an emerging field within Human–Robot Interaction (HRI), with growing applications in health and emotional wellbeing. Social robots can interact with humans through multimodal channels, increasing the sense of social presence and involvement during the interaction [2].

Several studies have explored the use of social robots for the regulation of stress and anxiety, especially in the context of guided breathing exercises. For example, it has been observed that the interaction with robots can reduce psychophysiological responses associated with stress and improve the emotional experience in comparison with traditional interfaces [3, 4].

More recently, work in this area has explored guided breathing systems based on social robots, as well as mechanisms of physiological entrainment, in which the rhythms of the robot can help synchronise the breathing of the user and facilitate physiological regulation [5, 6, 7, 8, 9]. These approaches suggest that the physical presence of an interactive agent can have a differential impact with respect to screen-based interfaces.

4.3 Use of HRV as a physiological indicator

For the study of stress, subjective measures have traditionally been the main source of evaluation. However, in recent years there has been increasing interest in the use of objective physiological measures that make it possible to characterise the response to stress more precisely.

In this context, heart rate variability (HRV) has been established as one of the most used indicators for the evaluation of the activity of the autonomic nervous system. In particular, parameters such as the RMSSD make it

possible to estimate the balance between the sympathetic and parasympathetic systems, associated with states of stress and relaxation respectively [10].

The use of HRV, together with other physiological measures such as electrodermal activity or the breathing pattern, makes it possible to obtain a view that complements the subjective measures of perceived stress [11].

5 Objectives and hypotheses

5.1 Main objective

The main objective of this study is to compare the effectiveness of two interaction modalities, one based on a social robot and the other based on a conventional digital interface, in the physiological regulation of stress.

Specifically, the aim is to obtain a first approximation of which of the two conditions is more effective in reducing the physiological stress response and promoting relaxation states during guided breathing exercises.

5.2 Secondary objectives

- To analyse preliminary differences in physiological variables related to stress between the robot condition and the screen condition, especially heart rate variability (HRV) and measures derived from electroencephalography (EEG), such as indicators of cortical activation and cognitive load.
- To compare the subjective perception of stress reported by the participants through questionnaires with the objective measures obtained from the physiological signals (HRV and EEG), with the aim of exploring the coherence between subjective and objective measures in an exploratory context.
- To explore the relationship and possible correlation between the physiological indicators (HRV and EEG) and the scores of the questionnaires on perceived stress and user experience, to identify preliminary trends according to the interaction modality.

5.3 Hypotheses

5.3.1 Null hypothesis (H0)

H0: There will be no significant differences between the social robot condition and the digital interface condition in any of the stress measures, either physiological (HRV) or subjective (questionnaires).

5.3.2 Alternative hypotheses

- **H1:** The participants who interact with the social robot will show a greater reduction of perceived stress than those in the screen-based condition.
- **H2:** The social robot condition will show higher HRV values, indicating a better physiological regulation of stress, than the digital interface condition.
- **H3:** There will be a partial dissociation between subjective and physiological measures, so that the reduction of stress perceived in the questionnaires will be proportionally greater than the improvement observed in the physiological measures.
- **H4:** There will be a significant correlation between HRV values and perceived stress scores, indicating coherence between objective and subjective measures.
- **H5:** The robot condition will generate a greater sense of emotional support and involvement during the exercise.

6 Experimental design

6.1 Type of study

The study follows a randomised between-subjects experimental design, in which each participant is assigned to a single experimental condition: social robot or computer interface.

All the participants follow exactly the same experimental protocol (questionnaires, measurement of physiological signals, stress induction and relaxation phase with biofeedback). The only difference between conditions is the modality used during the biofeedback phase (robot or computer).

This design makes it possible to attribute the observed differences exclusively to the type of interface used, reducing the influence of other experimental variables.

6.2 Experimental conditions

The study includes two main experimental conditions, which differ only in the interaction modality used during the relaxation and biofeedback phase.

6.2.1 Condition A – Computer

In this condition, the guided breathing and biofeedback exercises are done through a web application displayed on a computer. The interface shows simultaneously the target breathing rhythm and the real breathing of the participant, allowing them to adjust it in real time.

6.2.2 Condition B – Robot

In this condition, the same guided breathing and biofeedback exercises are led by Tyto, a zoomorphic social robot (inspired by an owl) designed specifically for this study, which reproduces the same visual and sound biofeedback mechanism as the computer condition.

This condition seeks to introduce an interactive physical agent with social presence capacity, while keeping a simple interaction with low cognitive load.

6.2.3 Task and duration

In both conditions, the participants do the same experimental protocol, which includes: baseline phase (rest), stress-induction phase and relaxation phase with biofeedback. The total duration of the session is approximately 50 to 60 minutes, depending on the time needed to prepare the sensors.

6.3 Variables

- Independent variables:
  - Interaction modality: Robot (Tyto) or computer (web application).
- Dependent variables:
  - Heart rate variability (HRV), with special emphasis on temporal indicators such as the RMSSD.
  - Heart rate and respiratory rate.
  - Electroencephalography (EEG) signals, such as indicators of cortical activation and cognitive load.
  - Questionnaires on stress and subjective experience.

7 Participants

7.1 Estimated number

The participation of between 12 and 20 participants is foreseen.

This study is conceived as a pilot (proof of concept) with the objective of exploring the feasibility of the experimental system and obtaining a first approximation of the effects of the interaction with a social robot versus a digital interface on stress regulation.

Given the exploratory nature of the work, the sample does not aim to allow robust statistical inferences on a large scale, but to identify preliminary trends, validate the experimental protocol and evaluate the feasibility of the physiological (HRV and EEG) and subjective measures.

7.2 Inclusion criteria

The inclusion criteria for participation in the study are:

- Adults over 18 years of age.
- Healthy participants, without relevant clinical diagnoses associated with severe stress, anxiety or depression disorders.
- Ability to understand and follow the instructions of the experimental protocol.
- Informed consent signed before participation.

7.3 Exclusion criteria

Participants who meet any of the following criteria will be excluded from the study:

- Severe cardiovascular disorders or medical conditions that may interfere with the HRV measures.
- A score of 10 or higher in the GAD-7 questionnaire, indicative of moderate or severe generalised anxiety.
- Relevant clinical psychological diagnoses that may affect the response to stress or emotional regulation.
- Use of medication that significantly alters heart rate variability (HRV), such as beta-blockers or other drugs with an autonomic impact.
- Neurological or sensory conditions that make it difficult to carry out the experimental protocol.
- A history of epilepsy or epileptic seizures (due to the use of EEG).
- Active wounds, injuries or irritations on the scalp.
- Known allergies to conductive gels or to materials used in the sensors.

7.4 Recruitment

The participants will be recruited by convenience sampling through:

- University students.
- Volunteers from the academic environment.
- Personal contacts of the researchers.

8 Experimental procedure

The experimental procedure is carried out in sequential phases to guarantee the correct collection of physiological and subjective data, as well as the wellbeing of the participant.

8.1 Reception of the participant (3 minutes)

- Verification of the inclusion and exclusion criteria.
- Assignment to the experimental condition (robot or computer).

8.2 Explanation of the study (5 minutes)

- Delivery and guided reading of the Participant Information Sheet (PIS).
- Presentation of the general objectives of the project.
- Explanation of the experimental tasks and resolution of any question that may arise.
- Detailed information about the duration and the general procedure of the session.

8.3 Informed consent (5 minutes)

- Signature of the informed consent document once the questions about the PIS are resolved.
- Confirmation of the voluntary participation of the subject.
- Information about data processing and the right to leave the study.

8.4 Placement of the sensors and preparation of the system (10–15 minutes)

- Placement of the heart rate monitor and the EEG cap.
- Verification of the signal quality and the impedance of the electrodes.
- Adjustment of the comfort of the participant to ensure that the sensors do not cause discomfort.

8.5 Pre-experiment questionnaires (10–15 minutes)

- Collection of basic demographic data.
- Evaluation of baseline generalised anxiety (GAD-7), for screening and exclusion purposes.
- Measurement of the prior attitude and acceptance toward technology and social robotics (NARS, ATI).

8.6 Baseline phase (5 minutes)

- Recording at rest of the following physiological variables:
  - Heart rate (BPM).
  - Respiratory rate (RR).
  - Electrical brain activity (EEG).
- Establishment of the individual physiological reference values of the participant.
- Recording of the momentary perceived stress (VAS, point T1).

8.7 Stress-induction phase (6 minutes)

- Execution of the Stroop Task, with time pressure and a penalty for errors.
- Objective: to provoke a physiological stress activation in a controlled and safe way.
- Recording of the momentary perceived stress (VAS, point T2), immediately after the task and before the relaxation phase.

8.8 Relaxation and biofeedback phase (10 minutes)

- Guided breathing exercises at a fixed rhythm of 6 breaths per minute (6 bpm), adapted to the assigned experimental condition:
  - Social robot (Tyto).
  - Digital interface on a screen (computer).
- Provision of real-time physiological feedback, showing both the target breathing rhythm and the real breathing of the participant, based on heart rate variability (HRV).
- Recording of the momentary perceived stress (VAS, point T3), at the end of the phase.

Methodological note: The fixed rhythm of 6 bpm has been chosen to guarantee the simplicity and reproducibility of the protocol between conditions [17, 18].

8.9 Post-experiment questionnaires (10–12 minutes)

- Measurement of the mental workload and the general usability of the system.
- Perception of the interactive experience (specific to the robot or the screen, depending on the condition).
- Subjective evaluation of the overall comfort of the session.

8.10 Qualitative interview (5–8 minutes)

- Collection of the general experience of the session in the participant's own words.
- Evaluation of the acceptance of the system and the comfort of the interaction.
- Collection of free comments and possible improvements of the interface.

The interview will be recorded in audio for its later transcription and qualitative analysis. The audio data will be completely anonymised through algorithmic coding (participant ID) and will not include names or any kind of direct identifying information.

8.11 Debriefing (3–5 minutes)

- Final and transparent explanation of the real objective of the study and of the stress mechanism.
- Resolution of any remaining doubt or concern of the participant.
- Reminder of the data usage protocol, stored securely and under strict confidentiality.

9 Devices used

During the study, different devices and experimental tools will be used for the measurement of physiological signals and for the human–robot interaction.

9.1 Tyto

Tyto is a zoomorphic social robot, inspired by the design of an owl, designed and developed by the principal investigator specifically for this project, taking as a base a robot previously developed in the framework of her Bachelor's thesis [24].

The robot is small and completely static, without any mechanical element that allows it to move. Its machinery consists of two circular screens that act as eyes, a rectangular screen located in the ventral area (where the biofeedback is visually represented), a pair of speakers (used to emit sound signals indicating the moments of inhalation and exhalation), an LED indicating a correct connection with the control web application, and a Raspberry Pi 4 processing unit.

The biofeedback system of the robot reproduces, with the highest possible fidelity, the same mechanism used in the computer interface condition, with the objective of isolating the interaction modality (robot vs. screen) as the only differential variable between conditions.

References: source code and report of the previous Bachelor's thesis: `https://hdl.handle.net/2117/377449`

9.2 Magene H603 heart rate monitor

The Magene H603 is a heart rate chest strap that allows the continuous recording of BPM (beats per minute) and the derivation of the respiratory rate (RR) from the variability of the cardiac signal.

Product reference: `https://es.magene.com/es/all-products/150-h603-heart-rate-monitor.html`

9.3 Enobio 8 EEG cap (Neuroelectrics)

The Enobio 8 is a portable 8-channel EEG system used for monitoring brain activity in research environments. It allows real-time recording and synchronisation with other physiological signals.

Product reference: `https://www.neuroelectrics.com/products/research/enobio/enobio8`

9.4 Hygiene and disinfection protocol of the devices

Since the biometric devices described above are multi-user and require direct and prolonged physical contact with the skin and scalp of the participants, a strict hygiene protocol is established. After the end of each experimental session and before receiving the next participant, the following cleaning and disinfection actions are carried out, to guarantee the highest dermatological safety and hygiene:

- **Heart rate monitor (Magene H603):** The elastic chest band is detached from the electronic sensor and washed by hand with lukewarm water and neutral dermatological soap. The sensor and the plastic contact electrodes are carefully cleaned and disinfected using damp wipes soaked in 70% isopropyl alcohol, guaranteeing the removal of sweat or skin grease residues.
- **EEG cap (Enobio 8):** The electrodes and the neoprene band are cleaned immediately after use to completely remove any residue of conductive gel. The electrodes are washed with lukewarm water and disinfected with a solution of isopropyl alcohol specific for medical material, making sure they are dry in the air before the next placement.
- **Surfaces and work environment:** The furniture (chair and table where the participant sits) and the elements of direct physical interaction (keyboard and mouse of the computer used for the Stroop task) are disinfected between sessions with quick-evaporating cleaning products.

10 Measures and questionnaires

This section describes the physiological measures and the questionnaires used in the experimental protocol, as well as the moment of administration of each instrument.

10.1 Physiological measures

During the whole experiment, the following physiological signals will be recorded continuously:

- **BPM (Beats Per Minute):** heart rate expressed in beats per minute as a general indicator of physiological activation.
- **HRV (Heart Rate Variability):** variability of the heart rate, with special attention to indicators such as the **RMSSD** and **SDNN**, used as markers of activation of the autonomic nervous system and emotional regulation.
- **RR (Respiratory Rate):** respiratory rate, especially relevant in the biofeedback phase.
- **EEG:** electroencephalographic activity for the analysis of states of attention, cognitive load and cortical activation.

10.2 Study questionnaires

The questionnaires will be administered at different moments of the experimental protocol to capture both the baseline state and the changes before and after the intervention.

Note on the language versions: Since no previously verified official translation into Catalan has been found for most of the questionnaires, the formal adaptation of the items, instructions and scales has been done entirely by the principal investigator **Helena Cadevall Soto**, and later reviewed by the project supervisors, **Linus Manubens-Gil** and **Marta Isabel Tarrés Puertas**, to ensure the rigour and fidelity of the translation.

Pre-experiment questionnaires

- **NARS (Negative Attitudes toward Robots Scale)** [12]
- **ATI Scale (Affinity for Technology Interaction)** [13]
- **GAD-7 (Generalized Anxiety Disorder Scale)** [14]

Momentary perceived stress measure

- **VAS (Visual Analogue Scale) of perceived stress, scale 0–10:** Administered at three moments: baseline (T1), post stress induction (T2), post biofeedback (T3). [15, 16]

Post-experiment questionnaires

- **RTLX (Raw NASA-TLX)** [21, 22]
- **SUS (System Usability Scale)** [19]
- **Godspeed Questionnaire Series:** Administered only in the robot condition [20].

11 Data management and protection

The processing of the personal data collected during this study complies with Regulation (EU) 2016/679, General Data Protection Regulation (GDPR), and with Organic Law 3/2018 on the Protection of Personal Data and guarantee of digital rights (LOPDGDD).

11.1 Hosting, custody and security of the information

- **Hosting and infrastructure:** All the digital data collected (physiological EEG and HRV records, questionnaire answers and audio files of the interviews) will be stored on a secure internal server of the Universitat Politècnica de Catalunya (UPC), based on the Nextcloud platform. The infrastructure of this server is hosted within the European Union (Germany), guaranteeing compliance with Regulation (EU) 2016/679 (General Data Protection Regulation, GDPR) and the information security policies of the UPC.

- **Access management:** Access to the data directory will be strictly restricted to the responsible researcher, Helena Cadevall Soto, and to the project supervisors, Dr. Marta Isabel Tarrés Puertas and Dr. Linus Manubens-Gil, through secure authentication credentials.
- **Anonymisation and coding:** An alphanumeric coding system will be used for each participant. This code will be the only identifier present in all the experimental data files. The table that relates the real identity of the participants with their alphanumeric code will be kept in a separate file, with restricted access and outside the set of analytical data.
- **Retention and deletion period:** The identifying data will be deleted once the Master's thesis and its evaluation process are finished, with a maximum period of two years from the end of the research. After this period, only the duly anonymised data necessary for scientific verification will be kept.
- **Publication and reuse:** The publication of the completely anonymised experimental data in open scientific data repositories (such as *OpenNeuro*) is foreseen, provided that it has been technically guaranteed that there is no reasonable risk of re-identification of the participants, following the current data protection regulations.

11.2 Data Protection Information Clause (UPC)

Table 1: Corporate data protection information clause.

Data controller	Universitat Politècnica de Catalunya (UPC).
Purpose of the processing	Management of the personal data for the carrying out of the academic research project based on respiratory biofeedback and social robotics.
Legal basis	Express and free consent given by the participant by signing the Informed Consent document (Art. 6.1.a of the GDPR).
Recipients of the data	The data will not be transferred to third parties outside the university, except for a legal obligation or in the case of scientific publications in an aggregated and anonymised form.
Rights of the participants	Access, rectification, erasure, objection, restriction of processing and portability of their data.
Retention period	The data will be kept for a maximum period of 2 years from the end of the project.
Data Protection Officer	Data Protection Service of the UPC. Email address: protecccio.dades@upc.edu.

12 Ethical considerations and risks

The Universitat Politècnica de Catalunya (UPC) has a research integrity code that establishes the measures necessary for carrying out experimental research with human participants. The research team will follow the principles established by the international directives applicable to research with human participants, including the Declaration of Helsinki.

12.1 Expected risks

- Mild and temporary stress during the stress-induction phase.
- Mild fatigue or discomfort derived from the cognitive tasks.
- Slight physical discomfort associated with the use of the physiological devices.

12.2 Mitigation measures

- **Voluntariness:** Completely voluntary participation and the possibility of leaving the study at any moment, without the need to justify the reason and without any consequence.
- **Supervision:** Continuous supervision of the state of the participant during all the phases of the experiment.

- **Interruption protocol:** Immediate interruption of the experimental session if the participant shows physical or emotional discomfort.
- **Interaction with the robot:** The interaction with the social robot does not involve relevant physical risks; during the experiment, the robot will remain completely static. There will be no displacements, mechanical movements or any type of physical contact with the participants on the part of the robot. The interaction between the participant and the robot takes place exclusively through visual (screens) and auditory (speakers) channels, without the direct physical contact between the participant and the device being necessary at any moment.
- **Privacy and nature of the EEG data:** The study uses an EEG system to obtain global physiological indicators (such as changes in attention or relaxation). The protocol does not apply cognitive neuroscience paradigms for decoding mental contents, or automatic learning techniques for inferring thoughts, memories or intentions. The recorded data do not allow the participant to be identified biometrically or specific information about their cognitive content to be extracted.

13 Expected results and impact

This study is expected to provide preliminary evidence about the effectiveness of human–robot interaction as a tool for stress regulation, providing relevant information about how the type of interactive agent (robot vs. digital interface) can influence the physiological response and the user experience during emotional regulation tasks.

Given that the robotic system used is a prototype designed specifically for this study, the results obtained will be interpreted as preliminary evidence associated with this specific device, without aiming to generalise them directly to social robots as a general category.

13.1 Limitations related to the robotic device

It should be noted that the social robot used in this study is a prototype of its own authorship, without a previous history of validation in human–robot interaction research. This implies two relevant limitations: (1) the results obtained are not directly generalisable to social robots in general, but specific to this device; and (2) a possible effect of novelty associated with the unique and non-standardised nature of the prototype cannot be ruled out. To mitigate this limitation, before the start of the data collection, technical functioning tests of the system will be carried out, to ensure the stability of the device during the experimental sessions.

In addition, it is acknowledged that the principal investigator has been responsible for both the design of the robot and the collection and analysis of the data, which introduces a potential role conflict. To mitigate this, the analysis of the physiological data (HRV, EEG) will be based on quantitative indicators and standardised processing protocols, minimising the margin of subjective interpretation.

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D.3 Participant Information Sheet

The Participant Information Sheet is given first in Catalan, which is the version delivered to the participants, and then in English.

D.3.1 Catalan Version

Helena Cadevall Soto

Full d'informació al participant

Títol de l'estudi

Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback

1 Objectiu de l'estudi

Aquest estudi té com a objectiu comparar si una interacció amb un robot social o amb una interfície convencional en ordinador resulta més adequada per guiar exercicis de respiració amb biofeedback orientats a la regulació de l'estrès.

A més, es compararan les mesures subjectives obtingudes mitjançant qüestionaris amb les mesures fisiològiques registrades (freqüència cardíaca i activitat electroencefalogràfica), amb l'objectiu d'explorar el potencial d'aquestes darreres com a eina objectiva d'avaluació de l'estrès.

Aquest estudi és una prova pilot (*proof of concept*) que servirà per valorar la viabilitat d'aquesta metodologia i orientar futurs estudis en l'àmbit de la robòtica social aplicada al benestar emocional.

2. Durada de la participació

La participació en aquest estudi consisteix en una única sessió experimental amb una durada total aproximada de 50 a 60 minuts.

Aquesta durada inclou totes les fases de l'experiment, des de l'explicació inicial fins al debriefing final, incloent la col·locació de sensors, la realització de qüestionaris i les tasques experimentals.

No es requereix cap seguiment posterior ni sessions addicionals.

3. Qui pot participar?

Poden participar persones que compleixin els següents criteris:

- Majors de 18 anys
- Sense diagnòstics clínics greus relacionats amb estrès, ansietat o depressió
- Sense condicions mèdiques o neurològiques que interfereixin amb les mesures fisiològiques
- Capacitat per entendre i seguir les instruccions de l'experiment
- Acceptació i signatura del consentiment informat

No poden participar persones amb condicions que puguin afectar les mesures fisiològiques o la resposta a l'estrès.

4. Què faràs durant l'estudi?

Totes les fases d'aquest estudi es duran a terme presencialment a les instal·lacions de l'Escola Politècnica Superior d'Enginyeria de Manresa (EPSEM) de la Universitat Politècnica de Catalunya (UPC).

La participació consisteix en una única sessió d'aproximadament 50-60 minuts estructurada de la següent manera:

1. Explicació inicial de l'estudi i resolució de dubtes.
2. Signatura del consentiment informat.
3. Col·locació de sensors no invasius (cinturó cardíac i casc EEG).
4. Ompliment de qüestionaris inicials.
5. Mesura de l'activitat en repòs (estat basal).
6. Tasca d'inducció d'estrès moderat (Stroop Task).
7. Fase de relaxació guiada amb biofeedback mitjançant la interacció assignada (robot social *Tyto* o interfície d'ordinador).
8. Ompliment de qüestionaris finals.
9. Realització d'una entrevista breu de valoració.
10. Debriefing final i cloenda de la sessió.

5. Dispositius utilitzats

Durant l'estudi s'utilitzaran dispositius no invasius:

- Casc EEG (Enobio 8) per registrar activitat cerebral
- Cinturó de freqüència cardíaca (Magene H603) per HRV i ritme cardíac
- Robot social *Tyto* (un robot de disseny propi, en forma d'òliba/mussol) o interfície en ordinador per la interacció

6. Beneficis i riscos

Beneficis: No hi ha beneficis directes per al participant, però la participació contribueix a la recerca científica.

Riscos o incomoditats possibles:

- Estrès lleu i temporal durant la tasca cognitiva
- Fatiga mental lleu
- Lleu incomoditat pels sensors
- Cansament per la durada de la sessió

No es preveuen riscos físics ni procediments invasius. El robot social romandrà completament estàtic durant tota la sessió i no hi haurà moviment ni contacte físic amb el participant. Les mesures fisiològiques registrades (ECG/HRV i EEG) s'utilitzaran exclusivament amb finalitats de recerca i no permeten inferir pensaments, records ni altres continguts mentals.

7. Participació voluntària

La participació és completament voluntària. Pots:

- Abandonar l'estudi en qualsevol moment
- No respondre a preguntes si no ho desitges
- Retirar el consentiment sense conseqüències

8. Confidencialitat i protecció de dades

Les dades de caràcter personal recollides en aquest estudi seran tractades d'acord amb el Reglament (UE) 2016/679 (Reglament General de Protecció de Dades - GDPR) i la Llei Orgànica 3/2018 (LOPDGDD).

- **Pseudonimització:** Les dades seran associades a un codi alfanumèric (ID de participant). La llista que vincula aquest codi amb la vostra identitat real es guardarà de forma separada i protegida per l'equip investigador.
- **Seguretat:** Les dades experimentals s'emmagatzemaran en servidors segurs amb accés restringit, i només l'equip investigador autoritzat hi tindrà accés.
- **Accés:** Només la investigadora principal i els supervisors del projecte tindran accés a les dades identificatives.
- **Termini de conservació:** Les dades identificatives es conservaran durant un màxim de dos anys després de la finalització de l'estudi. Posteriorment s'eliminaran, mantenint únicament les dades completament anonimitzades necessàries per justificar els resultats científics.

Informació bàsica sobre protecció de dades personals:

Responsable	Universitat Politècnica de Catalunya (UPC). Departament d'Enginyeria Minera, Industrial i TIC (EMIT).
Finalitat	Gestió de dades del projecte de recerca "Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback".
Legitimació	Consentiment de la persona interessada. Podeu revocar-lo en qualsevol moment contactant amb l'equip investigador.
Destinataris	Les dades no es comunicaran a terceres persones, llevat d'obligació legal. Els resultats de la recerca s'exposaran exclusivament en l'àmbit acadèmic del TFM de forma anònima.
Drets	Sol·licitar l'accés, rectificació, supressió o limitació del tractament enviant un correu a l'equip investigador. Més informació a: `https://www.upc.edu/normatives/ca/proteccio-de-dades/normativa-europea-de-proteccio-de-dades/drets`
Termini	Un màxim de 2 anys des de la finalització de la recerca (coincidint amb el tancament de l'avaluació acadèmica del TFM).
Delegat PD	Universitat Politècnica de Catalunya, Plaça Eusebi Güell, 6, edifici Vèrtex, 08034 Barcelona. Correu: `proteccio.dades@upc.edu`
Reclamació	Podeu presentar una reclamació davant l'Autoritat Catalana de Protecció de Dades (APDCAT) a: `https://apdcat.gencat.cat`

9. Publicació de les dades

Les dades experimentals completament anonimitzades podran publicar-se en repositoris científics de dades obertes amb la finalitat de facilitar la reproducció i reutilització dels resultats per part de la comunitat científica. En cap cas es publicarà informació que permeti identificar els participants.

10. Ús d'imatges i enregistrament d'àudio

Es garanteix de forma expressa que durant el desenvolupament de la sessió experimental no es realitzarà cap mena de captura fotogràfica ni gravació de vídeo de les persones participants.

L'únic registre de caràcter multimèdia associat a aquest estudi consistirà en l'enregistrament exclusiu d'àudio de l'entrevista final (detallada a l'Apartat 4, punt 9). Aquest procediment té com a única finalitat metodològica facilitar la posterior transcripció textual i indexació de les respostes per a l'anàlisi qualitatiu.

L'arxiu d'àudio resultant s'emmagatzemarà en un sistema segur amb accés restringit i s'esborrarà de manera definitiva i irreversible un cop completat el procés de transcripció. El text indexat estarà associat exclusivament al codi alfanumèric de pseudonimització (ID de participant), assegurant que cap fragment textual pugui ser vinculat

a la identitat real del subjecte. L'execució d'aquest enregistrament depèn exclusivament de l'atorgament del consentiment exprés de la persona participant mitjançant la seva signatura en el corresponent Full de Consentiment Informat.

11. Retirada de dades

El participant pot sol·licitar la retirada de les seves dades en qualsevol moment. Si les dades ja han estat anonimitzades, podria no ser possible eliminar-les individualment.

12. Contacte

Per a qualsevol dubte, aclariment o informació addicional relacionada amb aquest estudi, podeu contactar amb l'equip investigador a través de les següents dades:

- **Investigadora principal:** Helena Cadevall Soto
  - Email: helena.cadevallsoto@ip-paris.fr
- **Supervisors de la recerca (UPC):**
  - Marta Isabel Tarrés Puertas (Email: marta.isabel.tarres@upc.edu — Telèfon del despatx: 938777282)
  - Linus Manubens-Gil (Email: linus.manubens@upc.edu — Telèfon del despatx: 938777391)
- **Departament i Centre responsable:** Departament d'Enginyeria Minera, Industrial i TIC (EMIT). Escola Politècnica Superior d'Enginyeria de Manresa (EPSEM), Universitat Politècnica de Catalunya (UPC).
- **Adreça postal:** Avinguda de les Bases de Manresa, 61-73, 08242 Manresa (Barcelona).

13. Consentiment i declaració del participant

Abans de procedir a la signatura del document annex anomenat *"CONSENTIMENT INFORMAT DE LA PERSONA PARTICIPANT"*, assegureu-vos que es compleixen els següents punts:

- Heu llegit i entès completament la informació facilitada en aquest document.
- Heu pogut formular totes les preguntes que heu considerat oportunes i l'equip investigador us ha resolt els dubtes de manera satisfactòria.
- Enteneu perfectament que la vostra participació és totalment voluntària i que us podeu retirar de l'estudi en qualsevol moment sense haver de donar explicacions.
- Accepteu de manera lliure i conscient participar en aquest estudi experimental.

Si esteu d'acord amb tots els punts anteriors i voleu formalitzar la vostra participació, us preguem que empliu i signeu les caselles corresponents en el document **CONSENTIMENT INFORMAT DE LA PERSONA PARTICIPANT** *que us lliurarà l'investigador a l'inici de la sessió.*

D.3.2 English Version

Helena Cadevall Soto

Participant Information Sheet

Study Title

Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback

1 Objective of the Study

The aim of this study is to compare whether an interaction with a social robot or with a conventional computer interface is more suitable for guiding breathing exercises with biofeedback aimed at stress regulation.

In addition, the subjective measures obtained through questionnaires will be compared with the recorded physiological measures (heart rate and electroencephalographic activity), with the aim of exploring the potential of the latter as an objective tool for stress assessment.

This study is a proof of concept that will help assess the feasibility of this methodology and guide future studies in the field of social robotics applied to emotional wellbeing.

2. Duration of Participation

Participation in this study consists of a single experimental session lasting approximately 50 to 60 minutes in total.

This duration includes all phases of the experiment, from the initial explanation to the final debriefing, including sensor placement, completion of questionnaires, and experimental tasks.

No follow-up or additional sessions are required.

3. Who Can Participate?

People who meet the following criteria may take part:

- Over 18 years of age
- No serious clinical diagnoses related to stress, anxiety, or depression
- No medical or neurological conditions that could interfere with the physiological measurements
- Ability to understand and follow the experiment's instructions
- Acceptance and signature of informed consent

People with conditions that could affect the physiological measurements or the stress response may not participate.

4. What Will You Do During the Study?

All phases of this study will take place in person at the facilities of the Escola Politècnica Superior d'Enginyeria de Manresa (EPSEM) of the Universitat Politècnica de Catalunya (UPC).

Participation consists of a single session lasting approximately 50–60 minutes, structured as follows:

1. Initial explanation of the study and clarification of any questions.
2. Signing of the informed consent form.
3. Placement of non-invasive sensors (heart rate chest strap and EEG headset).
4. Completion of initial questionnaires.
5. Measurement of resting activity (baseline state).
6. Moderate stress induction task (Stroop Task).
7. Guided relaxation phase with biofeedback through the assigned interaction (the *Tyto* social robot or the computer interface).
8. Completion of final questionnaires.
9. A brief evaluation interview.
10. Final debriefing and closing of the session.

5. Devices Used

Non-invasive devices will be used during the study:

- EEG headset (Enobio 8) to record brain activity
- Heart rate chest strap (Magene H603) for HRV and heart rate
- *Tyto* social robot (an in-house designed robot shaped like an owl) or computer interface for the interaction

6. Benefits and Risks

Benefits: There are no direct benefits for the participant, but participation contributes to scientific research.

Possible risks or discomforts:

- Mild and temporary stress during the cognitive task
- Mild mental fatigue
- Mild discomfort from the sensors
- Tiredness due to the length of the session

No physical risks or invasive procedures are anticipated. The social robot will remain completely static throughout the session, with no movement or physical contact with the participant. The recorded physiological measures (ECG/HRV and EEG) will be used exclusively for research purposes and do not allow inference of thoughts, memories, or other mental content.

7. Voluntary Participation

Participation is entirely voluntary. You may:

- Withdraw from the study at any time
- Decline to answer any questions you do not wish to
- Withdraw your consent without any consequences

8. Confidentiality and Data Protection

Personal data collected in this study will be processed in accordance with Regulation (EU) 2016/679 (General Data Protection Regulation - GDPR) and Organic Law 3/2018 (LOPDGDD).

- **Pseudonymisation:** Data will be associated with an alphanumeric code (participant ID). The list linking this code to your real identity will be stored separately and protected by the research team.
- **Security:** Experimental data will be stored on secure servers with restricted access, and only the authorised research team will have access to it.
- **Access:** Only the principal investigator and the project supervisors will have access to identifying data.
- **Retention period:** Identifying data will be kept for a maximum of two years after the study's completion. It will subsequently be deleted, retaining only the fully anonymised data necessary to support the scientific results.

Basic information on personal data protection:

Data Controller	Universitat Politècnica de Catalunya (UPC). Department of Mining, Industrial and ICT Engineering (EMIT).
Purpose	Management of data for the research project "Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback".
Legal basis	Consent of the data subject. You may revoke it at any time by contacting the research team.
Recipients	Data will not be disclosed to third parties, except where legally required. The research results will be presented exclusively within the academic scope of the Master's Thesis (TFM), on an anonymous basis.
Rights	You may request access, rectification, erasure, or restriction of processing by sending an email to the research team. More information at: `https://www.upc.edu/normatives/ca/proteccio-de-dades/normativa-europea-de-proteccio-de-dades/drets`
Retention period	A maximum of 2 years from the completion of the research (coinciding with the closure of the academic assessment of the TFM).
Data Protection Officer	Universitat Politècnica de Catalunya, Plaça Eusebi Güell, 6, edifici Vèrtex, 08034 Barcelona. Email: `proteccio.dades@upc.edu`
Complaints	You may file a complaint with the Catalan Data Protection Authority (APDCAT) at: `https://apdcat.gencat.cat`

9. Publication of Data

Fully anonymised experimental data may be published in open scientific data repositories in order to facilitate the reproduction and reuse of the results by the scientific community. Under no circumstances will information that could identify participants be published.

10. Use of Images and Audio Recording

It is expressly guaranteed that no photographic capture or video recording of participants will take place during the experimental session.

The only multimedia recording associated with this study will be the exclusive audio recording of the final interview (detailed in Section 4, point 9). This procedure has the sole methodological purpose of facilitating the subsequent textual transcription and indexing of responses for qualitative analysis.

The resulting audio file will be stored in a secure system with restricted access and will be permanently and irreversibly deleted once the transcription process is complete. The indexed text will be associated exclusively with the pseudonymisation alphanumeric code (participant ID), ensuring that no textual fragment can be linked to the

subject's real identity. The performance of this recording depends exclusively on the participant granting explicit consent through their signature on the corresponding Informed Consent Form.

11. Withdrawal of Data

The participant may request the withdrawal of their data at any time. If the data have already been anonymised, it may not be possible to remove them individually.

12. Contact

For any questions, clarifications, or additional information regarding this study, you may contact the research team using the following details:

- **Principal Investigator:** Helena Cadevall Soto
  - Email: helena.cadevallsoto@ip-paris.fr
- **Research Supervisors (UPC):**
  - Marta Isabel Tarrés Puertas (Email: marta.isabel.tarres@upc.edu — Office phone: 938777282)
  - Linus Manubens-Gil (Email: linus.manubens@upc.edu — Office phone: 938777391)
- **Responsible Department and Centre:** Department of Mining, Industrial and ICT Engineering (EMIT). Escola Politècnica Superior d'Enginyeria de Manresa (EPSEM), Universitat Politècnica de Catalunya (UPC).
- **Postal address:** Avinguda de les Bases de Manresa, 61-73, 08242 Manresa (Barcelona).

13. Consent and Participant's Declaration

Before signing the attached document entitled *"INFORMED CONSENT OF THE PARTICIPANT"*, please make sure that the following points are met:

- You have read and fully understood the information provided in this document.
- You have been able to ask all the questions you considered necessary, and the research team has satisfactorily addressed your doubts.
- You fully understand that your participation is entirely voluntary and that you may withdraw from the study at any time without having to give any explanation.
- You freely and knowingly agree to take part in this experimental study.

If you agree with all of the above points and wish to formalise your participation, please fill in and sign the corresponding boxes on the **INFORMED CONSENT OF THE PARTICIPANT** *document, which will be given to you by the researcher at the start of the session.*

D.4 Informed Consent Form

The blank informed consent form is given first in Catalan, which is the version signed by the participants, and then in English.

D.4.1 Catalan Version

CONSENTIMENT INFORMAT DE LA PERSONA PARTICIPANT

[Camps que ha d'emplenar l'investigador/la investigadora]

Títol del projecte de recerca: Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback

Naturalesa de l'experiment: Estudi experimental i pilot (de tipus *proof of concept*) aleatoritzat entre subjectes. Implica la interacció persona-tecnologia i el registre no invasiu de senyals fisiològiques i cognitives en una única sessió de laboratori.

Propòsit/finalitat: Avaluar i comparar l'efecte d'un robot social (Tyto) i d'una interfície convencional basada en pantalla en una sessió de respiració guiada amb biofeedback per a la regulació de l'estrès. També es pretén estudiar la relació entre les mesures subjectives d'estrès obtingudes mitjançant qüestionaris i les mesures fisiològiques (HRV i EEG), així com explorar el potencial d'aquestes últimes com a eina objectiva d'avaluació en estudis d'interacció persona-robot i persona-ordinador.

El mètode que s'utilitzarà i les formes en què es durà a terme: L'estudi es realitzarà en una única sessió individual de 50 a 60 minuts. El procediment consta de: 1) Col·locació de sensors no invasius (cinturó de freqüència cardíaca Magene H603 i casc EEG Enobio 8); 2) Resposta a qüestionaris inicials; 3) Fase de repòs per establir la línia de base; 4) Fase d'inducció d'estrès mitjançant una tasca cognitiva sota pressió temporal (*Stroop Task*); 5) Fase de relaxació amb exercicis de respiració guiada amb biofeedback en temps real (segons la condició assignada: robot o ordinador); 6) Qüestionaris postexperimentals i una breu entrevista final enregistrada exclusivament en àudio.

Tots els participants seguiran exactament el mateix protocol experimental. L'única diferència entre condicions serà la interfície utilitzada durant la fase de respiració guiada amb biofeedback (robot social o ordinador).

Inconvenients i riscos que es poden presentar: Els riscos associats a la participació s'avaluen com a molt baixos. Poden incloure un estrès lleu, general i estrictament temporal durant la tasca cognitiva d'activació, una lleu fatiga mental derivada d'aquesta tasca i una lleugera incomoditat física ocasionada per l'ús dels sensors (cinturó de freqüència cardíaca i casc EEG) durant la sessió. Com a mesura de mitigació, l'investigador supervisarà contínuament el desenvolupament de l'experiment i el participant podrà interrompre la sessió o fer pauses en qualsevol moment, sense necessitat de justificar la seva decisió i sense cap conseqüència.

El robot social romandrà completament estàtic durant tota la sessió i mostrarà els exercicis de respiració mitjançant una representació visual a la seva pantalla frontal i senyals sonors indicatius, sense realitzar moviments mecànics ni establir contacte físic amb el participant.

L'activitat electroencefalogràfica registrada mitjançant el dispositiu Enobio 8 s'utilitzarà exclusivament per obtenir indicadors fisiològics generals relacionats amb la resposta a l'estrès. Les dades registrades no permeten inferir pensaments, records, intencions ni altres continguts mentals dels participants.

[Camps que ha d'emplenar la persona participant]

La persona participant voluntària ha de llegir i contestar les preguntes següents amb atenció. Per participar en aquest projecte se li ha facilitat la documentació següent:

1. **Full d'Informació al Participant** (Document de 4 pàgines que detalla els objectius, el procediment, els dispositius utilitzats, els riscos/beneficis i els contactes de l'equip investigador).

1

	SÍ	NO
Heu llegit tota informació que us ha estat facilitada sobre aquest projecte?	☐	☐
Heu tingut l'oportunitat de preguntar i comentar qüestions sobre el projecte?	☐	☐
Heu rebut prou informació sobre aquest projecte?	☐	☐
Heu rebut respostes satisfactòries a totes les preguntes?	☐	☐
Heu comprès que sou lliure d'abandonar aquest projecte sense que aquesta decisió pugui ocasionar-vos cap perjudici?	☐	☐
– En qualsevol moment	☐	☐
– Sense donar-ne cap raó	☐	☐
Heu comprès els possibles riscos associats a la vostra participació en aquest projecte?	☐	☐
Esteu d'acord amb l'enregistrament exclusiu en àudio de l'entrevista final per a la seva posterior transcripció textual, tal com es detalla a l'Apartat 9 del Full d'Informació?	☐	☐
Esteu d'acord a participar-hi?	☐	☐
Rebreu algun tipus de compensació per participar-hi?	☐	☐

Quin/a investigador/a us ha parlat d'aquest projecte? (nom i cognoms):

_____

Nom i cognoms de la persona voluntària: **Signatura:**

_____ _____

Data: .....

En signar aquest document dono el consentiment a la UPC per al tractament de les dades de caràcter personal recollides en aquest formulari, tal com es descriu en la taula següent:

Informació de protecció de dades	
Responsable del tractament	Universitat Politècnica de Catalunya. Departament d'Enginyeria Minera, Industrial i TIC (EMIT). Web: `https://emit.upc.edu/`. Investigadora principal: Helena Cadevall Soto (`helena.cadevallsoto@ip-paris.fr`) Tutores / Supervisors (UPC): Marta Isabel Tarrés Puertas (`marta.isabel.tarres@upc.edu`) i Linus Manubens-Gil (`linus.manubens@upc.edu`).
Delegat de protecció dades	Universitat Politècnica de Catalunya, Plaça Eusebi Güell, 6, edifici Vèrtex, 08034 Barcelona (Espanya). Correu: `proteccio.dades@upc.edu`
Finalitat del tractament	F05.4 Gestió de dades de projectes de recerca i innovació. Gestió del consentiment informat de participació en una investigació sobre "Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback".
Legitimació	Consentiment de la persona interessada. Pot revocar el seu consentiment en qualsevol moment posant-se en contacte amb l'equip investigador a través dels correus indicats a la secció de Responsable del tractament.
Destinataris	Les dades identificatives no es comunicaran a terceres persones. Únicament podran compartir-se dades completament anonimitzades amb finalitats de recerca, reproducció científica o dipòsit en repositoris de dades obertes, d'acord amb la normativa vigent i sempre que no existeixi un risc raonable de reidentificació dels participants.
Drets de les persones	Sol·licitar l'accés, la rectificació o supressió de les dades, i la limitació del tractament. Oposar-se al tractament. Portabilitat de les dades. Més informació a: `https://www.upc.edu/normatives/ca/proteccio-de-dades/normativa-europea-de-proteccio-de-dades/drets`
Termini de conservació	Les dades identificatives es conservaran durant un màxim de dos anys des de la finalització de la recerca. Posteriorment s'eliminaran, mantenint únicament les dades completament anonimitzades necessàries per justificar els resultats científics i, si escau, publicar-les en repositoris científics de dades obertes. Més informació a: `https://www.upc.edu/normatives/ca/proteccio-de-dades/normativa-europea-de-proteccio-de-dades/politica-de-conservacio-de-les-dades-de-caracter-personal`
Reclamació	Si no us ha satisfet l'exercici dels vostres drets, podeu presentar una reclamació davant l'APDCAT a: `https://apdcat.gencat.cat`

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En cas que més endavant vulgueu fer alguna pregunta o comentari sobre aquest projecte, o bé si voleu revocar la vostra participació, poseu-vos en contacte amb:

Nom de l'investigador o investigadora:
Helena Cadevall Soto (Investigadora principal), sota la supervisió de Marta Isabel Tarrés Puertas i Linus Manubens-Gil.

Departament, facultat i adreça:
Departament d'Enginyeria Minera, Industrial i TIC (EMIT). Escola Politècnica Superior d'Enginyeria de Manresa (EPSEM), Universitat Politècnica de Catalunya (UPC). Avinguda de les Bases de Manresa, 61-73, 08242 Manresa (Barcelona).

Correu de contacte:
`helena.cadevallsoto@ip-paris.fr` / `marta.isabel.tarres@upc.edu` / `linus.manubens@upc.edu`

Telèfon/s de contacte:
938777282 (Marta Isabel Tarrés Puertas) / 938777391 (Linus Manubens-Gil)

Lloc, data i signatura de l'investigador/la investigadora:
Manresa, ...de .....de 2026.

Signatura:

D.4.2 English Version

INFORMED CONSENT OF THE PARTICIPANT

[Fields to be completed by the researcher]

Title of the research project: Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback

Nature of the experiment: Randomised, between-subjects experimental and pilot study (proof of concept type). It involves human-technology interaction and the non-invasive recording of physiological and cognitive signals in a single laboratory session.

Purpose/aim: To evaluate and compare the effect of a social robot (Tyto) and a conventional screen-based interface in a guided breathing session with biofeedback for stress regulation. It also aims to study the relationship between the subjective stress measures obtained through questionnaires and the physiological measures (HRV and EEG), as well as to explore the potential of the latter as an objective assessment tool in human-robot and human-computer interaction studies.

The method to be used and how it will be carried out: The study will take place in a single individual session lasting 50 to 60 minutes. The procedure consists of: 1) Placement of non-invasive sensors (Magene H603 heart rate chest strap and Enobio 8 EEG headset); 2) Completion of initial questionnaires; 3) Resting phase to establish the baseline; 4) Stress induction phase using a cognitive task under time pressure (Stroop Task); 5) Relaxation phase with guided breathing exercises using real-time biofeedback (according to the assigned condition: robot or computer); 6) Post-experimental questionnaires and a brief final interview recorded exclusively in audio.

All participants will follow exactly the same experimental protocol. The only difference between conditions will be the interface used during the guided breathing phase with biofeedback (social robot or computer).

Possible inconveniences and risks: The risks associated with participation are assessed as very low. They may include mild, general and strictly temporary stress during the cognitive activation task, mild mental fatigue resulting from this task, and slight physical discomfort caused by the use of the sensors (heart rate chest strap and EEG headset) during the session. As a mitigation measure, the researcher will continuously supervise the progress of the experiment, and the participant may stop the session or take breaks at any time, without having to justify their decision and without any consequences.

The social robot will remain completely static throughout the session and will display the breathing exercises through a visual representation on its front screen and indicative sound signals, without performing any mechanical movements or making physical contact with the participant.

The electroencephalographic activity recorded using the Enobio 8 device will be used exclusively to obtain general physiological indicators related to the stress response. The recorded data do not allow inference of thoughts, memories, intentions, or other mental content of the participants.

[Fields to be completed by the participant]

The voluntary participant must read and answer the following questions carefully. You have been provided with the following documentation to take part in this project:

1. **Participant Information Sheet** (4-page document detailing the objectives, procedure, devices used, risks/benefits, and the research team's contact details).

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	YES	NO
Have you read all the information provided to you about this project?	☐	☐
Have you had the opportunity to ask and discuss questions about the project?	☐	☐
Have you received sufficient information about this project?	☐	☐
Have you received satisfactory answers to all your questions?	☐	☐
Have you understood that you are free to withdraw from this project without this decision causing you any harm?	☐	☐
– At any time	☐	☐
– Without giving any reason	☐	☐
Have you understood the possible risks associated with your participation in this project?	☐	☐
Do you agree to the audio-only recording of the final interview for its subsequent textual transcription, as detailed in Section 9 of the Information Sheet?	☐	☐
Do you agree to take part?	☐	☐
Will you receive any kind of compensation for taking part?	☐	☐

Which researcher told you about this project? (first and last name):

_____

Full name of the volunteer: _____ **Signature:** _____

Date: .....

By signing this document, I give my consent to the UPC for the processing of the personal data collected in this form, as described in the following table:

Data protection information	
Data controller	Universitat Politècnica de Catalunya. Department of Mining, Industrial and ICT Engineering (EMIT). Website: `https://emit.upc.edu/`. Principal investigator: Helena Cadevall Soto (`helena.cadevallsoto@ip-paris.fr`) Supervisors (UPC): Marta Isabel Tarrés Puertas (`marta.isabel.tarres@upc.edu`) and Linus Manubens-Gil (`linus.manubens@upc.edu`).
Data Protection Officer	Universitat Politècnica de Catalunya, Plaça Eusebi Güell, 6, edifici Vèrtex, 08034 Barcelona (Spain). Email: `proteccio.dades@upc.edu`
Purpose of processing	F05.4 Management of data for research and innovation projects. Management of informed consent to take part in research on "Comparative Evaluation of Human–Robot and Display-Based Interaction for Stress Regulation using Physiological Biofeedback".
Legal basis	Consent of the data subject. You may revoke your consent at any time by contacting the research team through the emails indicated in the Data Controller section.
Recipients	Identifying data will not be disclosed to third parties. Only fully anonymised data may be shared for research purposes, scientific reproduction, or deposit in open data repositories, in accordance with current regulations and provided there is no reasonable risk of participant re-identification.
Rights of individuals	To request access, rectification, or erasure of the data, and restriction of processing. To object to the processing. Data portability. More information at: `https://www.upc.edu/normatives/ca/proteccio-de-dades/normativa-europea-de-proteccio-de-dades/drets`
Retention period	Identifying data will be kept for a maximum of two years from the completion of the research. It will subsequently be deleted, retaining only the fully anonymised data necessary to support the scientific results and, where applicable, to publish them in open scientific data repositories. More information at: `https://www.upc.edu/normatives/ca/proteccio-de-dades/normativa-europea-de-proteccio-de-dades/politica-de-conservacio-de-les-dades-de-caracter-personal`
Complaints	If you are not satisfied with the exercise of your rights, you may file a complaint with the APDCAT at: `https://apdcat.gencat.cat`

If you would like to ask a question or make a comment about this project at a later date, or if you wish to withdraw your participation, please contact:

Name of the researcher:
Helena Cadevall Soto (Principal Investigator), under the supervision of Marta Isabel Tarrés Puertas and Linus Manubens-Gil.

Department, school and address:
Department of Mining, Industrial and ICT Engineering (EMIT). Escola Politècnica Superior d'Enginyeria de Manresa (EPSEM), Universitat Politècnica de Catalunya (UPC). Avinguda de les Bases de Manresa, 61-73, 08242 Manresa (Barcelona).

Contact email:
`helena.cadevallsoto@ip-paris.fr` / `marta.isabel.tarres@upc.edu` / `linus.manubens@upc.edu`

Contact phone number(s):
938777282 (Marta Isabel Tarrés Puertas) / 938777391 (Linus Manubens-Gil)

Place, date and signature of the researcher:
Manresa, ... of ..... 2026.

Signature: